

2025

Colorado Statewide Inventory of Greenhouse Gas Emissions and Sinks

Chapter 3: Energy

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COLORADO

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Department of Public Health & Environment



2025 COLORADO GREENHOUSE GAS INVENTORY

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ACRONYMS AND ABBREVIATIONS

Greenhouse Gases

CH ₄	Methane
CO ₂	Carbon Dioxide
N ₂ O	Nitrous Oxide

Agencies and Organizations

ACR	American Carbon Registry
API	American Petroleum Institute
AQCC	Air Quality Control Commission
ECMC	Energy and Carbon Management Commission
EIA	Energy Information Administration
EPA	Environmental Protection Agency
FERC	Federal Energy Regulatory Commission
FHWA	Federal Highway Administration
GMI	Global Methane Initiative
GRI	Gas Research Institute
IPCC	Intergovernmental Panel on Climate Change
MSHA	Mine Safety and Health Administration
NAICS	North American Industry Classification System
PHMSA	Pipeline and Hazardous Materials Safety Administration
RCSPs	Regional Carbon Sequestration Partnerships

Other Abbreviations

AR5	IPCC Fifth Assessment Report
CCS	Carbon Capture and Storage
CEP	Clean Energy Plan
CHP	Clean Heat Plan
CFR	Code of Federal Regulations
CMOP	Coalbed Methane Outreach Program
CNG	Compressed Natural Gas
CO ₂ FFC	CO ₂ Emissions from Fossil Fuel Combustion
CO ₂ eq	CO ₂ equivalent
EOR	Enhanced Oil Recovery
FRD	Facility Reported Data
GHG	Greenhouse Gas
GHGRP	Greenhouse Gas Reporting Program
GWP	Global Warming Potential
I/M	Inspection and Maintenance
IPPU	Industrial Processes and Product Use
LDC	Local Distribution Company
LNG	Liquefied Natural Gas
LPG	Liquefied Petroleum Gas

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LULUCF	Land Use, Land-Use Change, and Forestry
MOVES	Motor Vehicle Emission Simulator
NESHAP	National Emission Standards for Hazardous Air Pollutants
NEUFFs	Non-Energy Use of Fossil Fuels
NEU	Non-Energy Use
NGPS	Natural Gas and Petroleum Systems
NI by State	National Inventory by State
NIR	National Inventory Report
NSPS	New Source Performance Standards
ONGAEIR	Oil and Natural Gas Annual Emissions Inventory Reporting
QA/QC	Quality Assurance and Quality Control
RCI	Residential, Commercial, and Industrial
RECs	Reduced Emissions Completions
SEDS	State Energy Data System
SIT	State Inventory Tool
SNG	Synthetic Natural Gas
TIGS	Transport, Injection, and Geologic Storage
U.S.	United States
VMT	Vehicle Miles Traveled
VOCs	Volatile Organic Compounds

UNITS

Mass

lb	pound	1 lb = 0.453592 kg
kg	kilogram	1 kg = 1,000 g = 2.20462 lb
Mg	megagram	1 Mg = 1,000 kg = 1 metric ton
Tg	teragram	1 Tg = 10 ⁶ Mg = 10 ⁹ kg
MT	metric ton, or "tonne"	1 MT = 1,000 kg = 1.10231 short tons
MMT	million metric tons	1 MMT = 1 Tg

Volume

gal	US gallon	1 gal = 3.78541 L
bbl	barrel	1 bbl = 42 gal = 158.987 L
MMbbl	million barrels	1 MMbbl = 106 bbl
scf	standard cubic feet*	
MMscf	million standard cubic feet*	1 MMscf = 10 ⁶ scf

Energy

Btu	British thermal unit	1 Btu = 0.293071 Wh
MMBtu	million Btu	1 MMBtu = 293.071 kWh
Wh	watt-hour	1 Wh = 3.41214 Btu
kWh	kilowatt-hour	1 kWh = 1,000 Wh
MWh	megawatt-hour	1 MWh = 1,000 kWh

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Temperature

°C	degree Celsius	$1\text{ }^{\circ}\text{C} = 33.8\text{ }^{\circ}\text{F}$
°F	degrees Fahrenheit	$1\text{ }^{\circ}\text{F} = -17.222\text{ }^{\circ}\text{C}$

Concentration

ppm	parts per million
-----	-------------------

Length

in	inch	$1\text{ in} = 2.54\text{ cm}$
cm	centimeter	$1\text{ cm} = 0.01\text{ m}$
m	meter	$1\text{ m} = 100\text{ cm}$
km	kilometer	$1\text{ km} = 1,000\text{ m}$

* Standard volume measures the amount of gas present in a given volume at a reference temperature and pressure. At different temperatures and pressures, 1 scf of a given gas will occupy different volumes, but will always have the same mass and the same number of molecules.

3. ENERGY

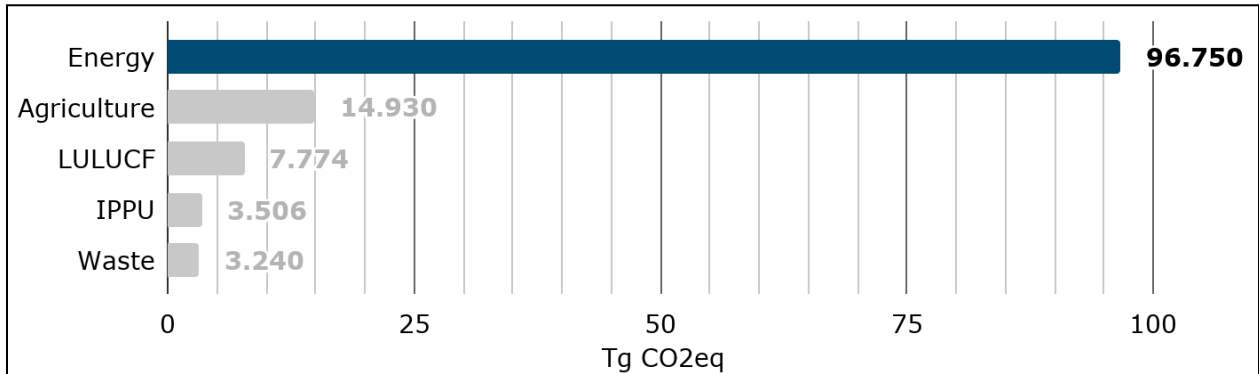
The Energy sector includes activities related to the production, transformation, and consumption of energy. These activities generate carbon dioxide (CO₂), methane (CH₄), and nitrous oxide (N₂O) emissions through a variety of pathways, including fossil fuel combustion, fugitive leaks, and intentional releases. The Energy sector in this Inventory includes the following subsectors: Coal Mining; CO₂ Transport, Injection, and Geologic Storage (CO₂ TIGS); Electric Power; Natural Gas and Petroleum Systems (NGPS); Non-Energy Use of Fossil Fuels (NEUFFs); Residential, Commercial, and Industrial (RCI) Fuel Use; and Transportation.

Biogenic CO₂ emissions are not included in the Energy sector. In accordance with IPCC methodological guidelines, these emissions are captured in the net CO₂ flux from carbon stock changes under the Land Use, Land-Use Change, and Forestry (LULUCF) sector (Chapter 6). However, biogenic CH₄ and N₂O emissions from biomass combustion are included in this sector, as they are not accounted for in the LULUCF sector.

All emissions are reported in teragrams (Tg) of GHG or CO₂ equivalent (CO₂eq) calculated using 100-year Global Warming Potential (GWP) values from the Intergovernmental Panel on Climate Change (IPCC) [Fifth Assessment Report](#) (AR5), unless noted otherwise. Chapter 1 discussed the use of GWP values to express emissions in CO₂eq. For all categories with estimated emissions, this chapter outlines their methodology, data sources, uncertainty, quality assurance and quality control (QA/QC) steps, comparisons to the [prior Inventory \(2023\)](#), and planned improvements. This Inventory utilized several data sources and methodologies, including the Environmental Protection Agency’s (EPA) State Inventory Tool (SIT) Modules, National Inventory Report (NIR), and National Inventory by State (NI by State). This Inventory also utilized fuel consumption from the Energy Information Administration (EIA) [State Energy Data System \(SEDS\)](#) and facility-reported data (FRD) under the EPA and Colorado’s Greenhouse Gas Reporting Programs (GHGRPs).

In 2023, Energy emissions were 96.750 Tg CO₂eq, accounting for 76.7% of statewide emissions, including LULUCF, and 81.7% excluding LULUCF. As Figure 3.1 shows, emissions from the Energy sector were more than six times larger than any other subsector in 2023.

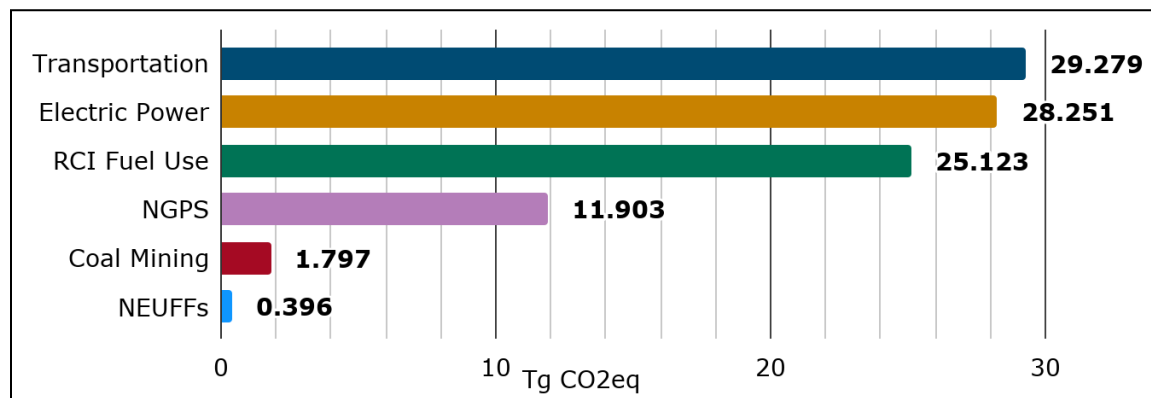
Figure 3.1. 2023 Statewide Emissions by Sector.



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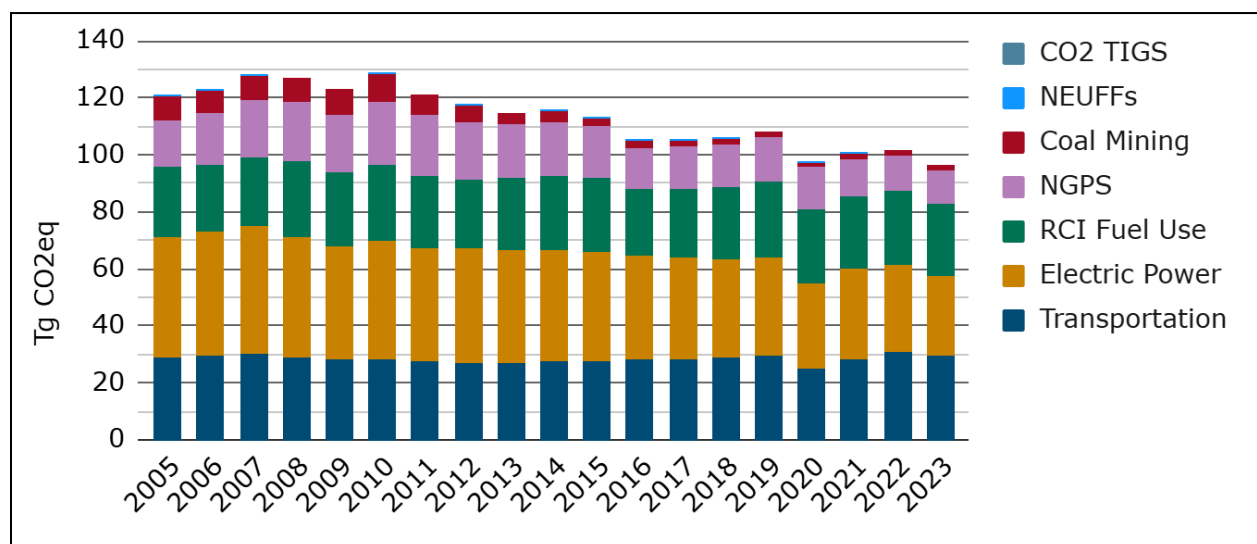
As shown in Figure 3.2, the Energy sector's 2023 emissions were largely from Transportation, Electric Power, and RCI Fuel Use, which collectively represented around 82.653 Tg CO₂eq or about 85% of 2023 Energy emissions. Transportation, Electric Power, and RCI Fuel Use accounted for 30.3% (29.279 Tg CO₂eq), 29.2% (28.251 Tg CO₂eq), and 26.0% (25.123 Tg CO₂eq) of Energy emissions in 2023, respectively. NGPS emissions accounted for 12.3% (11.903 Tg CO₂eq), and Coal Mining emissions accounted for 1.9% (1.797 Tg CO₂eq) of the Energy sector in 2023. Emissions from NEUFFs and CO₂ TIGS accounted for just 0.4% (0.396 Tg CO₂eq) and less than 0.01% (less than 0.0001 Tg CO₂eq), respectively. Because CO₂ TIGS emissions were de minimis, they are not included in the figure below.

Figure 3.2. 2023 Energy Emissions by Subsector.



Historically, the same three subsectors (Transportation, Electric Power, and RCI Fuel Use) accounted for the majority of Energy emissions, as shown in Figure 3.3. From 2005 through 2022, Electric Power emissions were higher than emissions from any other subsector, accounting for 30% to 35% of the sector's total emissions. However, Transportation became the highest-emitting subsector in 2023, surpassing emissions from Electric Power. Table 3.1 details the Energy sector and subsector emissions in select years.

Figure 3.3. Historical Energy Sector Emissions by Subsector.



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Table 3.1. Energy Sector Emissions by Subsector (Tg CO₂eq).

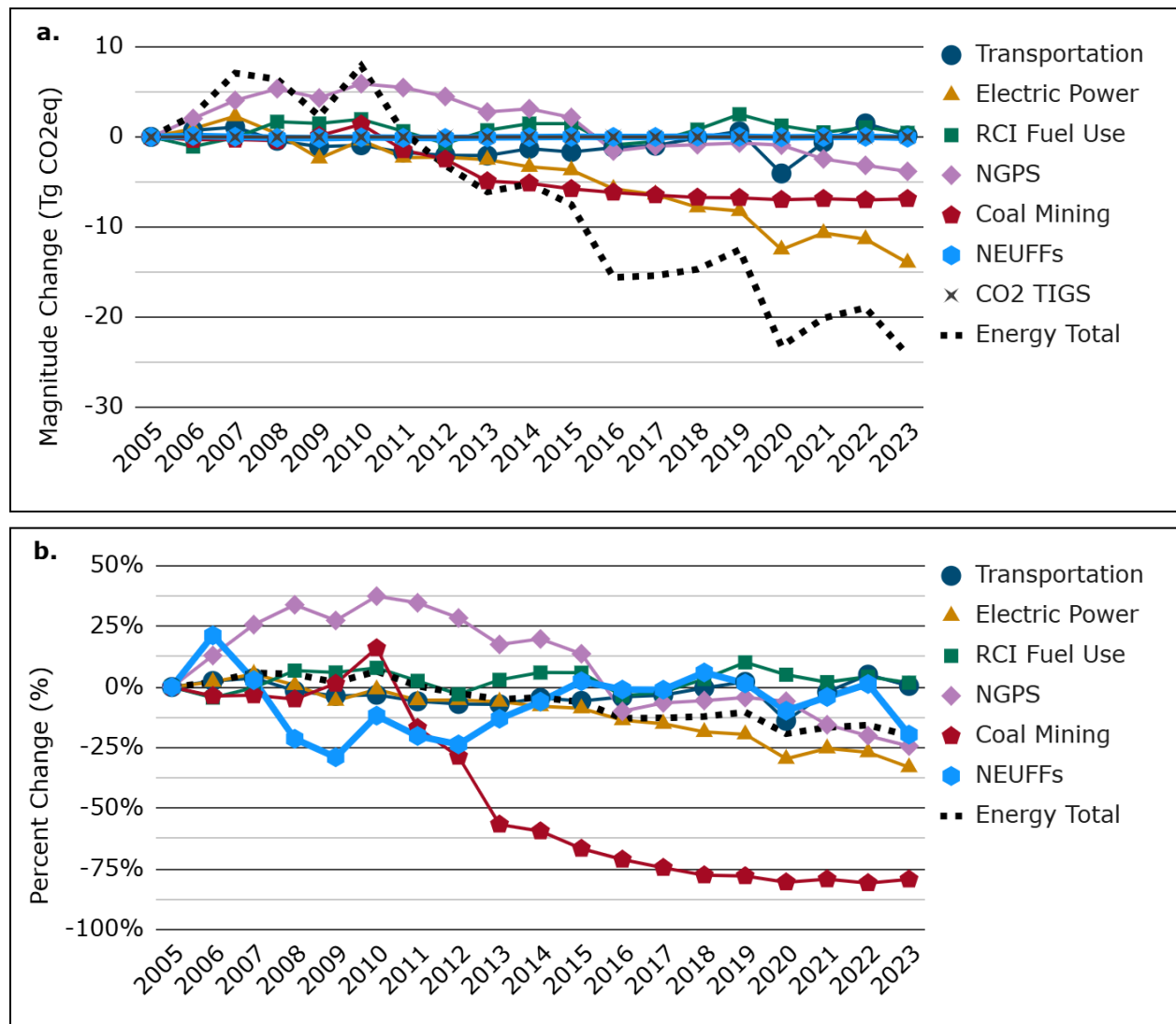
Subsector	2005	2010	2015	2020	2023
CO ₂ TIGS	0.000	+	+	+	+
Coal Mining	8.672	10.064	2.896	1.694	1.797
Electric Power	42.204	41.808	38.507	29.711	28.251
NGPS	15.733	21.631	17.891	14.814	11.903
NEUFFs	0.492	0.434	0.504	0.444	0.396
RCI Fuel Use	24.680	26.635	26.142	25.937	25.123
Transportation	29.138	28.212	27.470	25.077	29.279
Energy Sector Total	120.920	128.784	113.410	97.678	96.750

+ Emissions do not exceed 0.0005 Tg CO₂eq.

From 2005 through 2023, Energy emissions decreased by 24.170 Tg CO₂eq or 20.0%. In the same period, Electric Power emissions decreased by 13.953 Tg CO₂eq (33%). This decrease is most likely a result of the gradual replacement of coal-fired generation by renewable energy and natural gas. Emissions from Coal Mining decreased by 6.875 Tg CO₂eq (79%) since 2005, but decreased by 8.267 Tg CO₂eq (82.1%) since their peak in 2010. This change was likely driven by statewide mine closures. Emissions from NGPS and NEUFFs both decreased by 20% from 2005 through 2023, corresponding to reductions of 3.830 Tg CO₂eq and 0.097 Tg CO₂eq, respectively. Transportation and RCI Fuel Use were the only two Energy subsectors to see increased emissions from 2005 through 2023. Transportation emissions increased by 0.141 Tg CO₂eq (0.5%), and RCI Fuel Use emissions increased by 0.443 Tg CO₂eq (1.8%). Within RCI Fuel Use, Industrial Fuel Use emissions decreased while Residential and Commercial Fuel Use emissions increased. Figure 3.4 details the magnitude and percent changes since 2005 for the Energy sector and subsectors.

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Figure 3.4. (a) Magnitude and (b) Percent Changes in Energy Sector Emissions by Subsector Since 2005.

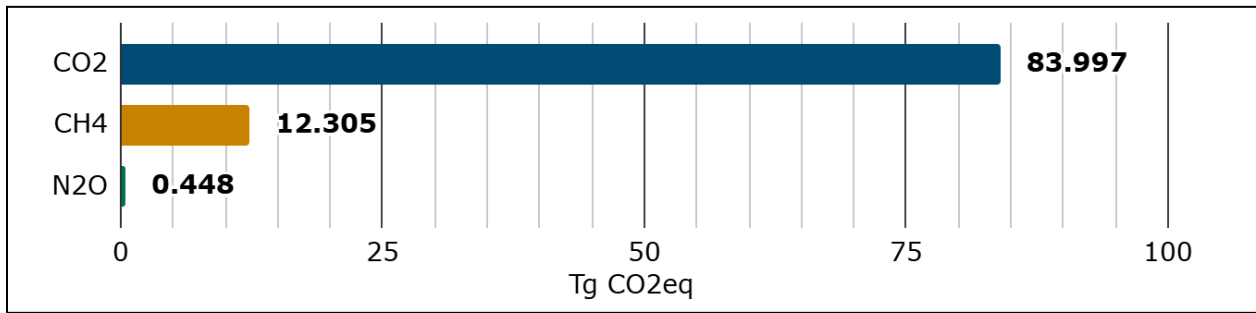


As shown in Figure 3.5, 2023 Energy CO₂ emissions were 83.997 Tg CO₂eq, CH₄ emissions were 12.305 Tg CO₂eq, and N₂O emissions were 0.448 Tg CO₂eq, respectively accounting for 86.8%, 12.7%, and 0.5% of the sector.

CO₂ emissions were primarily driven by subsectors with fossil fuel combustion. Transportation, Electric Power, and RCI Fuel Use respectively represented 34.4%, 33.5%, and 29.2% of 2023 Energy sector CO₂ emissions. The remaining 2.8% of CO₂ emissions were from NGPS, Coal Mining, and NEUFFs. NGPS accounted for most of the CH₄ emissions (81.0%), followed by Coal Mining (14.0%) and RCI Fuel Use (4.4%). Transportation and Electric Power together contributed less than 1%. N₂O emissions were largely associated with Transportation (72.4%) and Electric Power (21.4%), with smaller emissions from RCI Fuel Use (6.0%) and from NGPS (0.2%). Coal Mining is the only subsector that does not include N₂O emissions, whereas CO₂ TIGS and NEUFFs include only CO₂ emissions.

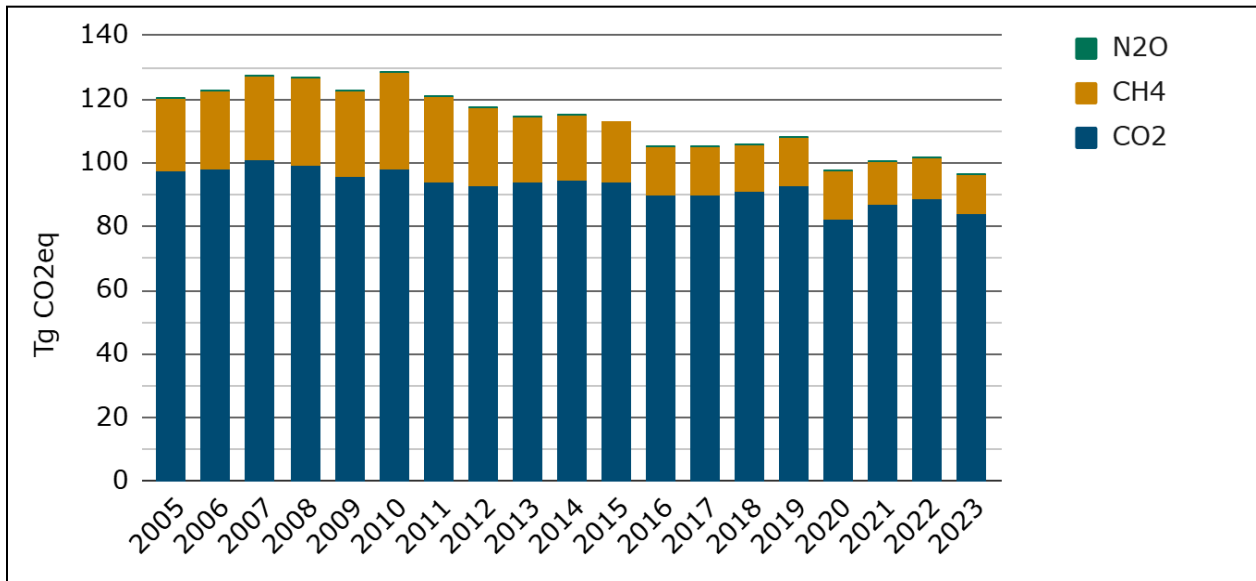
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Figure 3.5. 2023 Energy Sector Emissions by GHG.



From 2005 through 2023, the Energy sector was consistently dominated by CO₂ emissions, with smaller contributions from CH₄ and minimal contributions from N₂O, as shown in Figure 3.6. CO₂ emissions accounted for 76.2% to 87.0% of the Energy sector’s total emissions across the time series. CH₄ emissions accounted for 12.5% to 23.4% of the sector, and that share has been declining as CH₄ emissions have decreased. N₂O emissions accounted for up to 0.7% of the sector. Table 3.2 provides Energy emissions by GHG and subsector for select years.

Figure 3.6. Historical Energy Sector Emissions by GHG.



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Table 3.2. Energy Sector Emissions by GHG and Subsector (Tg CO₂eq).

GHG	Subsector	2005	2010	2015	2020	2023
CO ₂	CO ₂ TIGS	0.000	+	+	+	+
CO ₂	Coal Mining	0.287	0.201	0.133	0.064	0.077
CO ₂	Electric Power	42.023	41.582	38.304	29.568	28.123
CO ₂	NGPS	1.746	2.161	2.733	2.339	1.929
CO ₂	NEUFFs	0.492	0.434	0.504	0.444	0.396
CO ₂	RCI Fuel Use	24.252	25.961	25.303	25.223	24.557
CO ₂	Transportation	28.443	27.734	27.081	24.740	28.915
CO₂ Subtotal		97.245	98.074	94.058	82.379	83.997
CH ₄	Coal Mining	8.385	9.862	2.763	1.630	1.720
CH ₄	Electric Power	0.045	0.061	0.053	0.043	0.033
CH ₄	NGPS	13.986	19.469	15.157	12.475	9.973
CH ₄	RCI Fuel Use	0.400	0.641	0.804	0.688	0.539
CH ₄	Transportation	0.063	0.046	0.040	0.037	0.040
CH₄ Subtotal		22.879	30.079	18.817	14.873	12.305
N ₂ O	Electric Power	0.135	0.164	0.150	0.099	0.096
N ₂ O	NGPS	+	0.001	0.001	0.001	0.001
N ₂ O	RCI Fuel Use	0.028	0.033	0.035	0.027	0.027
N ₂ O	Transportation	0.632	0.433	0.348	0.299	0.324
N₂O Subtotal		0.796	0.631	0.535	0.426	0.448
Energy Sector Total		120.920	128.784	113.410	97.678	96.750

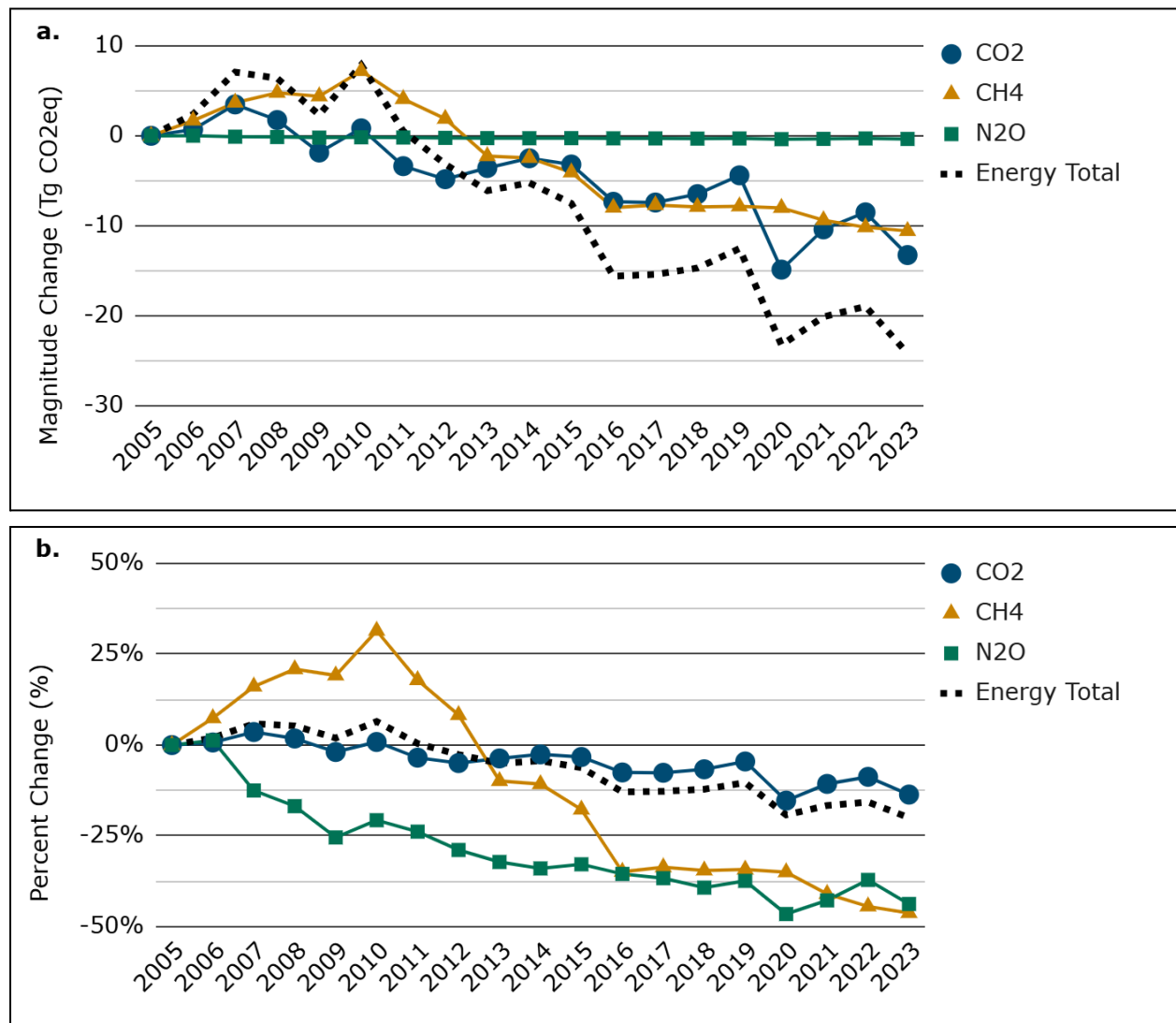
As illustrated in Figure 3.7, CO₂ emissions decreased by 13.248 Tg CO₂eq (13.6%) from 2005 through 2023 in the Energy sector. This change was primarily driven by a decrease in CO₂ emissions from the Electric Power subsector, which decreased by 13.901 Tg CO₂eq (33.1%). CO₂ emissions from NEUFFs and Coal Mining decreased by 0.097 Tg CO₂eq (19.6%) and 0.210 Tg CO₂eq (73.2%), respectively. In contrast, CO₂ emissions from Transportation, RCI Fuel Use, and NGPS increased by 0.472 Tg CO₂eq (1.7%), 0.305 Tg CO₂eq (1.3%), and 0.183 Tg CO₂eq (10.5%), respectively.

Energy CH₄ emissions decreased by 10.574 Tg CO₂eq (46.2%) from 2005 through 2023, mainly due to a reduction in Coal Mining emissions, which fell by 6.665 Tg CO₂eq (79.5%). CH₄ emissions from NGPS, Transportation, and Electric Power decreased by 4.013 Tg CO₂eq (28.7%), 0.023 Tg CO₂eq (36.5%), and 0.013 Tg CO₂eq (27.8%), respectively. CH₄ emissions from RCI Fuel Use increased by 0.139 Tg CO₂eq (34.8%).

Energy N₂O emissions decreased by 0.348 Tg CO₂eq (43.7%) from 2005 through 2023, mostly driven by the 0.307 Tg CO₂eq (48.7%) reduction in Transportation emissions. N₂O emissions from Electric Power and RCI Fuel Use decreased by 0.040 Tg CO₂eq (48.7%) and 0.001 Tg CO₂eq (4.3%). In contrast, N₂O emissions from NGPS increased by less than 0.0005 Tg CO₂eq (82.9%).

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Figure 3.7. (a) Magnitude and (b) Percent Changes in Energy Sector Emissions by GHG Since 2005.



Not Occurring/Not Estimated

Some activities in the Energy sector are not included in this Inventory because they do not occur in Colorado or because emissions were not estimated. The Division reviewed emissions estimates and methodologies in the SIT, NIR, NI by State, and FRD. Internal documentation and reports from operating facilities in Colorado using the relevant North American Industry Classification System (NAICS) were also reviewed to ensure that no activities were omitted. For activities that may be occurring but lack emissions estimates, the Division will continue to research the operations and identify potential datasets for inclusion in future inventories, where applicable.

Emissions from the Incineration of Waste would typically be included in the Energy sector, but they do not occur in Colorado. A review of known landfills and NAICS code 562213 Solid Waste

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Combustors and Incinerators confirmed that no such activities exist within the state. Emissions estimates across all reviewed sources were zero for the entire time series. While waste incineration is not separately reported under the state or federal GHGRPs, any related incidental combustion would be addressed under [Subpart C, General Stationary Fuel Combustion](#), and no such incidental combustion was reported in Colorado.

The CO₂ Transport, Injection, and Geologic Storage category is a new addition to this Inventory. These subsectors account for fugitive emissions from CCS projects, including transport systems, equipment at the injection site, and end containment once the CO₂ is sequestered.¹ However, emissions from the Injection and Geologic Storage subsectors were not estimated due to insufficient data. A review of the SIT, NI by State, NIR, and FRD confirmed that while estimates exist for CO₂ Transport, these sources did not provide emissions estimates for injection and storage in Colorado. Furthermore, emissions from Injection and Geologic Storage are based on data reported under [Subpart RR, Geologic Sequestration of Carbon Dioxide](#). Because Colorado had no Subpart RR reporters throughout the entire time series, no emissions were included for these specific subsectors in the NI by State, and they were likely not included in Colorado emissions estimates.

¹ [2025 NIR](#), Chapter 3, Section 3.9, CO₂ Transport, Injection, and Geological Storage, Introduction

3.1. Electric Power

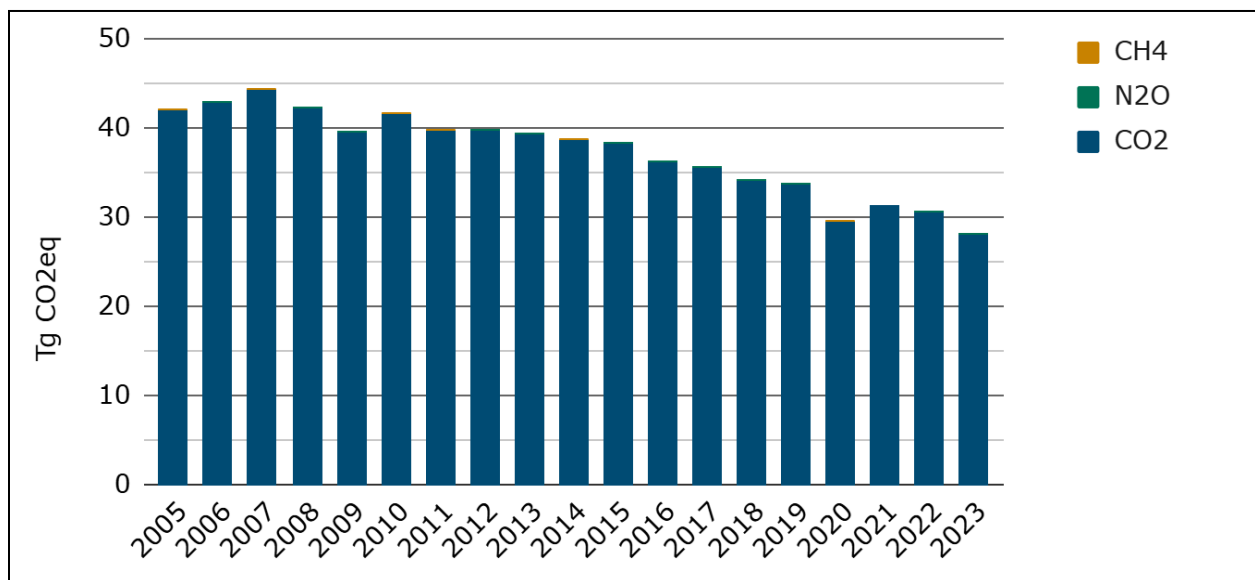
The Electric Power subsector represents direct CO₂, CH₄, and N₂O emissions from electricity generation. This subsector consists of electric utilities and independent power producers whose primary business is the production of electricity and are categorized under the NAICS 2022 codes within the Utilities Sector 22.² This Inventory does not currently disaggregate Electric Power emissions into further categories, such as fuel types. Biogenic CO₂ emissions (CO₂ emissions from biomass) are not included here and instead are captured in the net CO₂ flux from carbon stock changes in the LULUCF sector (Chapter 6).

While some commercial and industrial facilities produce electricity for their own use onsite, their primary business is not the production of electricity. Emissions associated with electricity generation for onsite use at these facilities are not included here and instead are captured under [RCI Fuel Use \(Section 3.2\)](#) in association with the primary purpose of their business.

In 2023, Electric Power emissions were 28.251 Tg CO₂eq, accounting for 29.2% of the Energy sector. Electric Power emissions accounted for about 23.9% of statewide emissions when LULUCF is excluded and 22.4% of statewide emissions when LULUCF is included.

From 2005 through 2023, Electric Power emissions decreased by 13.953 Tg CO₂eq (33.1%). Emissions peaked in 2007 at 44.494 Tg CO₂eq, shown in Figure 3.8, and have generally declined since then. Emissions in 2023 were the lowest since 2005.

Figure 3.8. Historical Electric Power Emissions by GHG.



² [U.S. Census, North American Industry Classification System \(NAICS\)](#)

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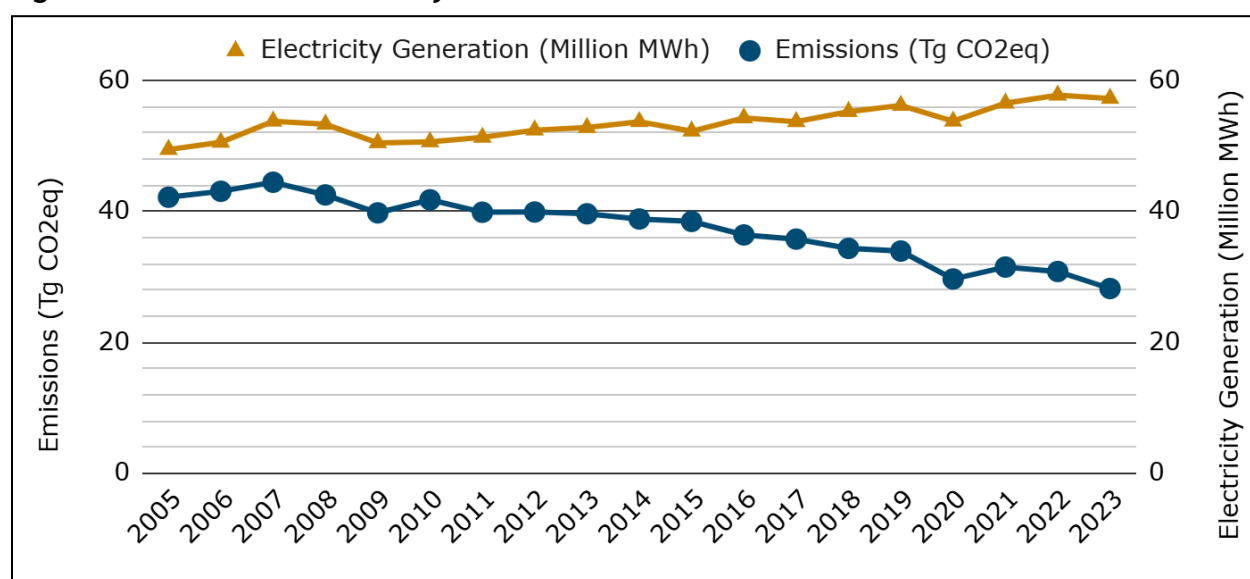
More than 99% of Electric Power emissions were CO₂ from 2005 through 2023. Over this period, CO₂ emissions decreased by 13.901 Tg CO₂eq (33.1%). Because CO₂ represents the vast majority of the subsector total, the overall decline in Electric Power emissions is driven almost entirely by the reduction in CO₂. N₂O and CH₄ also decreased from 2005 through 2023, by 0.040 Tg CO₂eq (29.3%) and 0.013 Tg CO₂eq (27.8%), respectively. Table 3.3 details Electric Power emissions by GHG for select years.

Table 3.3. Electric Power Emissions by GHG (Tg CO₂eq).

GHG	Subsector	2005	2010	2015	2020	2023
CO ₂	Electric Power	42.023	41.582	38.304	29.568	28.123
CH ₄	Electric Power	0.045	0.061	0.053	0.043	0.033
N ₂ O	Electric Power	0.135	0.164	0.150	0.099	0.096
Electric Power Total		42.204	41.808	38.507	29.711	28.251

Emissions from Electric Power have decreased despite an increase in electricity generation, as shown in Figure 3.9 below. These diverging trends are primarily due to the gradual replacement of coal-fired electricity generation with renewable energy and natural gas. From 2005 through 2023, electricity generation increased from about 49.5 million megawatt hours MWh (MWh) to 57.3 million MWh, an increase of nearly 16%. In 2005, about 4% of electricity was generated from renewable fuels, compared to 37% in 2023.³ As generation increases and emissions decrease, the emissions intensity of electricity generation declines as well. In 2005, the emissions intensity of electricity generation was about 0.853 Tg CO₂eq/million MWh, and in 2023 the intensity was 0.493 Tg CO₂eq/million MWh. Annual electricity generation data are provided in Table D.1 in Appendix D.

Figure 3.9. Historical Electricity Generation and Electric Power Emissions.



³ [EIA Electricity Data Browser](#), Colorado Electric Power Sector generation. Data as of April 9, 2026.

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3.1.1. Methodology

This Inventory estimated Electric Power emissions using FRD and the NI by State. Data reported to the EPA and Colorado GHGRPs were prioritized and supplemented with data reported to the EIA using [Form EIA-923](#).

Electricity generating facilities that are subject to the requirements of the Acid Rain Program must report emissions under [40 CFR Part 98, Subpart D, Electricity Generation](#) to the EPA and Colorado GHGRPs. These facilities also report under [40 CFR, Part 75](#). Facilities that generate electricity but do not report emissions under Subpart D may be required to report emissions under [Subpart C, General Stationary Fuel Combustion Sources](#). Subpart C includes a reporting threshold of 25,000 metric tons (0.025 Tg) CO₂eq, meaning facilities with emissions below this amount do not need to report. Subpart D does not include a reporting threshold. The Division reviewed FRD from Subparts C and D to determine the relevance and usability for estimating Electric Power emissions. This included review of NAICS codes, other reported subparts, and electricity generating unit details, as well as cross-checking with other datasets (such as EIA).

All Subpart D emissions reported from 2010 through 2023 were included. NAICS codes for Subpart D facilities were [221112 \(Fossil Fuel Electric Power Generation\)](#) and [221330 \(Steam and Air-Conditioning Supply\)](#). Though 221330 is not traditionally associated with electricity generation, the facility was confirmed as a power plant from other reported details and validated in the EIA dataset.

Subpart C emissions reported from 2010 through 2023 were included with the following NAICS codes: 221112, 221330, [221117 \(Biomass Electric Power Generation\)](#), and [221119 \(Other Electric Power Generation\)](#). If a facility reported both Subpart C and D emissions, only the Subpart D emissions were considered “electric power” emissions; the Subpart C emissions reflect other combustion, such as boilers or heaters. Some facilities reported Subpart C emissions from electricity generation for on-site use. These emissions would be captured in RCI Fuel Use (Commercial or Industrial) rather than the Electric Power subsector.

Subpart D and C emissions data for Electric Power facilities were prioritized. However, when gaps existed, they were filled using data collected by the EIA through Form EIA-923. The Form EIA-923 collects detailed electric power data (monthly and annually) on electricity generation, fuel consumption, fossil fuel stocks, and receipts at the power plant and prime-mover levels. Emissions were calculated using reported fuel consumption and emission factors from the [2025 GHG Emissions Factors Hub](#). The EIA data was pared down using NAICS codes and by comparing it with the GHGRP facilities identified as relevant to the Electric Power sector (i.e., not commercial or industrial facilities that generate electricity for their own on-site use only). The calculations and the gap-filling were facility- and GHG-specific. In some cases, Electric Power facilities were present in the EIA dataset but not in the EPA or Colorado GHGRP datasets. This was most common for small facilities that fell below the reporting threshold. When this occurred, those emissions were included.

After developing the complete emissions dataset by facility and GHG for 2010 through 2023, emissions for 2005 through 2009 were estimated by GHG using Equation 5.1 from the IPPC

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2006 Guidelines.⁴ Equation 5.1 creates a consistent time series by aligning older estimates with newer estimates during years where data overlap; the GHGRP and EIA emissions data for 2010 through 2023 were overlapped with the NI by State data for 2010 through 2023. The overlap method, compared to a linear or other simple extrapolation to 2005, allowed for annual variability and incorporation of expected trends in emissions. The total Electric Power emissions in this Inventory represent the sum of GHG-specific emissions for 2005 through 2009 and the sum of GHG-specific emissions by facility for 2010 through 2023.

3.1.2. Uncertainty

This Inventory does not quantify the uncertainty associated with estimated emissions from Electric Power. However, uncertainty exists in the estimates, driven by reporting thresholds, gaps in GHGRP data, and inconsistencies or gaps in EIA data. Additional uncertainty was introduced in the overlap method with the NI by State for years before GHGRP data were available (2005 through 2009).

The NIR provides uncertainty ranges for national emissions estimates from combustion. The EPA performed Monte Carlo uncertainty analyses by end-use sector and primary fuel type for CO₂ emissions. Uncertainty levels for the national 2023 Electric Power CO₂ emissions from Natural Gas ranged from -3% to +5%. Uncertainty levels for the national 2023 Electric Power CO₂ emissions from petroleum ranged from -4% to +7%. Uncertainty levels for the national 2023 Electric Power CO₂ emissions from coal ranged from -4% to +10%. The primary sources of uncertainty identified were fuel consumption by United States (U.S.) territories and carbon contents of fuels and products.⁵

For CH₄ and N₂O, uncertainty was assessed only for stationary combustion in total, not by end-use sector or fuel. Uncertainty levels for the national 2023 CH₄ and N₂O emissions from stationary combustion ranged from -34% to +125% and -23% to +51%, respectively. The uncertainties associated with the CH₄ and N₂O emission estimates were greater than those associated with CO₂ estimates, largely due to the use of emission factors (rather than carbon content) that represented only a limited subset of combustion conditions.⁶ Uncertainty in the state-level emissions estimates in this Inventory is likely more constrained due to the reliance on facility-specific calculations.

3.1.3. QA/QC

For each emissions source category, the Division applied standard QA/QC procedures, including reviews of data connections and imports, unit conversions, GWP calculations, naming conventions, and data formatting.

The Division also reviewed FRD for accuracy, reasonability, and order of magnitude. This review included comparisons of FRD emissions with emissions estimates in the NI by State to

⁴ [IPCC, 2006 Guidelines, Volume 1, Chapter 5, Time Series Consistency](#)

⁵ [2025 NIR](#), Chapter 3, Section 3.1, CO₂ from Fossil Fuel Combustion, Uncertainty

⁶ [2025 NIR](#), Chapter 3, Section 3.1, CH₄ and N₂O from Stationary Combustion, Uncertainty

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further validate the estimates. Electric Power emissions in this Inventory and in the NI by State followed very similar trends and were within 1.237 Tg CO₂eq or 4.2% each other.

The EPA's GHGRP reporting tool includes built-in verification processes for facility reports to identify potential errors and ensure that submitted data are accurate, complete, and consistent. After submission, the EPA employs additional statistical, algorithm, and reporting trends checks, and may follow up with facilities to resolve mistakes.⁷

3.1.4. Recalculations and Comparisons to Prior Inventory

This Inventory (2025) and the prior Inventory (2023) both prioritized FRD, using GHGRP and EIA data, and overlapped emissions with the NI by State for earlier years. However, the gap-filling methods and scope of the methodologies were slightly different. The specific facilities included in the analysis differed slightly between the two Inventories. This Inventory included facilities (and their emissions) that reported only to the EIA but not under Subpart C or D, as long as they were considered power plants. The overlap method also differed between the two Inventories. This Inventory overlapped emissions by GHG, while the prior Inventory overlapped total emissions. Despite these changes, the emissions in the two Inventories were very similar. From 2005 to 2020, the emissions estimates in this Inventory ranged from 0.020 Tg CO₂eq (less than 0.1%) to 0.527 Tg CO₂eq (1.3%) higher than those in the prior Inventory.

3.1.5. Planned Improvements

The Division will continue to prioritize using FRD to estimate emissions and evaluate methods to improve the accuracy of estimates in years when FRD were unavailable. The Division will also make efforts to further disaggregate Electric Power emissions, such as by fuel type, and to provide context on imported and exported electricity to and from Colorado.

3.2. Residential, Commercial, and Industrial Fuel Use

The Residential, Commercial, and Industrial (RCI) Fuel Use subsector includes estimated CO₂, CH₄, and N₂O emissions from the residential, commercial, and industrial combustion of fuels, including coal, natural gas, petroleum, and wood. This covers fuel use by end users for a variety of applications, including heating homes and businesses, running engines, and supporting industrial processes. The CO₂ emissions from wood combustion are considered biogenic and are captured within the net carbon flux of the LULUCF sector; CH₄ and N₂O emissions from wood combustion are included in this subsector.

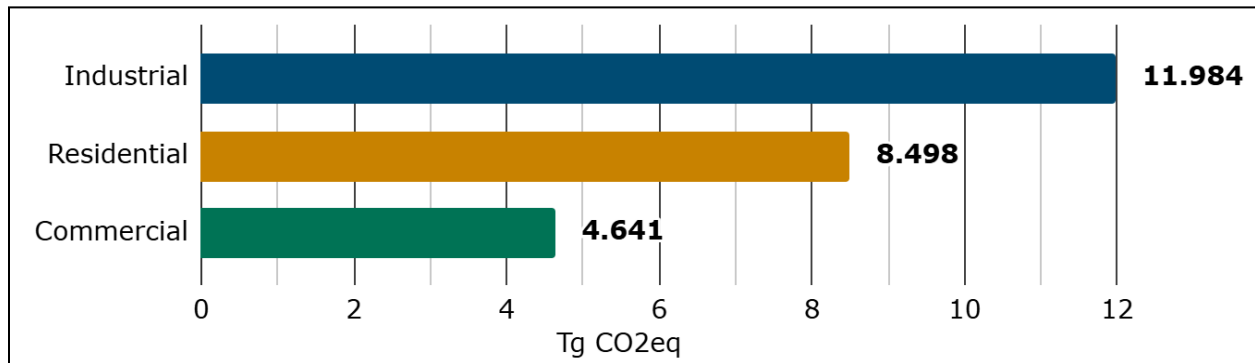
In 2023, RCI Fuel Use emissions were 25.123 Tg CO₂eq, accounting for 26.0% of Energy emissions. As shown in Figure 3.10, the Industrial Fuel Use category had the highest emissions, 11.984 Tg CO₂eq, accounting for 47.7% of RCI Fuel Use emissions. 2023 Residential Fuel Use emissions were 8.498 Tg CO₂eq (accounting for 33.8% of RCI Fuel Use emissions), and

⁷ [GHGRP Methodology and Verification | US EPA](#)

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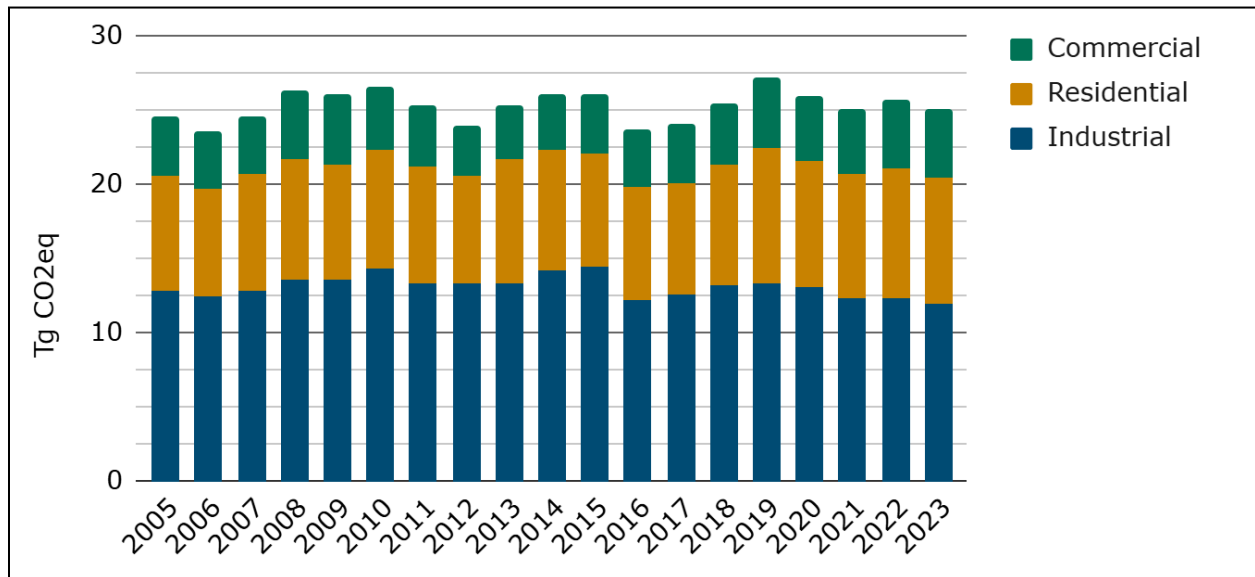
Commercial Fuel Use emissions were 4.641 Tg CO₂eq (accounting for 18.5% of RCI Fuel Use emissions).

Figure 3.10. 2023 RCI Fuel Use Emissions by Category.



As shown in Figure 3.11, emissions from RCI Fuel Use categories remain relatively stable over time, ranging from 23.608 Tg CO₂eq in 2006 to 27.192 Tg CO₂eq in 2019. Industrial Fuel Use had the highest emissions across the time series, ranging from 11.984 Tg CO₂eq in 2023 to 14.435 Tg CO₂eq in 2015. Commercial Fuel Use emissions were lowest of the three categories, ranging from 3.376 Tg CO₂eq in 2012 to 4.747 Tg CO₂eq in 2009. Emissions from the Residential Fuel Use category ranged from 7.176 Tg CO₂eq in 2012 to 9.143 Tg CO₂eq in 2019.

Figure 3.11. Historical RCI Fuel Use by Category.



Between 2005 and 2023, emissions from the RCI Fuel Use increased by 0.443 Tg CO₂eq (1.8%). This increase was primarily driven by the Residential Fuel Use category, which saw emissions increase by 0.784 Tg CO₂eq (10.2%) in 2023 relative to 2005 levels. Commercial Fuel Use emissions increased by 0.515 Tg CO₂eq (12.5%) between 2005 and 2023, while Industrial Fuel Use emissions declined by 0.856 Tg CO₂eq (6.7%). Across Residential, Commercial, and Industrial Fuel Use, most emissions were from natural gas.

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Table 3.4 presents RCI Fuel Use emissions by GHG and category for selected years. CO₂, CH₄, and N₂O emissions were relatively consistent across the time series, with some fluctuations. From 2005 through 2023, CO₂ accounted for most of the RCI Fuel Use emissions, and most CO₂ emissions were from Industrial Fuel Use. Industrial Fuel Use was also the largest contributor of CH₄ emissions across the time series. N₂O emissions were relatively small and more evenly distributed across the three categories.

Table 3.4. RCI Fuel Use Emissions by GHG and Category (Tg CO₂eq).

GHG	Category	2005	2010	2015	2020	2023
CO ₂	Residential	7.626	7.869	7.533	8.497	8.420
CO ₂	Commercial	4.101	4.253	4.015	4.286	4.614
CO ₂	Industrial	12.526	13.839	13.755	12.439	11.523
CO₂ Subtotal		24.252	25.961	25.303	25.223	24.557
CH ₄	Residential	0.077	0.101	0.114	0.063	0.068
CH ₄	Commercial	0.020	0.022	0.024	0.020	0.021
CH ₄	Industrial	0.303	0.518	0.666	0.604	0.450
CH₄ Subtotal		0.400	0.641	0.804	0.688	0.539
N ₂ O	Residential	0.012	0.015	0.017	0.011	0.011
N ₂ O	Commercial	0.005	0.006	0.005	0.005	0.005
N ₂ O	Industrial	0.011	0.012	0.013	0.011	0.011
N₂O Subtotal		0.028	0.033	0.035	0.027	0.027
RCI Fuel Use Total		24.680	26.635	26.142	25.937	25.123

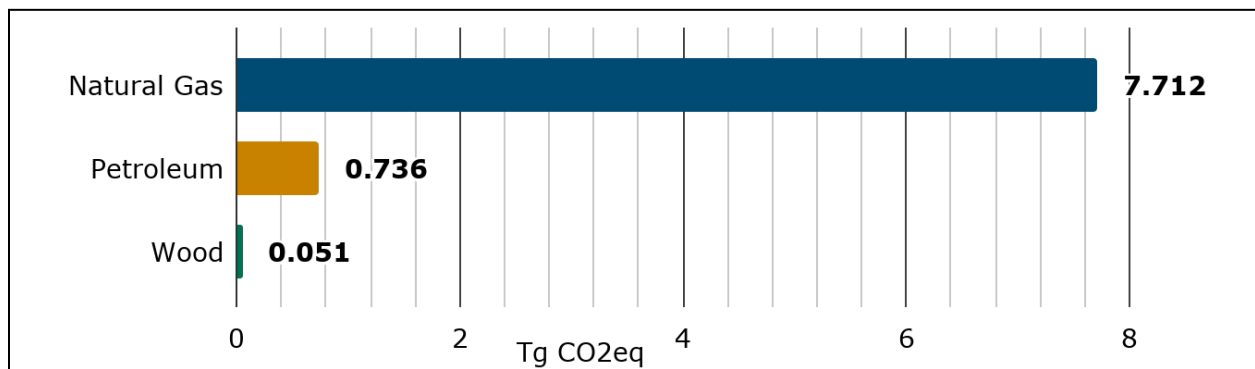
3.2.1. Residential Fuel Use

Residential Fuel Use emissions result from the combustion of fuels, including coal, natural gas, petroleum, and wood, for use in homes, such as lighting, appliance use, and heating or cooling. The petroleum category encompasses a variety of specific fuels, including distillate fuel, hydrocarbon gas liquids, and kerosene. For most fuels, CO₂, CH₄, and N₂O emissions were estimated. For wood combustion, the Inventory only estimates CH₄ and N₂O emissions. The CO₂ emissions from wood are considered biogenic and are captured within the net carbon flux of the LULUCF sector.

In 2023, Residential Fuel Use emissions were 8.498 Tg CO₂eq, accounting for 33.8% of RCI Fuel Use emissions and 8.8% of Energy emissions. As shown in Figure 3.12, emissions from natural gas were 7.712 Tg CO₂eq (accounting for 90.7% of Residential Fuel Use emissions), and emissions from petroleum were 0.736 Tg CO₂eq (accounting for 8.7% of Residential Fuel Use emissions) in 2023. Emissions from wood were 0.051 Tg CO₂eq (0.6%), and emissions from coal were zero in 2023.

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Figure 3.12. 2023 Residential Fuel Use Emissions by Fuel Type.

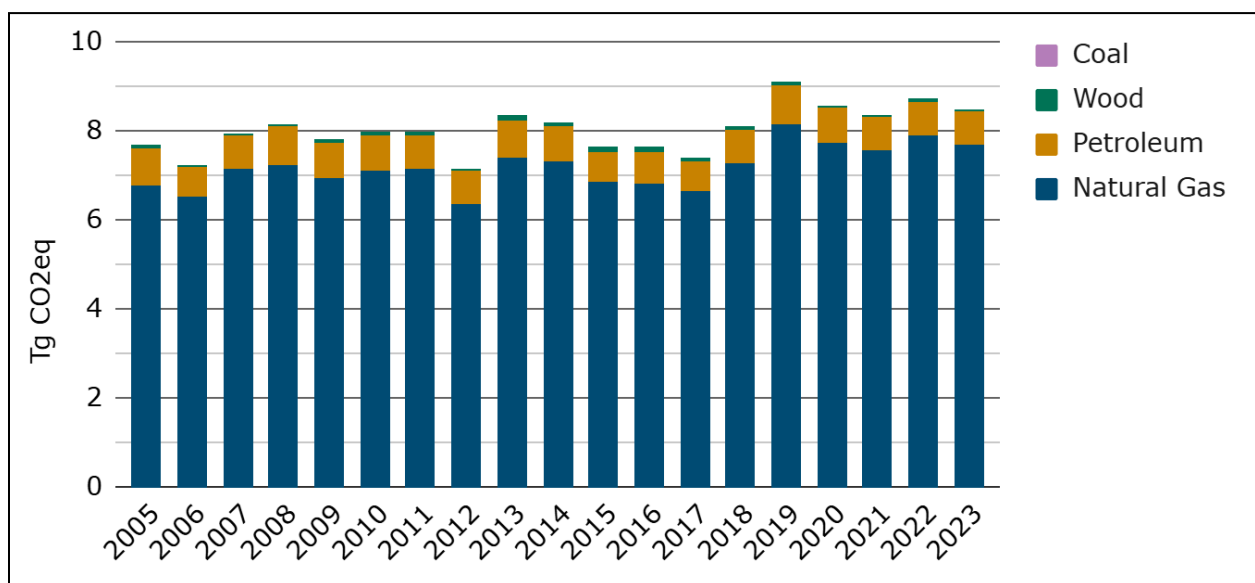


Emissions less than 0.0005 Tg CO₂eq are not included in the figure.

As shown in Figure 3.13, Residential Fuel Use emissions have fluctuated since 2005, peaking at 9.143 Tg CO₂eq in 2019. Emissions in 2023 were 0.784 Tg CO₂eq (10.2%) above 2005 emissions and 0.645 Tg CO₂eq (7.0%) below 2019 peak emissions.

Natural gas accounted for the highest emissions among all fuel types in the Residential subsector, with emissions fluctuating but generally increasing since 2005. Residential Fuel Use emissions from natural gas peaked at 8.172 Tg CO₂eq in 2019, and by 2023, natural gas emissions were 0.922 Tg CO₂eq (13.6%) higher than 2005 levels. Petroleum emissions, mostly from hydrocarbon gas liquids, remained relatively stable across the time series, though 2023 emissions were about 0.103 Tg CO₂eq (12.2%) below 2005 levels. Wood emissions generally increased from 2005 until a peak of 0.120 Tg CO₂eq in 2019, but fell to 0.046 Tg CO₂eq in 2020. By 2023, emissions were 0.011 Tg CO₂eq (17.5%) below 2005 levels. Coal emissions were 0.025 Tg CO₂eq in 2005, then declined through 2007 and remained zero from 2008 through 2023.

Figure 3.13. Historical Residential Fuel Use Emissions by Fuel Type.



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Table 3.5 below details Residential Fuel Use emissions by GHG and fuel type, highlighting that the majority of emissions are CO₂ and from natural gas use. In 2023, 99.1% of emissions from Residential Fuel Use were CO₂. As of 2023, CO₂ emissions had increased by about 0.794 Tg CO₂eq, or 10.4%, since 2005. The changes in CO₂ emissions were largely due to increased natural gas consumption, which increased by about 13.8% (about 17,598 billion British thermal units (Btu)). CH₄ and N₂O emissions, on the other hand, decreased over the same timeframe, though the magnitudes of their reductions were very small compared to changes in CO₂. CH₄ emissions decreased by 0.010 Tg CO₂eq (12.5%) while N₂O emissions decreased less than 0.001 Tg CO₂eq (9.4%).

Table 3.5. Residential Fuel Use Emissions by GHG and Fuel Type (Tg CO₂eq).

GHG	Fuel Type	2005	2010	2015	2020	2023
CO ₂	Coal	0.023	0.000	0.000	0.000	0.000
CO ₂	Natural Gas	6.770	7.085	6.836	7.726	7.689
CO ₂	Petroleum	0.833	0.784	0.697	0.771	0.731
CO₂ Subtotal		7.626	7.869	7.533	8.497	8.420
CH ₄	Coal	0.002	0.000	0.000	0.000	0.000
CH ₄	Natural Gas	0.017	0.018	0.017	0.019	0.019
CH ₄	Petroleum	0.004	0.003	0.003	0.003	0.003
CH ₄	Wood	0.054	0.080	0.093	0.041	0.045
CH₄ Subtotal		0.077	0.101	0.114	0.063	0.068
N ₂ O	Coal	+	0.000	0.000	0.000	0.000
N ₂ O	Natural Gas	0.003	0.003	0.003	0.003	0.003
N ₂ O	Petroleum	0.002	0.002	0.002	0.002	0.002
N ₂ O	Wood	0.007	0.010	0.012	0.005	0.006
N₂O Subtotal		0.012	0.015	0.017	0.011	0.011
Residential Fuel Use Total		7.715	7.985	7.663	8.571	8.498

+ Emissions do not exceed 0.0005 Tg CO₂eq.

3.2.1.1. Methodology

This Inventory used the EPA's SIT Modules, with updates, to estimate combustion emissions from Residential Fuel Use. The CO2FFC Module was used to estimate CO₂ emissions from the combustion of residential fuels. The Stationary Combustion Module was used to estimate CH₄ and N₂O emissions from residential fuel combustion.

The SIT Modules available at the time of this analysis only included data through 2022. To estimate 2023 emissions, the Division incorporated updated consumption data from [EIA SEDS](#) and extrapolated inputs as needed. Appendix D details the mapping of fuel codes in SEDS to fuel names in the SIT Modules. Default inputs were used where available, including emission factors, carbon content coefficients, and an assumed combustion efficiency of 100%. Where the inputs were variable across the time series, 2023 values were assumed equal to 2022 values.

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Within each Module, emissions were generally calculated as the product of fuel consumption and emission factors, along with conversion factors and GWP values as needed. The emission factors in the CO2FFC Module represent carbon content of the fuel in pounds carbon per million Btu (lbs C/million Btu); the calculated carbon emissions are then converted to CO₂ emissions. For many fuels, including natural gas, petroleum fuels, and coal, the carbon content changed slightly over time. From 2005 through 2023, the average carbon content coefficients for residential natural gas, coal, and distillate fuel were 31.86, 57.32, and 44.62 lbs C/million Btu, respectively.

The emission factors for CH₄ and N₂O were constant across all fuels throughout the time series. Residential CH₄ emission factors for coal and wood were the highest among fuels, at 0.301 and 0.285 metric tons CH₄ per billion Btu (MT CH₄/billion Btu), respectively. Residential CH₄ emission factors were 0.00475 MT CH₄/billion Btu for natural gas and 0.01002 MT CH₄/billion Btu for petroleum (including distillate fuel, kerosene, and hydrocarbon gas liquids). Residential N₂O emission factors for wood and coal were the highest among fuels, at 0.0038 and 0.0015 MT N₂O/billion Btu, respectively. Residential N₂O emission factors were 0.00009 MT N₂O/billion Btu for natural gas and 0.006 MT N₂O/billion Btu for petroleum (including distillate fuel, kerosene, and hydrocarbon gas liquids).

As shown in Table 3.6, natural gas was the most widely consumed residential fuel. Residential coal consumption was reported as zero after 2007.

Table 3.6. Residential Fuel Use (Billion Btu).

Fuel Type	2005	2010	2015	2020	2023
Coal	237	0	0	0	0
Natural Gas	127,677	133,463	129,193	146,020	145,275
Petroleum	13,205	12,455	11,034	12,203	11,600
Wood	6,832	9,985	11,711	5,087	5,633
Residential Total	147,951	155,903	151,938	163,310	162,508

There are no reporting programs under Colorado's or EPA's GHGRP for residential fuel consumption or emissions. Subpart NN data are only representative of natural gas, assuming all natural gas supplied by local distribution companies (or gas utilities) in Colorado is combusted in Colorado. The NI by State methods are very similar to those of the SIT, but the SIT provided the flexibility to make modifications and ensured consistency with the methodology and data used for Commercial and Industrial Fuel Use emissions.

3.2.1.2. Uncertainty

The SIT Modules do not quantify uncertainty, but each Module provides a brief narrative about uncertainty. Uncertainty may be introduced through any of the inputs, including fuel consumption, carbon content values, and combustion efficiency. CH₄ and N₂O emissions largely depend on consumption volumes, the type, size, and vintage of the combustion device, the maintenance and operation of the equipment, and the type of pollution control technology.

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The NIR provides uncertainty estimates for national emissions from combustion, which likely provide a reasonable approximation of the uncertainty in emissions estimates presented in this Inventory, as the carbon contents and emission factors were generally the same. In the SIT Module, the carbon content represents the maximum amount of carbon emitted per unit of energy released when assuming 100% combustion efficiency. Note that, in this context, the quantity of energy released refers to the total energy contained in the fuel, whereas the quantity of energy converted to a useful form after the fuel is combusted depends on equipment efficiency.

The EPA performed Monte Carlo uncertainty analyses by end-use sector and primary fuel type for CO₂. Uncertainty in the national 2023 residential CO₂ emissions ranged from -3% to +7% for natural gas combustion and from -6% to +5% for petroleum combustion. Uncertainty in national CO₂ emissions from coal combustion was not estimated because emissions did not occur in 2023. The primary sources of uncertainty identified were attribution of fuel consumption to sectors and carbon contents of fuels and products.⁸

For CH₄ and N₂O, uncertainty was assessed only for stationary combustion in total, not by end-use sector or fuel. Uncertainty in national 2023 CH₄ and N₂O emissions from stationary combustion ranged from -34% to +125% and -23% to +51%, respectively. The greater uncertainty in CH₄ and N₂O emission estimates compared to CO₂ estimates is largely due to the use of emission factors (rather than carbon content), which represent only a limited subset of combustion conditions.⁹

3.2.1.3. QA/QC

For each emissions source category, the Division applied standard QA/QC procedures, including reviews of data connections and imports, unit conversions, GWP calculations, naming conventions, and data formatting.

The Division also compared the emissions estimates from this Inventory using the SIT to the emissions estimates provided in the NI by State. Between the two, emissions estimates in this Inventory ranged from 0.077 Tg CO₂eq (1.0%) to 0.135 Tg CO₂eq (1.6%) higher than those in the NI by State. These differences can mostly be attributed to the differing methods for accounting for imports and exports, and for attributing fuels to sectors. The NIR estimated national fuel consumption primarily using the [Monthly Energy Review](#), then the NI by State disaggregated consumption to states using SEDS and other data sources. The SIT and this Inventory used SEDS directly for fuel consumption. While QA/QC procedures are not identified for each SIT Module, their methodologies and emissions estimates generally align with those of the NI by State, which does have established QA/QC procedures.

3.2.1.4. Recalculations and Comparisons to Prior Inventory

This Inventory and the prior Inventory used the same methods, relying on the SIT CO2FFC and Stationary Combustion Modules, to estimate Residential Fuel Use emissions. Although the SIT

⁸ [2025 NIR](#), Chapter 3, Section 3.1, CO₂ from Fossil Fuel Combustion, Uncertainty

⁹ [2025 NIR](#), Chapter 3, Section 3.1, CH₄ and N₂O from Stationary Combustion, Uncertainty

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Modules formed the basis of the methodology for both Inventories, the underlying SIT data were updated by the EPA. Additionally, this Inventory incorporated updated SEDS data, allowing the SIT calculations to extend to 2023.

From 2005 to 2020, emissions in the two Inventories follow a similar trend with small differences in their estimates. The estimates in this Inventory range from 0.050 Tg CO₂eq or 0.6% lower to less than 0.001 Tg CO₂eq or 0.01% higher than the estimates in the prior Inventory. The majority of those differences stem from differences in wood emissions estimates. These differences are primarily attributed to updates in the SEDS methodology, which adjusts Residential wood consumption to exclude estimated quantities obtained at no cost.¹⁰

3.2.1.5. Planned Improvements

The Division will review the possibility of using FRD, the NI by State, or other new data sources to estimate the emissions from Residential Fuel Use, such as Colorado-specific emission factors based on local and regional utilities, along with reported data under the EPA's GHGRP [Subpart NN, Suppliers of Natural Gas and Natural Gas Liquids](#). Coordination and cooperation with other relevant external agencies, along with their data, could improve data quality and help estimate residential combustion emissions more accurately.

3.2.2. Commercial Fuel Use

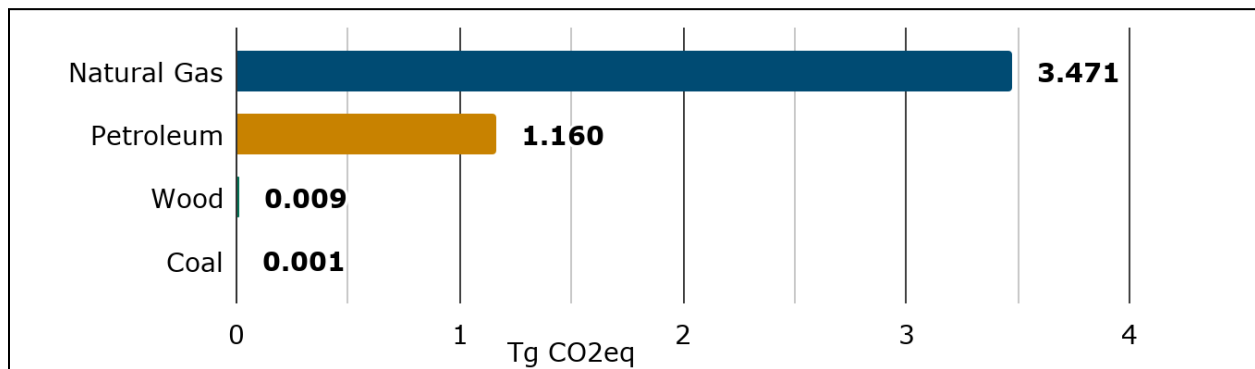
Commercial Fuel Use emissions result from the combustion of fuels used in businesses, such as for lighting, appliances, and heating or cooling. This fuel use releases CO₂, CH₄, and N₂O emissions. Commercial activities utilize several fuel types, including coal, natural gas, petroleum, and wood. Petroleum includes a variety of fuels, such as distillate fuel, hydrocarbon gas liquids, kerosene, motor gasoline, and residual fuel. For wood combustion, the Inventory only estimates CH₄ and N₂O emissions in this category. The CO₂ emissions from wood are considered biogenic and are captured within the net carbon flux of the LULUCF sector.

In 2023, Commercial Fuel Use emissions were 4.641 Tg CO₂eq, accounting for 18.5% of RCI Fuel Use emissions and 4.8% of Energy emissions. As shown in Figure 3.14, 2023 emissions from natural gas were 3.471 Tg CO₂eq (accounting for 74.8% of Commercial Fuel Use emissions), and emissions from petroleum were 1.160 Tg CO₂eq (accounting for 25.0% of Commercial Fuel Use emissions). Emissions from wood were 0.009 Tg CO₂eq (0.2%) and emissions from coal were 0.001 Tg CO₂eq (0.01%) in 2023.

¹⁰ [EIA Prices and expenditures technotes 2023: consumption adjustment](#)

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Figure 3.14. 2023 Commercial Fuel Use Emissions by Fuel Type.



As Figure 3.15 illustrates, Commercial Fuel Use emissions fluctuated from 2005 through 2023, with a peak of 4.747 Tg CO₂eq in 2009. Emissions in 2023 were 0.515 Tg CO₂eq (12.5%) above 2005 levels and 0.106 Tg CO₂eq (2.2%) below the 2009 peak.

Natural gas accounted for the highest emissions among all fuel types in the commercial sector, fluctuating over the time series and peaking at 3.554 Tg CO₂eq in 2008. Petroleum emissions, primarily from the use of gasoline and distillate fuel, generally increased over the time series, with the lowest emissions in 2007 at 0.320 Tg CO₂eq and the peak emissions in 2019 at 1.183 Tg CO₂eq. Emissions from coal primarily occurred before 2012, peaking at 0.662 Tg CO₂eq in 2008. Between 2012 and 2023, coal emissions remained consistently low, never exceeding 0.03 Tg CO₂eq, and reached zero in 2018 and 2019. Wood emissions remained less than 0.015 Tg CO₂eq from 2005 through 2023.

Figure 3.15. Historical Commercial Fuel Use Emissions by Fuel Type.

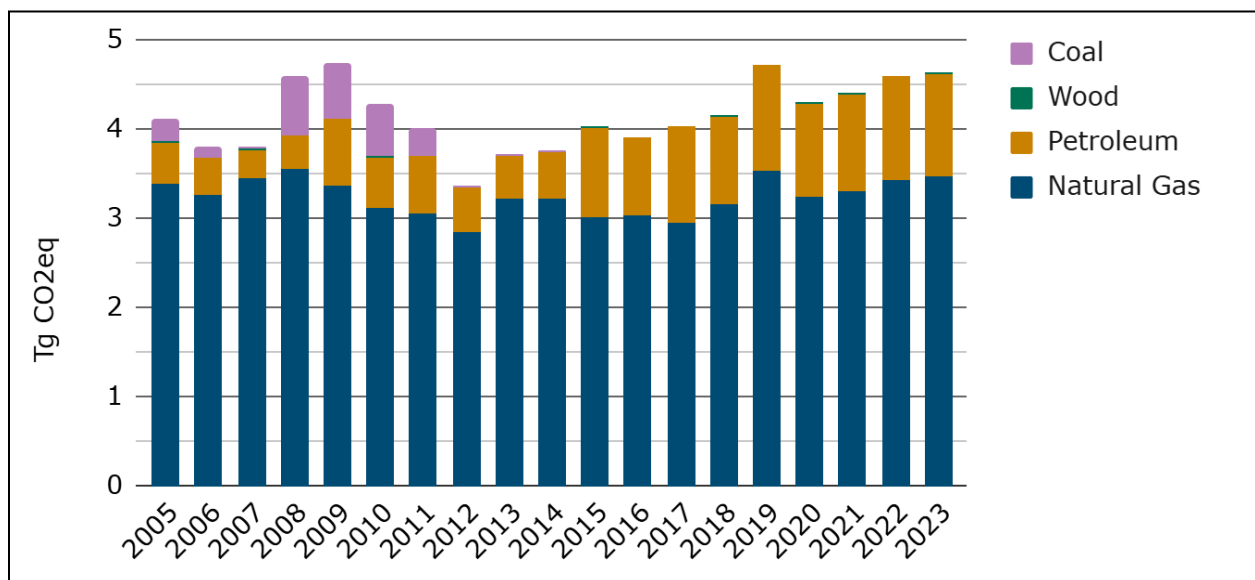


Table 3.7 below details Commercial Fuel Use emissions by GHG and fuel type, highlighting that the majority of emissions are CO₂ and from natural gas. In 2023, 99.4% of Commercial Fuel Use emissions were from CO₂. CO₂ emissions have increased by about 0.513 Tg CO₂eq

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since 2005, or 12.5%. The changes in CO₂ emissions are largely due to an increase in petroleum consumption, which increased by about 151.8% from 2005 through 2023 (about 9,931 billion Btu). CH₄ emissions have increased by 0.002 Tg CO₂eq (7.9%) since 2005. N₂O emissions did not change significantly over the time series, with 2023 emissions increasing less than 0.001 Tg CO₂eq (9.5%) compared to 2005.

Table 3.7. Commercial Fuel Use Emissions by GHG and Fuel Type (Tg CO₂eq).

GHG	Fuel Type	2005	2010	2015	2020	2023
CO ₂	Coal	0.260	0.572	0.007	0.003	0.001
CO ₂	Natural Gas	3.383	3.113	3.015	3.240	3.461
CO ₂	Petroleum	0.458	0.568	0.992	1.043	1.153
CO₂ Subtotal		4.101	4.253	4.015	4.286	4.614
CH ₄	Coal	0.001	0.002	+	+	+
CH ₄	Natural Gas	0.008	0.008	0.008	0.008	0.009
CH ₄	Petroleum	0.002	0.002	0.004	0.004	0.005
CH ₄	Wood	0.009	0.010	0.013	0.008	0.008
CH₄ Subtotal		0.020	0.022	0.024	0.020	0.021
N ₂ O	Coal	0.001	0.002	+	+	+
N ₂ O	Natural Gas	0.002	0.001	0.001	0.001	0.002
N ₂ O	Petroleum	0.001	0.001	0.002	0.002	0.003
N ₂ O	Wood	0.001	0.001	0.002	0.001	0.001
N₂O Subtotal		0.005	0.006	0.005	0.005	0.005
Commercial Fuel Use Total		4.126	4.281	4.044	4.312	4.641

+ Emissions do not exceed 0.0005 Tg CO₂eq.

3.2.2.1. Methodology

This Inventory used the EPA's SIT Modules, with updates, to estimate combustion emissions from Commercial Fuel Use. The CO2FFC Module was used to estimate CO₂ emissions from the combustion of commercial fuels. The Stationary Combustion Module was used to estimate CH₄ and N₂O emissions from the combustion of commercial fuels.

The SIT Modules available at the time of this analysis included data only through 2022. To estimate 2023 emissions, the Division incorporated updated consumption data from [EIA SEDS](#) and extrapolated inputs as needed. Appendix D details the mapping of fuel codes in SEDS to fuel names in the SIT Modules. Default inputs were used where available, including emission factors, carbon content coefficients, and an assumed combustion efficiency of 100%. Where the inputs were variable across the time series, 2023 values were assumed equal to 2022 values.

Within each Module, emissions were generally calculated as the product of fuel consumption and emission factors, along with conversion factors and GWP values as needed. The emission factors in the CO2FFC Module represent carbon content of the fuel in pounds carbon per

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million Btu (lbs C/million Btu); the calculated carbon emissions are then converted to CO₂ emissions. For many fuels, including natural gas, petroleum fuels, and coal, the carbon content changed slightly over time. From 2005 through 2023, the average carbon content coefficients for commercial natural gas, coal, and distillate fuel were 31.86, 57.32, and 44.62 lbs C/million Btu, respectively.

The emission factors for CH₄ and N₂O were constant values across the time series. Commercial CH₄ emission factors for wood were the highest among fuels, at 0.28487 MT CH₄/billion Btu. Commercial CH₄ emission factors were 0.00475 MT CH₄/billion Btu for natural gas, and 0.01002 MT CH₄/billion Btu for coal and petroleum (distillate fuel, kerosene, motor gasoline, and hydrocarbon gas liquids). Commercial N₂O emission factors for wood and coal were the highest among fuels, at 0.0038 and 0.0015 MT N₂O/billion Btu, respectively. Commercial N₂O emission factors were 0.00009 MT N₂O/billion Btu for natural gas and 0.006 MT N₂O/billion Btu for petroleum (distillate fuel, kerosene, motor gasoline, and hydrocarbon gas liquids).

As shown in Table 3.8, natural gas was the most widely consumed commercial fuel in the commercial subsector.

Table 3.8. Commercial Fuel Use (Billion Btu).

Fuel Type	2005	2010	2015	2020	2023
Coal	2,720	6,055	76	32	6
Natural Gas	63,809	58,642	56,980	61,238	65,384
Petroleum	6,543	7,954	14,037	14,912	16,474
Wood	1,096	1,298	1,569	1,012	1,012
Commercial Total	74,168	73,949	72,662	77,194	82,876

Some FRD are available for Commercial Fuel Use under Subpart C, but data coverage is incomplete and inconsistent due to reporting thresholds. Subpart NN data are only representative of natural gas, assuming all natural gas supplied in Colorado by local distribution companies (or gas utilities) is combusted in Colorado. The NI by State methods are very similar to those of the SIT, but the SIT provided the flexibility to make modifications and ensured consistency with the methodology and data used for Residential and Industrial Fuel Use emissions.

3.2.2.2. Uncertainty

The SIT Modules do not quantify uncertainty, but each Module provides a brief narrative about uncertainty. Uncertainty may be introduced through any of the inputs, including fuel consumption, carbon content values, and combustion efficiency. CH₄ and N₂O emissions largely depend on consumption volumes, the type, size, and vintage of the combustion device, the maintenance and operation of the equipment, and the type of pollution control technology.

The NIR provides uncertainty estimates for national emissions from combustion, which likely provide a reasonable approximation of the uncertainty in emissions estimates presented in

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this Inventory, as the carbon contents and emission factors were generally the same. In the SIT Module, the carbon content represents the maximum amount of carbon emitted per unit of energy released when assuming 100% combustion efficiency. Note that, in this context, the quantity of energy released refers to the total energy contained in the fuel, whereas the quantity of energy converted to a useful form after the fuel is combusted depends on equipment efficiency.

The EPA performed Monte Carlo uncertainty analyses by end-use sector and primary fuel type for CO₂. Uncertainty in the national 2023 commercial CO₂ emissions ranged from -3% to +7% for natural gas, -5% to +5% for petroleum, and -5% to +15% for coal. The primary sources of uncertainty identified were attribution of fuel consumption to sectors and carbon contents of fuels and products.¹¹

For CH₄ and N₂O, uncertainty was assessed only for stationary combustion in total, not by end-use sector or fuel. Uncertainty in national 2023 CH₄ and N₂O emissions from stationary combustion ranged from -34% to +125% and -23% to +51%, respectively. The greater uncertainty in CH₄ and N₂O emission estimates compared to CO₂ estimates is largely due to the use of emission factors (rather than carbon content), which represent only a limited subset of combustion conditions.¹²

3.2.2.3. QA/QC

For each emissions source category, the Division applied standard QA/QC procedures, including reviews of data connections and imports, unit conversions, GWP calculations, naming conventions, and data formatting.

The Division also compared the emissions estimates from this Inventory using the SIT to the emissions estimates provided in the NI by State. Between the two, emissions estimates in this Inventory ranged from 0.080 Tg CO₂eq (1.9%) lower to 0.173 Tg CO₂eq (4.4%) higher than those in the NI by State. These differences can mostly be attributed to the differing methods for accounting for imports and exports, and for attributing fuels to sectors. For several fuels, the NI by State disaggregated national emissions to the state level using a variety of datasets, including the [Monthly Energy Review](#). Similarly, while QA/QC procedures are not identified for each SIT Module, their methodologies and emissions estimates align with those of the NI by State, which does have established QA/QC procedures.

3.2.2.4. Recalculations and Comparisons to Prior Inventory

This Inventory and the prior Inventory used the same methods, relying on the SIT CO2FFC and Stationary Combustion Modules, to estimate Commercial Fuel Use emissions. Although the SIT Modules formed the basis of the methodology for both Inventories, the underlying SIT data were updated by the EPA. Additionally, this Inventory incorporated updated SEDS data, allowing the SIT calculations to extend to 2023.

¹¹ [2025 NIR](#), Chapter 3, Section 3.1, CO₂ from Fossil Fuel Combustion, Uncertainty

¹² [2025 NIR](#), Chapter 3, Section 3.1, CH₄ and N₂O from Stationary Combustion, Uncertainty

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From 2005 to 2020, emissions in the two Inventories followed the same trends with small differences in their estimates. The estimates in this Inventory range from 0.007 Tg CO₂eq or 0.2% lower to less than 0.001 Tg CO₂eq or 0.02% higher than the estimates in the prior Inventory.

3.2.2.5. Planned Improvements

The Division will review the possibility of using FRD, the NI by State, or a combination of the two to estimate the emissions from commercial combustion. This may include developing Colorado-specific emission factors based on data from local and regional utilities. Coordination and cooperation with other relevant external agencies and their data, and data submitted under the BPS rule as published, could improve data quality, assist in developing commercial combustion emissions factors, and lead to more accurate emissions estimates. This supporting data may be utilized along with data submitted under the EPA's GHGRP [Subpart NN, Suppliers of Natural Gas and Natural Gas Liquids](#), or from electric utilities. The Division will further assess the feasibility of including CH₄ slip emissions in commercial combustion estimates, should sufficient data on commercial-sector engines become available.

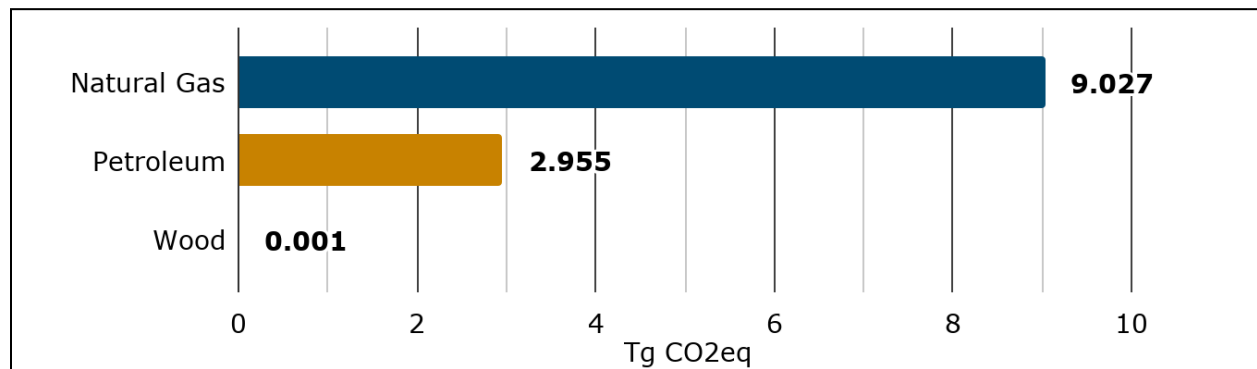
3.2.3. Industrial Fuel Use

The combustion of fuels for industrial uses, such as heating and cooling, powering engines, or producing steam for industrial processes, releases CO₂, CH₄, and N₂O emissions. Several fuels may have industrial uses, including coal, natural gas, petroleum, and wood. Petroleum includes a variety of fuels, such as: distillate fuel (diesel), asphalt and road oil, aviation gasoline blending components, crude oil, feedstocks, kerosene, hydrocarbon gas liquids, lubricants, motor gasoline, motor gasoline blending components, miscellaneous petroleum products, petroleum coke, pentanes plus, residual fuel, still gas, special naphthas, unfinished oils, and waxes. For most fuels, CO₂, CH₄, and N₂O emissions were estimated. For wood combustion, the Inventory only estimates CH₄ and N₂O emissions. The CO₂ emissions from wood are considered biogenic and are captured within the LULUCF sector's net carbon flux.

In 2023, Industrial Fuel Use emissions were 11.984 Tg CO₂eq, accounting for 47.7% of RCI Fuel Use emissions and 12.4% of Energy emissions. As shown in Figure 3.16, 2023 emissions from natural gas were 9.027 Tg CO₂eq (accounting for 75.3% of Industrial Fuel Use emissions) and emissions from petroleum were 2.955 Tg CO₂eq (accounting for 24.7% of Industrial Fuel Use emissions). Emissions from wood accounted for less than 0.001 Tg CO₂eq or 0.1%, and emissions from coal were zero in 2023.

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Figure 3.16. 2023 Industrial Fuel Use Emissions by Fuel Type.



Emissions less than 0.0005 Tg CO₂eq are not included in the figure.

As shown in Figure 3.17, Industrial Fuel Use emissions have fluctuated since 2005, with a peak of 14.435 Tg CO₂eq in 2015. Emissions in 2023 were 0.856 Tg CO₂eq (6.7%) below 2005 emissions and 2.451 Tg CO₂eq (17.0%) below 2015 peak emissions. The distribution of emissions by fuel type was largely consistent from 2005 through 2023: natural gas accounted for 72.1% to 81.1%, and petroleum for 18.9% to 27.9% of Industrial Fuel Use emissions. Petroleum emissions primarily come from distillate fuel, though they also include emissions from motor gasoline, still gas, and petroleum coke, among others. Emissions from coal were zero in most years, which is explained in more detail below, though they peaked at 0.387 Tg CO₂eq in 2014. Emissions from wood were de minimis from 2005 through 2023, peaking at just over 0.001 Tg CO₂eq in 2022.

Figure 3.17. Historical Industrial Fuel Use Emissions by Fuel Type.

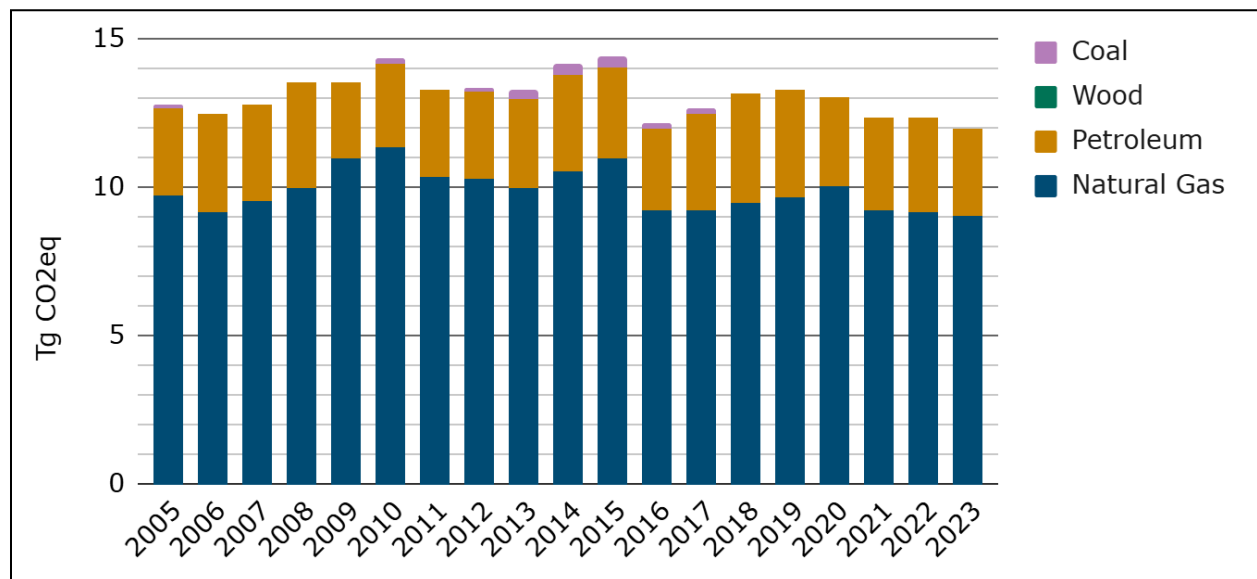


Table 3.9 below details Industrial Fuel Use emissions by GHG and fuel type, highlighting that the majority of emissions are CO₂ and from natural gas. In 2023, 96.2% of Industrial Fuel Use emissions were CO₂. CO₂ emissions decreased since 2005 by about 1.003 Tg CO₂eq or 8.0%. CH₄ emissions have increased since 2005 by 0.147 Tg CO₂eq or 48.7%. N₂O emissions did not change

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significantly over the time series (less than 0.001 Tg CO₂eq). The changes in CO₂ emissions are largely due to a 8.0% decrease in natural gas consumption from 2005 through 2023 (about 14,711 billion Btu). CH₄ emissions from natural gas increased despite overall lower natural gas consumption due to increased consumption by engines in the midstream segment of oil and gas operations. However, CH₄ emissions from natural gas peaked at 0.661 Tg CO₂eq in 2015 and have generally decreased since then.

Table 3.9. Industrial Fuel Use Emissions by GHG and Fuel Type (Tg CO₂eq).

GHG	Fuel Type	2005	2010	2015	2020	2023
CO ₂	Coal	0.141	0.210	0.356	0.000	0.000
CO ₂	Natural Gas	9.454	10.864	10.357	9.424	8.577
CO ₂	Petroleum	2.931	2.766	3.042	3.015	2.946
CO₂ Subtotal		12.526	13.839	13.755	12.439	11.523
CH ₄	Coal	+	0.001	0.001	0.000	0.000
CH ₄	Natural Gas	0.299	0.514	0.661	0.601	0.446
CH ₄	Petroleum	0.003	0.003	0.003	0.003	0.003
CH ₄	Wood	+	+	+	+	+
CH₄ Subtotal		0.303	0.518	0.666	0.604	0.450
N ₂ O	Coal	0.001	0.001	0.001	0.000	0.000
N ₂ O	Natural Gas	0.004	0.005	0.005	0.004	0.004
N ₂ O	Petroleum	0.006	0.006	0.006	0.006	0.006
N ₂ O	Wood	+	+	+	+	0.001
N₂O Subtotal		0.011	0.012	0.013	0.011	0.011
Industrial Fuel Use Total		12.840	14.369	14.435	13.054	11.984

+ Emissions do not exceed 0.0005 Tg CO₂eq.

This Inventory further disaggregates Industrial Fuel Use emissions from natural gas consumption within the upstream and midstream segments of the oil and gas supply chain (see Section 3.5 for the NGPS process or non-combustion emissions). Natural gas emissions are segmented by equipment type, as detailed in Table 3.10 below. The emissions from engines and turbines include estimates of CH₄ slip. The emissions from midstream engines also include estimates of CH₄ from crankcase venting. CH₄ slip primarily refers to unburned fuel escaping through engine exhaust, while crankcase venting refers to gases escaping from the engine cylinder into the crankcase, which is then vented into the atmosphere. The “Other Industrial” segment emissions in the table refer to all industrial emissions from natural gas that were not attributed to the upstream or midstream segments, while “Other” equipment in the upstream and midstream segments includes heaters, boilers, and reboilers. Emissions from industrial natural gas consumption by GHG, segment, and equipment type are detailed in Table D.3 in Appendix D.

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Table 3.10. Emissions from Industrial Natural Gas Use by Segment and Equipment Type (Tg CO₂eq).

Segment	Equipment Type	2005	2010	2015	2020	2023
Upstream	Engines	0.268	0.489	0.572	0.274	0.269
Upstream	Turbines	+	+	+	+	+
Upstream	Other	0.627	1.146	1.342	0.642	0.631
Upstream Subtotal		0.895	1.634	1.915	0.916	0.900
Midstream	Engines	1.474	2.538	3.297	3.115	2.278
Midstream	Turbines	0.386	0.665	0.863	0.816	0.596
Midstream	Other	0.347	0.598	0.777	0.734	0.537
Midstream Subtotal		2.208	3.801	4.937	4.665	3.411
Other Industrial	All Equipment	6.654	5.947	4.171	4.449	4.717
Other Industrial Subtotal		6.654	5.947	4.171	4.449	4.717
Industrial Natural Gas Use Total		9.757	11.382	11.023	10.029	9.027

+ Emissions do not exceed 0.0005 Tg CO₂eq.

3.2.3.1. Methodology

This Inventory used the EPA's SIT Modules, with modifications and updates, to estimate combustion emissions from all industrial fuels except natural gas. The CO2FFC Module was used to estimate CO₂ emissions from the combustion of industrial fuels. The Stationary Combustion Module was used to estimate CH₄ and N₂O emissions from the combustion of industrial fuels. Within each Module, emissions were generally calculated as the product of fuel consumption and emission factors.

The SIT Modules available at the time of this analysis included data only through 2022. To estimate 2023 emissions, the Division incorporated updated consumption data from [EIA SEDS](#) and extrapolated inputs as needed. Appendix D details the mapping of fuel codes in SEDS to fuel names in the SIT Modules. Default inputs were used where available, including emission factors, carbon content coefficients, and an assumed combustion efficiency of 100%. Where the inputs were variable across the time series, 2023 values were assumed equal to 2022 values. For many fuels, including natural gas, petroleum fuels, and coal, the carbon content changed slightly over time. From 2005 through 2023, the average carbon content coefficients for industrial natural gas, coal, and distillate fuel were 31.86, 57.25, and 44.62 lbs C/million Btu, respectively. The emission factors for CH₄ and N₂O were constant across all fuels throughout the time series. Default CH₄ emission factors for industrial natural gas, coal, petroleum fuels, and wood were 0.00095, 0.01002, 0.00301, and 0.02849 MT CH₄/billion Btu, respectively. Default N₂O emission factors for industrial natural gas, coal, petroleum fuels, and wood were 0.00009, 0.00150, 0.00060, and 0.00380 MT N₂O/billion Btu, respectively.

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To incorporate updated fuel consumption data, additional steps were needed to avoid double-counting in SEDS data for hydrocarbon gas liquids, motor gasoline, and coal. The Division followed the SIT guidance, as follows:

- Hydrocarbon gas liquids: subtract the consumption of pentanes plus, as it is accounted for separately
- Motor gasoline: subtract the consumption of ethanol used in motor gasoline
- Coal: subtract the consumption of industrial coal used to make synthetic natural gas¹³

Table 3.11. Industrial Fuel Consumption (billion Btu).

Fuel Type	2005	2010	2015	2020	2023
Coal	1,501	2,233	3,785	0	0
Petroleum	64,395	67,011	64,029	65,252	64,132
Natural Gas	182,765	208,973	199,528	184,050	168,054
Wood	177	240	361	467	613
Industrial Total	248,838	278,457	267,703	249,769	232,799

For emissions from natural gas, the Division followed the SIT's basic methodology but incorporated additional data and disaggregated the upstream and midstream segments of oil and gas operations. The Division also included an additional analysis to account for emissions from crankcase venting in midstream engines that burn natural gas.

Industrial natural gas consumption was disaggregated to the upstream and midstream segments using consumption data for plant fuel and lease fuel.¹⁴ Plant fuel consumption occurs at natural gas processing plants within the midstream segment, so 100% of the plant fuel consumption was allocated to the midstream segment. Lease fuel consumption occurs at facilities in both the upstream and midstream segments (excluding natural gas processing plants). The Division reviewed data reported to the federal GHGRP under [Subpart W, Petroleum and Natural Gas Systems](#) to distinguish emissions associated with the upstream segment (i.e., production) and midstream segment (i.e., gathering and boosting). Based on the proportions of upstream and midstream combustion emissions, 44.54% of lease fuel consumption was allocated to the upstream segment, and 55.46% of lease fuel consumption was allocated to the midstream segment. The sum of plant and lease fuel consumption was subtracted from the total industrial natural gas consumption; the remainder was allocated to “other” industrial uses outside upstream and midstream oil and gas operations. “Other” industrial uses refer to any industrial combustion of natural gas other than upstream or midstream operations, such as cement production, glass manufacturing, or hydrogen production.

¹³ The [EIA dataset on supplemental gaseous fuel supplies](#) used as the source for synthetic natural gas includes propane-air, coke oven gas, refinery gas, biomass gas, and manufactured gas commingled and distributed with natural gas, in addition to SNG. The data are not disaggregated by fuel type and are presented only as total supplemental gaseous fuels. By subtracting out all supplemental gaseous fuels as a proxy for synthetic natural gas, consumption of and resulting emissions from coal for industrial purposes are underestimated.

¹⁴ [EIA, Colorado Natural Gas Summary](#)

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In addition to the segment disaggregation, natural gas consumption was disaggregated by use in engines, turbines, and other unit types (i.e., heaters, reboilers, or boilers), as shown in Table 3.12. This disaggregation was completed using data submitted under Subpart W and to the Division related to the counts and types of fuel combustion equipment and their associated GHG emissions. Because different equipment types produce different emission levels per unit of fuel, the reported CO₂ emissions were used to scale fuel use rather than the number of equipment types.

Table 3.12. Percentage of Natural Gas Consumption by Combustion Equipment Type and Segment.

Equipment Type	Upstream	Midstream
Engines	27.97%	62.17%
Turbines	0.02%	19.89%
Other	72.02%	17.93%

Note: totals in the table may not sum to 100% due to rounding.

The fuel used by engines was further divided by engine type based on data reported through the [Oil and Natural Gas Annual Emissions Inventory Reporting \(ONGAEIR\) program](#), as detailed in Table 3.13 below. CH₄ emission factors, presented in kilograms (kg) per million Btu in Table 3.14, were proportionately applied for 2-stroke lean burn, 4-stroke lean burn, and 4-stroke rich burn engines in the upstream and midstream oil and gas segments. The same method of attributing fuel use by reported CO₂ emissions was applied to generate the distribution of engine types in the upstream and midstream segments. Data were not available to disaggregate turbines or other equipment. The CH₄ emission factors for engines and turbines reflect emissions from combustion and CH₄ slip. The CH₄ emission factor for other equipment reflects only combustion emissions.

Table 3.13. Percentage of Natural Gas Consumption in Engines by Engine Type and Segment.

Equipment Type	Upstream	Midstream
Engines: 2-stroke lean burn	2.642%	3.221%
Engines: 4-stroke lean burn	26.399%	73.806%
Engines: 4-stroke rich burn	70.958%	22.972%

Table 3.14. Oil and Gas Combustion Equipment Types and Associated CH₄ Emission Factors.

Equipment Type	CH ₄ Emission Factor (kg CH ₄ /million Btu)
Engines: 2-stroke lean burn	0.658
Engines: 4-stroke lean burn	0.522
Engines: 4-stroke rich burn	0.045
Turbines	0.004
Other	0.00095

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After disaggregating the natural gas consumption by segment and equipment type (as available), emissions were calculated following the general methodology established in the SIT CO2FFC and Stationary Combustion Modules (emissions equal the product of consumption and an emission factor). Non-energy consumption of natural gas was assumed to be zero for the upstream and midstream segments; in Colorado, natural gas consumed in upstream and midstream operations is most likely to be burned as a fuel rather than used for non-energy uses, such as a feedstock for other processes. Emissions from “other” industrial consumption of natural gas (i.e., not upstream or midstream consumption) did account for non-energy consumption.¹⁵ As with other fuels, default inputs were used where available, and where inputs varied across the time series, 2023 values were assumed equal to 2022 values.

Though the disaggregation of natural gas consumption was primarily implemented to incorporate equipment-specific CH₄ emission factors, the disaggregated consumption data were also used to estimate CO₂ and N₂O emissions from natural gas combustion following the SIT methodologies. The carbon content or carbon “emission factor” for natural gas varied across the years; the average from 2005 through 2023 was 31.86 pounds of carbon/million Btu of natural gas.¹⁶ This carbon content translates to an average of about 52.98 metric tons CO₂/billion Btu of natural gas. The CO2FFC Module assumes a 100% combustion efficiency, per IPCC guidelines. The default N₂O emission factor applied to all natural gas consumption was 0.000090 metric tons N₂O/billion Btu of natural gas.

The Division also conducted an analysis to estimate crankcase venting emissions from engines burning natural gas in the midstream segment. The calculations aligned with the crankcase venting methodology established in [Subpart W](#) and used operating parameters as reported to the Division for [midstream combustion](#). Crankcase venting emissions were estimated only for engines greater than 130 horsepower using Equation W-46 as the product of crankcase venting emission factors, engines’ operating hours, and a conversion factor (kg to metric tons). The emission factor applied to engines in the gathering and boosting segment was 0.083 kg CH₄/hour, and the emission factor applied to all other engines was 0.11 kg CH₄/hour. Operating hours, fuel use, and engine data (horsepower) were submitted by midstream operators in Colorado to the Division for operating year 2021. This data was used to identify operators in the gathering and boosting segment, categorize engines by horsepower, and estimate operating hours for midstream engines burning natural gas. Operating hours were estimated using the ratio of reported natural gas consumption to natural gas consumption allocated to the midstream segment as described above.

Although some FRD related to industrial fuel use and combustion are available under Subparts C and NN in Colorado, the emissions that could be attributed to industrial combustion would be incomplete. Subpart NN emissions are only representative of natural gas combustion, assuming all natural gas supplied by local distribution companies (or gas utilities) in-state is

¹⁵ These emissions are captured under the Non-Energy Use of Fossil Fuels subsector.

¹⁶ Carbon content values in the SIT were based on carbon content data values published in the NIR. The NIR determined carbon content of pipeline-quality natural gas using a regression analysis on pipeline and flared natural gas carbon content data. Annex 2.2 of the 2025 NIR details the methodology for estimating the carbon content of fossil fuels.

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combusted in-state. Subpart C emissions are not representative of industrial fuel use in its entirety because facilities subject to Subpart C may be subject to a reporting threshold. The NI by State methods are very similar to those of the SIT, but the SIT provided the flexibility to modify emission factors and calculations for specific operations within the Industrial category.

3.2.3.2. Uncertainty

The SIT Modules do not quantify uncertainty, but each Module provides a brief narrative about uncertainty. Uncertainty may be introduced through any of the inputs, including fuel consumption, carbon content values, and combustion efficiency. CH₄ and N₂O emissions largely depend on fuel consumption volumes, the type, size, and vintage of the combustion device, the maintenance and operation of the equipment, and the type of pollution control technology.

The NIR provides uncertainty estimates for national emissions from combustion, which likely provide a reasonable approximation of the uncertainty in emissions estimates presented in this Inventory, as the carbon contents and emission factors were generally the same. In the SIT Module, the carbon content represents the maximum amount of carbon emitted per unit of energy released when assuming 100% combustion efficiency. Note that in this context, the quantity of energy released means the total energy contained in the fuel, as opposed to the quantity of energy converted to a useful form after the fuel is combusted, which depends on equipment efficiency. Additional uncertainty was introduced in the SIT through its attempts to prevent double-counting of emissions from coal combustion. By using total supplemental fuels as a proxy for SNG, the SIT likely underestimated industrial emissions from coal use. Based on reported coal consumption by industrial facilities in Colorado, emissions may be underestimated by up to 0.3 Tg CO₂eq in a given year.

The EPA performed Monte Carlo uncertainty analyses by end-use sector and primary fuel type for CO₂. Uncertainty in the national 2023 Industrial CO₂ emissions ranged from -3% to +7% for natural gas, -23% to +23% for petroleum, and -5% to +16% for coal. The primary sources of uncertainty identified were attribution of fuel consumption to sectors and carbon contents of fuels and products.¹⁷

For CH₄ and N₂O, uncertainty was assessed only for stationary combustion in total, not by end-use sector or fuel. Uncertainty in national 2023 CH₄ and N₂O emissions from stationary combustion ranged from -34% to +125% and -23% to +51%, respectively. The greater uncertainty in CH₄ and N₂O emission estimates compared to CO₂ estimates is largely due to the use of emission factors (rather than carbon content), which represent only a limited subset of combustion conditions.¹⁸

¹⁷ [2025 NIR](#), Chapter 3, Section 3.1, CO₂ from Fossil Fuel Combustion, Uncertainty

¹⁸ [2025 NIR](#), Chapter 3, Section 3.1, CH₄ and N₂O from Stationary Combustion, Uncertainty

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3.2.3.3. QA/QC

For each emissions source category, the Division applied standard QA/QC procedures, including reviews of data connections and imports, unit conversions, GWP calculations, naming conventions, and data formatting.

The Division also compared the emissions estimates from this Inventory using the SIT to the emissions estimates provided in the NI by State. Between the two, emissions estimates in this Inventory ranged from 1.037 Tg CO₂eq (7.6%) lower to 0.335 Tg CO₂eq (2.8%) higher than those in the NI by State. These differences can mostly be attributed to the differing methods for accounting for imports and exports, and for attributing fuels to sectors. The national emissions were estimated using consumption data from a variety of datasets, including the [Monthly Energy Review](#), then attributed to sectors and states using SEDS and other proxy data. Similarly, while QA/QC procedures are not identified for each SIT Module, their methodologies and emissions estimates align with those of the NI by State, which does have established QA/QC procedures.

3.2.3.4. Recalculations and Comparisons to Prior Inventory

This Inventory and the prior Inventory used primarily the same methods, relying on the SIT CO2FFC and Stationary Combustion Modules, and incorporating additional analyses to account for crankcase venting emissions and updated CH₄ emission factors for natural gas. Although the SIT Modules formed the basis of the methodology for both Inventories, the underlying SIT data were updated by the EPA. Additionally, this Inventory incorporated updated SEDS data, allowing the SIT calculations to extend to 2023.

The methodology for estimating crankcase venting emissions from natural gas-burning engines in the midstream segment changed slightly. The prior Inventory estimated operating hours using the ratio of reported natural gas consumption by engines to total natural gas consumption published by EIA, while this Inventory estimated operating hours using the ratio of reported natural gas consumption by engines to total reported natural gas consumption. This change did not significantly impact the results, but it was considered a more accurate representation of the midstream segment reporters.

The common historical time series between the two Inventories was 2005 through 2020. Across this time series, emissions in the two Inventories follow a similar trend, with fluctuations since 2005 and emissions peaks in 2010 and 2015. The estimates in this Inventory ranged from 0.005 (less than 0.1%) to 0.438 Tg (3.5%) higher than those in the prior Inventory. These differences are largely attributable to updates in the SEDS fuel consumption data, notably for natural gas and coal.

3.2.3.5. Planned Improvements

The Division will review options for greater incorporation of FRD and disaggregation of emissions, particularly for fuels beyond natural gas and expanding crankcase venting emissions beyond midstream estimates. The Division will evaluate reported data as an input to emissions estimates or for validation purposes against other data sets, such as those from

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the EIA. This evaluation will include a particular review of industrial coal consumption data. While the methodology employed in this and previous inventories addresses possible double counting energy consumption and resulting emissions from coal and synthetic natural gas, the Division will consider changing that methodology to better account for known activity and emissions in the state. The Division will also consider inclusion of emissions from other new source types or modifications to emission factors to better align with reported data, research studies, aerial monitoring, and updated methods utilized by the Division in other programs.

3.3. Transportation

The Transportation subsector includes CO₂, CH₄, and N₂O emissions from the combustion of fossil- and biomass-based fuels used by transportation and mobile sources in Colorado. These emissions primarily come from on-road vehicles, but also from off-road vehicles and mobile sources (e.g., construction and agricultural equipment and recreational vehicles) and non-road vehicles (e.g., boats, locomotives, and aircraft).

CO₂ emissions from fuel combustion are best calculated using the amount of fuel consumed and the associated carbon content. This simple calculation follows the stoichiometric assumption and the understanding that combustion converts nearly all the carbon in a fuel into CO₂.

For non-CO₂ emissions, however, a variety of factors may influence their release. Although there is virtually no CH₄ in either gasoline or distillate (diesel) fuel, CH₄ is emitted as a combustion product that is influenced by fuel composition, combustion conditions, and control technologies. While "lean" air-fuel ratios (excess oxygen) reduce CH₄, they unfortunately increase nitrogen oxide pollutants. Modern vehicles use three-way catalysts to balance these trade-offs. CH₄ output peaks during cold starts, low-speed operation, aggressive driving, or poor engine tuning.

N₂O production is less understood but occurs via two primary pathways: in-cylinder formation and catalytic after-treatment. With in-cylinder formation, N₂O is created during combustion, but it is usually destroyed by post-flame reactions. In catalytic after-treatment, the primary source of N₂O, levels spike significantly during catalyst warm-up (specifically around 360°C). Once the catalyst exceeds 460°C, N₂O emissions drop as conditions in the device effectively neutralize the gas, creating O₂ and N₂.

As shown in Table 3.15, due to data limitations, this Inventory presents CO₂ emissions by fuel only, but CH₄ and N₂O emissions by fuel and vehicle type (see the Methodology discussion below).¹⁹ Additionally, there are small differences in how certain fuels are represented between CO₂ and non-CO₂ emissions.

¹⁹ Non-CO₂ emissions by vehicle are available in [this Inventory's data set](#).

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Five fuels (motor gasoline, distillate fuel, jet fuel, aviation gasoline, and residual fuel) are all represented across each of the three GHGs.²⁰ In addition to those five fuels, CO₂ emissions were estimated from natural gas and hydrocarbon gas liquids (comprising liquefied petroleum gases (LPGs) and many other heavier-than-methane hydrocarbons), and non-CO₂ emissions are estimated from “alternative fuels” which include compressed and liquefied natural gas (CNG and LNG), methanol, ethanol, and LPG, but not other hydrocarbon gas liquids.

For each fuel, non-CO₂ emissions were estimated for a variety of vehicle and mobile equipment types, including:

- Motor gasoline: boats, construction equipment, farm equipment, heavy-duty vehicles, heavy-duty utility equipment, light-duty trucks, motorcycles, passenger cars, small utility equipment, and snowmobiles.
- Distillate fuel: construction equipment, farm equipment, heavy-duty buses, heavy-duty vehicles, heavy-duty utility equipment, light-duty trucks, locomotives (trains), and passenger cars.
- Aviation gasoline and jet fuel: aircraft.
- Alternative fuels: buses, heavy-duty vehicles, and light-duty vehicles.

Table 3.15. Representation of Fuels, Vehicle Types, and GHGs in Transportation Emissions.

Fuel(s)	Vehicle Type(s) (CH ₄ and N ₂ O only)	GHG(s)
Alternative Fuels*	Buses	CH ₄ , N ₂ O
Alternative Fuels*	Heavy-Duty Vehicles	CH ₄ , N ₂ O
Alternative Fuels*	Light-Duty Vehicles	CH ₄ , N ₂ O
Aviation Gasoline	Aircraft	CO ₂ , CH ₄ , N ₂ O
Distillate Fuel	Construction Equipment	CO ₂ , CH ₄ , N ₂ O
Distillate Fuel	Farm Equipment	CO ₂ , CH ₄ , N ₂ O
Distillate Fuel	Heavy-Duty Buses	CO ₂ , CH ₄ , N ₂ O
Distillate Fuel	Heavy-Duty Vehicles	CO ₂ , CH ₄ , N ₂ O
Distillate Fuel	Light-Duty Trucks	CO ₂ , CH ₄ , N ₂ O
Distillate Fuel	Locomotive	CO ₂ , CH ₄ , N ₂ O
Distillate Fuel	Heavy-duty Utility Equipment	CO ₂ , CH ₄ , N ₂ O
Distillate Fuel	Passenger Cars	CO ₂ , CH ₄ , N ₂ O
Motor Gasoline	Boats	CO ₂ , CH ₄ , N ₂ O
Motor Gasoline	Construction Equipment	CO ₂ , CH ₄ , N ₂ O
Motor Gasoline	Farm Equipment	CO ₂ , CH ₄ , N ₂ O
Motor Gasoline	Heavy-Duty Vehicles	CO ₂ , CH ₄ , N ₂ O
Motor Gasoline	Light-Duty Trucks	CO ₂ , CH ₄ , N ₂ O
Motor Gasoline	Motorcycles	CO ₂ , CH ₄ , N ₂ O
Motor Gasoline	Heavy-duty Utility Equipment	CO ₂ , CH ₄ , N ₂ O
Motor Gasoline	Small Utility Equipment	CO ₂ , CH ₄ , N ₂ O
Motor Gasoline	Passenger Cars	CO ₂ , CH ₄ , N ₂ O

²⁰ Though residual fuel was included in the methods, consumption and emissions were zero from 2005 through 2023. Thus, the results do not include discussion of residual fuel.

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Fuel(s)	Vehicle Type(s) (CH ₄ and N ₂ O only)	GHG(s)
Motor Gasoline	Snowmobile	CO ₂ , CH ₄ , N ₂ O
Hydrocarbon Gas Liquids**	N/A	CO ₂
Jet Fuel	Aircraft	CO ₂ , CH ₄ , N ₂ O
Natural Gas***	N/A	CO ₂
Residual Fuel	Boats	CO ₂ , CH ₄ , N ₂ O
Residual Fuel	Locomotives	CO ₂ , CH ₄ , N ₂ O

* Alternative fuels include CNG, LNG, methanol, ethanol, and LPG.

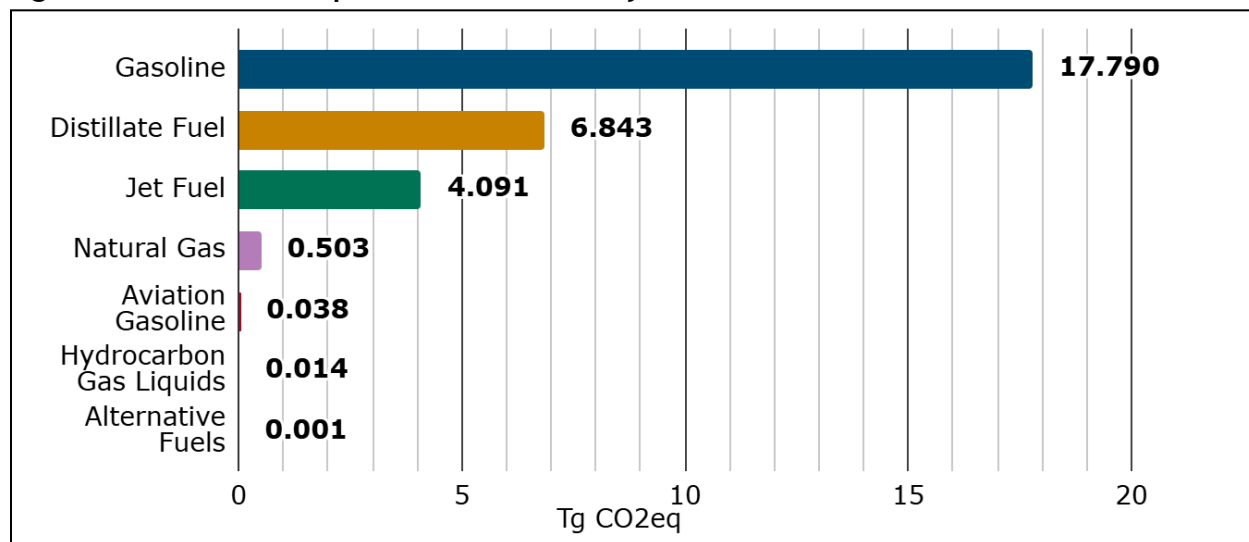
** Hydrocarbon gas liquids include hydrocarbons heavier than CH₄.

*** Natural gas includes CNG and LNG.

To avoid double-counting of CO₂ in the LULUCF sector, biofuels and the biofuel portions of blended fuels should be excluded from Transportation CO₂ estimates. This Inventory backed out most biofuels, particularly methanol, ethanol, and the ethanol portions of motor gasoline blends (see Section 3.3.5. Planned Improvements for discussion of biofuels that were not backed out). Also, international bunker fuels are excluded from fuel consumption estimates and thus Transportation emissions (see [Section 3.3.1.1, International Bunker Fuels](#), for more information).

In 2023, Transportation emissions were 29.279 Tg CO₂eq, representing 30.3% of the Energy sector and about 24.7% of Colorado's statewide emissions when LULUCF is excluded.²¹ This means that Transportation was the largest subsector emission source in that year. The majority of these emissions (60.8%) resulted from the combustion of gasoline (17.790 Tg CO₂eq), while distillate fuel and jet fuel accounted for 23.4% (6.843 Tg CO₂eq) and 14.0% (4.091 Tg CO₂eq), as shown in Figure 3.18. Together, these three fuels accounted for more than 98% of Transportation emissions in 2023.

Figure 3.18. 2023 Transportation Emissions by Fuel.



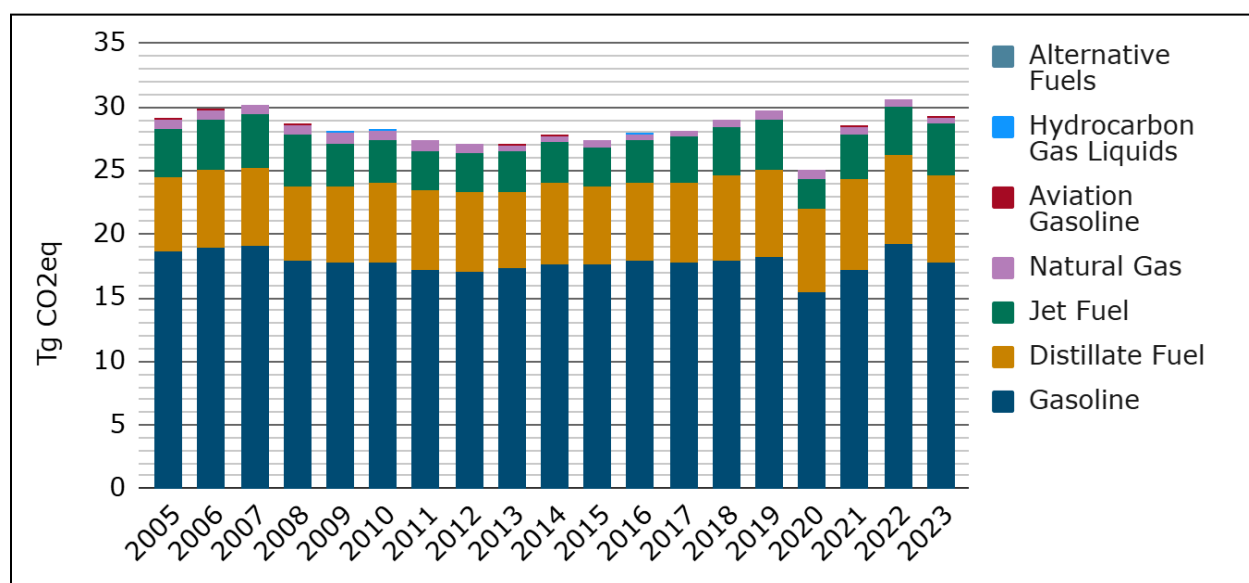
²¹ Transportation represented 23.2% of statewide emissions in 2023 when LULUCF was included.

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Excluding the period during the start of the COVID-19 pandemic (2020 and 2021), when personal travel saw the largest single-year reduction in recent history, between 2005 and 2023, each fuel accounted for relatively consistent shares of Transportation emissions, as illustrated in Figure 3.19. Motor gasoline emissions, primarily from on-road vehicles (including passenger cars and light-duty trucks) have historically accounted for 61% to 64% of the subsector emissions, while emissions from distillate fuel, which result from a variety of vehicle types (including light and heavy-duty vehicles, locomotives, farm equipment, and construction equipment), contributed between 20% and 23%, or around one-third of typical motor gasoline emissions.

After motor gasoline and distillate fuel, jet fuel was the next largest source of emissions. Jet fuel's annual contributions ranged from 11% to 14%. CO₂ emissions from natural gas contributed just 2% to 3%. In contrast, aviation gasoline emissions contributed historically marginal shares at less than 0.3%. The sum of emissions from hydrocarbon gas liquids (CO₂) and alternative fuels (CH₄ and N₂O) has historically been de minimis, together contributing less than 0.1% of total Transportation emissions in any given year.

Figure 3.19. Historical Transportation Emissions by Fuel.



Transportation emissions in 2023 increased by 0.141 Tg CO₂eq (0.5%) relative to 2005 levels. As reflected in Figure 3.19, emissions reduced modestly between 2007 and 2013, before starting to climb through 2019. After 2020, emissions rebounded quickly, peaking in 2022 before easing slightly in 2023. It is noteworthy that the lowest and highest emission levels, for both the Transportation subsector and the motor gasoline category, occurred within two years of each other (2020 and 2022). Further, since 2005, Colorado's population has grown by about 28%, VMT has increased by about 14%, and passenger traffic at Denver International Airport

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has grown by nearly 80%.^{22,23,24} That overall emissions have remained relatively steady points to improvements in average fuel efficiency across vehicle types, including through transitions towards electric vehicles, and uptake of alternative biomass-based fuels.

The overall increase in Transportation emissions in 2023 relative to 2005 was due to increases in emissions from distillate fuel and jet fuel (1.303 Tg CO₂eq) that were larger than reductions in emissions from all other fuels (1.163 Tg CO₂eq), predominantly motor gasoline.

Distillate fuel emissions have undergone small “bust-boom” cycles averaging about every 4 years. Nevertheless, the emissions have steadily grown over the time series. In 2023, distillate fuel use emissions were 1.029 Tg CO₂eq (17.7%) higher than those in 2005.

Jet fuel emissions declined sharply after 2008, possibly due to technological improvements in aircraft design that allowed for improved fuel efficiency in response to peak prices.²⁵ From 2009 through 2015, emissions from jet fuel were between 14% and 20% lower than 2005 levels. Emissions then started to climb through 2023, reaching 0.252 Tg CO₂eq (6.6%) above those in 2005. However, emissions have not risen above a 2007 peak level of 4.136 Tg CO₂eq. This is particularly notable given the significant growth in domestic passenger flight traffic.

Motor gasoline remains the subsector's largest emission source, but has also seen the most significant decrease. By 2023, motor gasoline emissions fell by 0.896 Tg CO₂eq (4.8%) relative to 2005. This contributed the most to offsetting increases in distillate and jet fuel emissions.

Natural gas has also contributed to the overall reduction in Transportation emissions, remaining consistently below 2005 levels since 2012. Emissions from natural gas reached their minimum of 0.421 Tg CO₂eq in 2016 before rebounding to near-2005 levels in 2019 and then reducing to 2023 levels of 0.503 Tg CO₂eq (0.231 Tg CO₂eq, 31.5%, lower than in 2005). Aviation gasoline emissions have remained below 2005 levels since 2007. While these emissions were 0.008 Tg CO₂eq (18.0%) lower in 2023 than in 2005, they were lowest in 2017. Due to the small scale of these emissions, the impact on total sector reductions is minimal. Emissions from hydrocarbon gas liquids decreased by 0.005 Tg CO₂eq (25.6%) in 2023 since 2005. Unlike aviation gasoline, hydrocarbon gas liquids emissions showed no clear upward or downward trajectory, fluctuating above and below 2005 levels throughout the period with a de minimis overall impact on the inventory.

During the peak impacts from the COVID-19 pandemic in 2020, and relative to 2019 levels, motor gasoline and jet fuel emissions decreased by 15% and 40%, respectively, while distillate fuel emissions saw only a 3% decrease. The reduced motor gasoline and jet fuel consumption reflects the fact that large shares of people traveled less for work and leisure, while freight

²² [U.S. Census Bureau, Population Tables](#)

²³ [Federal Highway Administration, Highway Statistics Series](#)

²⁴ [U.S. Department of Transportation, Bureau of Transportation Statistics, Airport Statistics](#). 2005 saw about 40 million passengers while 2023 was about 71 million. As of 2025, traffic exceeded 75 million passengers, equaling a nearly 90% growth in that metric since 2005.

²⁵ [U.S. Department of Transportation, Bureau of Transportation Statistics, Dietraveledsel and Jet Fuel Prices](#)

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traffic increased to provide for more at-home services. After the pandemic, emissions from both motor gasoline and jet fuel rebounded. From the historic minimum in 2020, motor gasoline emissions then increased by about 25% to peak in 2022 at 19.249 Tg CO₂eq. Likewise, jet fuel emissions increased by 74% from the 2020c minimum of 2.353 Tg CO₂eq to the most recent peak in 2023 of 4.091 Tg CO₂eq.

Regardless of the fuel consumed, Transportation GHG emissions were overwhelmingly composed of CO₂ (about 98% to 99%). CH₄ and N₂O contributed a very small fraction of the emissions on a CO₂eq basis, with N₂O emissions up to 10 times larger than those from CH₄. Detailed Transportation emissions are shown by GHG and fuel in Table 3.16 for select years. Table 3.16 also includes emissions from international bunker fuels, but they are presented for reference only, as they are excluded from official Transportation emissions reported here, which cover only domestic travel per IPCC guidelines (see the Methodology discussion below).

In 2023, CO₂ emissions were 28.915 Tg CO₂eq, accounting for 98.8% of all Transportation emissions. N₂O emissions accounted for 0.322 Tg CO₂eq (1.1%), and CH₄ emissions were 0.042 Tg CO₂eq (0.1%). From 2005 through 2023, CO₂ emissions increased by 0.472 Tg CO₂eq (1.7%), while emissions from N₂O and CH₄ saw significant decreases. N₂O showed the largest percentage decrease, with a reduction of 0.310 Tg CO₂eq (49.1%). The percentage decrease in CH₄ emissions was also significant, totaling 0.021 Tg CO₂eq (32.7%).

Table 3.16. Transportation Emissions by GHG Group and Fuel (Tg CO₂eq).

GHG	Fuel	2005	2010	2015	2020	2023
CO ₂	Aviation Gasoline	0.045	0.040	0.029	0.037	0.037
CO ₂	Distillate Fuel	5.748	6.261	6.082	6.488	6.676
CO ₂	Hydrocarbon Gas Liquids	0.019	0.006	0.037	0.009	0.014
CO ₂	Jet Fuel	3.805	3.312	3.031	2.333	4.055
CO ₂	Motor Gasoline	18.092	17.341	17.383	15.244	17.630
CO ₂	Natural Gas	0.734	0.774	0.519	0.629	0.503
CO₂ Subtotal		28.443	27.734	27.081	24.740	28.915
CH ₄	Alternative Fuels	+	+	0.001	0.001	0.001
CH ₄	Aviation Gasoline	0.001	0.001	0.001	0.001	0.001
CH ₄	Distillate Fuel	0.007	0.007	0.010	0.012	0.015
CH ₄	Jet Fuel	0.003	0.002	0.002	0.002	0.003
CH ₄	Motor Gasoline	0.052	0.035	0.027	0.022	0.023
CH₄ Subtotal		0.063	0.046	0.040	0.037	0.042
N ₂ O	Alternative Fuels	+	+	+	+	+
N ₂ O	Aviation Gasoline	+	+	+	+	+
N ₂ O	Distillate Fuel	0.059	0.069	0.110	0.138	0.174
N ₂ O	Jet Fuel	0.031	0.027	0.025	0.019	0.033
N ₂ O	Motor Gasoline	0.542	0.337	0.213	0.142	0.115

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GHG	Fuel	2005	2010	2015	2020	2023
<i>N2O Subtotal</i>		<i>0.542</i>	<i>0.337</i>	<i>0.213</i>	<i>0.142</i>	<i>0.115</i>
Transportation Total		28.443	27.734	27.081	24.740	28.915
CO2	<i>International Bunker Fuels</i>	1.241	1.369	1.422	0.787	1.545
CH4	<i>International Bunker Fuels</i>	0.001	0.001	0.001	0.001	0.001
N2O	<i>International Bunker Fuels</i>	0.010	0.011	0.012	0.006	0.013

+ Emissions do not exceed 0.0005 Tg CO₂eq.

Though not incorporated into this Inventory, the Colorado Department of Transportation recently released the [Statewide Transportation and Greenhouse Gas Reduction Accomplishments Report](#), which provides additional insight into trends in transportation emissions, population growth, and vehicle miles traveled.

3.3.1. Methodology

This Inventory used the EPA's SIT Modules and their methods, with modifications and updates, to estimate Transportation combustion emissions. The CO2FFC Module was used to estimate CO₂ emissions by fuel, and the Mobile Combustion Module was used to estimate CH₄ and N₂O emissions by fuel and vehicle type. The SIT Modules available at the time of this analysis included data only through 2022. The Division incorporated more current consumption data to calculate all historical CO₂ emissions through 2023 and extrapolated CH₄ and N₂O emissions to estimate their 2023 values. The Division considered using the NI by State instead of the SIT because it provides emissions estimates through 2023. However, the SIT was selected over the NI by State because it provides greater levels of disaggregation.

The general process for estimating Transportation emissions involves four steps. First, activity data are gathered, including fuel consumption (for CO₂ and off-road/non-road non-CO₂) and vehicle miles traveled (VMT) by fuel, vehicle type, vehicle age, and emission control technology (on-road non-CO₂). Second, fuel consumption for international travel (known as international bunker fuels) is backed out from total energy consumption values (for CO₂ and off-road/non-road non-CO₂). In Colorado, jet fuel is the only international bunker fuel. Third, biofuels and biofuel portions of blended fuels are backed out for CO₂ emission calculations to avoid double-counting of emissions that are accounted for in the LULUCF sector (Chapter 6). Finally, CO₂ and non-CO₂ emissions are calculated.

The SIT's CO2FFC Module was used to estimate CO₂ emissions as the product of fuel consumption, emission factors, and carbon contents for each fuel, not accounting for transportation mode or vehicle type. Because the SIT estimated CO₂ emissions only through 2022, the Division incorporated more current consumption data from the [EIA's SEDS](#) throughout the full time series (2005 through 2023), as highlighted in Table 3.17. Appendix D details the mapping of fuel codes in SEDS to fuel names in the SIT Modules. Note that motor gasoline here is ethanol-free motor gasoline to avoid double-counting of biogenic CO₂ in the LULUCF sector, and jet fuel is adjusted to exclude international bunker fuels. Aviation gasoline was not associated with any international consumption. The Division used the default

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factors provided by the CO2FFC Module for 2005 through 2022, including emission factors, carbon content coefficients, and an assumed combustion efficiency of 100%. The 2023 factors were assumed equal to those for 2022.

Table 3.17. Transportation Fuel Consumption by Fuel (in billion Btu)

Fuel	2005	2010	2015	2020	2023
Aviation Gasoline	655	581	422	529	537
Distillate Fuel	76,952	84,348	82,010	87,515	90,046
Hydrocarbon Gas Liquids	297	93	594	149	221
Jet Fuel*	52,676	45,850	41,955	32,292	56,136
Motor Gasoline**	255,347	243,860	246,208	215,723	249,483
Natural Gas	13,841	14,589	9,808	11,886	9,498
Transportation Total	399,768	389,321	380,997	348,094	405,921

Residual Fuel consumption is excluded because it is zero across the time series.

* Jet fuel is adjusted to exclude international bunker fuels.

** Ethanol is excluded from Motor Gasoline based on SIT recommendations to reduce possible double-counting of fuels across sectors.

The SIT's Mobile Combustion Module was used with default inputs to estimate CH₄ and N₂O emissions for 2005 through 2022, and recent trends in emissions were then used to estimate emissions in 2023. While the CO2FFC Module's methodologies provide emissions estimates by fuel only, the Mobile Combustion Module estimates non-CO₂ emissions by fuel and vehicle type. Vehicle types are classified as either on-road, off-road, or non-road. The methodologies for estimating emissions from these three transportation modes generally include computing the product of activity data and emission factors, but the nature of these inputs for on-road differs from those for off-road and non-road. While on-road calculations rely on annual VMT, factoring in vehicle type, age, and emission control technology, the off-road and non-road methods are based on fuel consumption by fuel and applicable vehicle type.

The Mobile Combustion Module considers several on-road vehicle types that combust only motor gasoline, only distillate fuel, or both. Passenger cars, light-duty trucks, and heavy-duty vehicles combust both motor gasoline and distillate fuel. Heavy-duty buses combust only distillate fuel, and motorcycles combust only motor gasoline. Emissions for these vehicles were estimated using VMT and emission factors that are vehicle-, fuel-, and emission control technology-specific. The Mobile Combustion Module also considers "alternative fuel vehicles" (light-duty, heavy-duty, and buses) that burn methanol, ethanol, LPG, LNG, or CNG. Emissions for these vehicles were also estimated using VMT and emission factors, but these are only vehicle- and fuel-specific and do not consider control technologies.

As shown in Table 3.18, annual on-road VMT is compiled by the [Federal Highway Administration](#) (FHWA) based on vehicle type and fuel. The Mobile Combustion Module apportions these annual VMT data to each emission control technology based on the estimated distributions of vehicle model years in operation and, in turn, the associated distributions of control technologies employed in each model year. VMT from alternative fuel vehicles is

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backed out from on-road VMT totals by vehicle type, so their activities are not double-counted. Emissions are then calculated as the product of VMT (in miles) and emission factors (in grams GHG per mile). While total VMT data were available through 2023 from the FHWA, model year and control technology distributions in the SIT were only available through 2022, thus limiting the SIT's ability to proportion VMT as needed for estimating CH₄ and N₂O emissions to 2022. The Division used these historical estimates as-is. 2023 emissions by vehicle type and fuel were estimated by linearly extrapolating from 2013 through 2022 emissions, excluding 2020 as that year was a clear outlier due to the COVID-19 pandemic. 2013 was selected as the start year for the linear regression because CH₄ and N₂O emission trends were both relatively consistent after that.

Table 3.18. Vehicles Miles Traveled by Vehicle Type and Fuel (Million Miles)

Vehicle Type	Fuel	2005	2010	2015	2020	2022
Heavy-duty Buses	Distillate Fuel	107	207	250	219	267
Heavy-duty Vehicles	Distillate Fuel	3,101	4,323	4,337	4,900	4,832
Heavy-Duty Trucks	Motor Gasoline	474	199	209	196	802
Light-Duty Trucks	Distillate Fuel	485	397	413	470	498
Light-Duty Vehicles	Distillate Fuel	146	160	175	198	218
Light-Duty Trucks	Motor Gasoline	16,131	9,458	9,825	10,791	11,058
Light-Duty Vehicles	Motor Gasoline	27,354	31,902	34,908	31,573	35,857
Motorcycles	Motor Gasoline	164	293	320	295	401
Total VMT in Colorado		47,962	46,940	50,437	48,642	53,935

The Mobile Combustion Module also considers off-road vehicles (farm equipment, construction equipment, small- and heavy-duty utility equipment, and snowmobiles) and non-road vehicles (aircraft, locomotives, and boats). CH₄ and N₂O emissions for both off-road and non-road vehicles were calculated using fuel consumption and fuel-specific emission factors for each GHG. For aircraft, the Division maintained consistency with the CO2FFC Module by utilizing updated SEDS fuel consumption data for 2005 through 2023 and adjusting for international bunker fuels as previously described in the CO2FFC Module discussion above. Fuel consumption data for aircraft (aviation gasoline and jet fuel) were in Btu, while all other vehicle types were in gallons. Because all the off-road and non-road emission factors were in units of grams of GHG per unit mass of fuel, energy content ratios (kg fuel per MMBtu) were applied for aviation, while density factors (kg per gal) were used for all other cases to align units for estimating emissions. Emission factors for aircraft and locomotives were static across the time series, while those for the other off-road and non-road vehicles were annualized.

Emissions for each vehicle type, fuel, and year were then the product of fuel consumption, unit conversion, and emission factor. Because aircraft fuel consumption relied on updated SEDS, its emissions were estimated through 2023. For the other vehicle classes, fuel consumption data were not consistently available. Thus, as with on-road estimates, the SIT limited its CH₄ and N₂O emissions estimates for them through 2022. In these cases, the

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Division estimated 2023 emissions by linearly extrapolating trends from 2013 through 2022, excluding 2020 because that year was a clear outlier due to the COVID-19 pandemic. 2013 was selected as the start year for the linear regression because CH₄ and N₂O emission trends were both relatively consistent after that.

3.3.1.1. International Bunker Fuels

Emissions resulting from the combustion of fuels used for international transport activities that originate in Colorado, termed “international bunker fuels”, are not included in emissions totals. Instead, International Bunker Fuels are reported as a memo item and accounted separately based on the location of fuel sales. This approach distinguishes global transportation emissions from national emissions, recognizing the complex international nature of these activities.²⁶ The NIR considers two transport modes under the IPCC definition of international bunker fuels, aviation and marine.²⁷ However, only bunker jet fuel is subtracted in Colorado because international bunkering of residual and distillate fuels does not occur in the state, as these fuels are mainly used on marine cargo ships. To account for international bunker fuels, then, only jet fuel consumption data were adjusted before calculating emissions, as shown in Table 3.17. Specifically, jet fuel bunker consumption estimates were subtracted from total jet fuel consumption, and transportation emissions were then calculated based on the adjusted fuel consumption. While jet fuel was categorized as either naphtha-type or kerosene-type prior to 2005, all Jet Fuel in this Inventory only includes kerosene, as SEDS now classifies naphtha-type Jet Fuel under “Industrial sector, Other petroleum”.²⁸

3.3.2. Uncertainty

The SIT Modules do not quantify uncertainty, but the NIR provides uncertainty ranges for national emissions estimates from combustion. The uncertainty in the estimates presented in this Inventory is likely similar to the national uncertainty, as the carbon contents, emission factors, and fuel consumption were generally the same.

The EPA performed Monte Carlo uncertainty analyses by end-use sector and primary fuel for CO₂. Uncertainty in the national 2023 Transportation CO₂ emissions from natural gas ranged from -3% to 7%. Uncertainty in national 2023 Transportation CO₂ emissions from petroleum ranged from -6% to 6%.²⁹ The primary sources of uncertainty identified were fuel consumption by U.S. territories, international bunker fuels, and the carbon content of fuels, the fraction of fuel oxidized, and the products.

For CH₄ and N₂O, uncertainty was assessed only for mobile combustion in total, not by end-use sector or fuel. Uncertainty in national 2023 CH₄ and N₂O emissions from mobile combustion ranged from -4% to +30% and -7% to +22%, respectively. The uncertainties associated with the CH₄ and N₂O emission estimates were greater than those associated with CO₂ estimates,

²⁶ [IPPC, CO₂, CH₄, AND N₂O EMISSIONS FROM TRANSPORTATION-WATER-BORNE NAVIGATION](#)

²⁷ [2025 NIR](#), Chapter 3, Section 3.10, International Bunker Fuels, Introduction

²⁸ [EIA Transportation sector energy consumption estimates, 1960-2023, United States](#)

²⁹ [2025 NIR](#), Chapter 3, Section 3.1, CO₂ from Fossil Fuel Combustion, Uncertainty

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largely due to the use of emission factors (rather than carbon content) that represented only a limited subset of combustion conditions.³⁰ Uncertainties surrounding emission factors are relatively high because emissions vary with ambient temperature, vehicle speed, gasoline volatility, and other factors. Values for these factors can vary significantly depending on many variables, such as driving conditions and vehicle characteristics. Furthermore, the 2023 emissions estimates for CH₄ and N₂O likely carry greater uncertainty because a large portion of the emissions were estimated by extrapolating historical trends rather than relying on activity data for that year.

The more complex methodology for estimating CH₄ and N₂O in the SIT Mobile Combustion Module introduces additional layers of uncertainty. Unlike the fuel-based CO₂ calculations, these estimates relied on activity data (VMT) and a variety of emission factors. Because VMT data is gathered annually by the FHWA using methods that vary by state, the degree of uncertainty in state activity data can fluctuate. Further uncertainty arises when the Mobile Combustion Module allocates state VMT across vehicle types using national averages, which may not fully capture Colorado-specific consumer preferences.

3.3.3. QA/QC

For each emissions source category, the Division applied standard QA/QC procedures, including reviews of data connections and imports, unit conversions, GWP calculations, naming conventions, and data formatting.

The Division also compared the emissions estimates from this Inventory using the SIT to the estimates provided in the NI by State. Total Transportation emissions from both sources were found to be similar in magnitude and followed similar trends over time. Between the two, emissions estimates in this Inventory ranged from 1.655 Tg CO₂eq (5.1%) lower to 0.401 Tg CO₂eq (1.4%) higher than those in the NI by State. These differences are primarily due to differing allocation methodologies: while the EPA allocates national fuel totals based on FHWA fuel sales data (purchase-based), this Inventory uses EIA SEDS data.

For this inventory, the Division utilized the SIT CO2FFC and Mobile Combustion Modules, incorporating updated data from the EIA SEDS through 2023. The Division compared fuel usage, emission factors, and total emissions with data from other sources. These comparisons included a review of the NI by State, [Motor Vehicle Emission Simulator](#) (MOVES) 4, MOVES 5, and SEDS emission estimates. While all sources showed similar trends across the time series, the Division ultimately elected to utilize the SIT for consistency with other categories in this Inventory that also leverage similar source data and to take advantage of its greater levels of disaggregation. Recalculations using the updated SEDS data were conducted externally, and the SIT tool was used to verify these values, ensuring that both the CO2FFC and Mobile Combustion Modules utilized the same underlying data.

Finally, the Division identified and corrected a discrepancy regarding the fuel density factor for locomotive fuel in the Mobile Combustion Module to reflect the correct value.

³⁰ [2025 NIR](#), Chapter 3, Section 3.1, CH₄ and N₂O from Mobile Combustion, Uncertainty

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3.3.4. Recalculations and Comparisons to Prior Inventory

This Inventory and the prior Inventory used mostly the same methods, both relying on the SIT CO2FFC and Mobile Combustion Modules. However, instead of relying on the SIT's default fuel consumption data, which was limited to 1990 through 2022 estimates from the 2024 SEDS, this Inventory incorporated more current 2025 SEDS and select FHWA fuel consumption data, allowing the SIT calculations to extend to 2023. The updated activity data also included minor revisions to some historical estimates, which contributed to small differences between the two inventories. Additionally, the EPA updates other activity data, such as VMT, vehicle age, mileage accumulation, control technology, and other distributions, in each SIT release, but the revisions to historical values between the 2023 and 2025 SITs result in de minimis changes to historical emissions estimates. Consequently, emissions estimated in this Inventory are very similar to those estimated in the prior Inventory, ranging from 0.001 Tg CO₂eq (0.002%) lower to 0.039 Tg CO₂eq (0.2%) higher than previously estimated across their common time series (2005 through 2020). These small changes stem almost exclusively from updated estimates of distillate and jet fuel between the 2022 and 2025 SEDS.

3.3.5. Planned Improvements

For future inventories, the Division intends to refine the Transportation subsector by increasing the level of disaggregation of CO₂ emissions to the vehicle level and assessing both “top-down” and “bottom-up” methods for estimating emissions. A primary component of this strategy is to explore using the EPA's [MOVES](#) to estimate on-road emissions. Unlike fuel-based top-down approaches, MOVES enables more granular “bottom-up” analysis by accounting for the phase-in of federal emissions standards, specific vehicle and equipment activity, local meteorological conditions, such as temperature and humidity, and the impact of regional emission-control activities, including inspection and maintenance programs. The “top-down” approach, which predominantly relies on fuel consumption and VMT data, will help constrain the “bottom-up” model approach using MOVES. The Division hopes these two approaches will complement each other and provide greater collective insights.

Coordination and cooperation with other relevant external agencies, including but not limited to the Colorado Department of Transportation, will be prioritized, particularly to help develop input data and emission factors for the MOVES model. Should the Division leverage MOVES, it will likely limit its application to on-road vehicles, including cars, trucks, and buses, and potentially specific non-road categories such as construction and lawn equipment. For sources not covered by MOVES, such as aircraft and locomotives, the Division will continue to evaluate data and methodologies to improve accuracy and representation (possibly by aircraft type). Ultimately, these efforts aim to better align inventory estimates across other state programs while providing more actionable information.

Finally, future inventories will better account for biofuels. The SIT states that consumption data and information on biodiesel and renewable diesel were incomplete at the state level, and states are to provide this information if available. Although these biofuels were not backed out of the distillate fuel consumption values for this Inventory, future inventories will

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aim to do so. By not backing out this consumption, this Inventory likely overestimated CO₂ emissions from distillate fuel, though the Division expects the impact to be relatively small (less than 1%).

3.4. Coal Mining

The Coal Mining subsector includes estimated direct CH₄ and CO₂ emissions in Colorado related to the production of coal at active mines and ongoing CH₄ leakage from abandoned coal mines. The origins of these emissions are different from those calculated in the Electric Power (Section 3.1) and RCI Fuel Use (Section 3.2) subsectors, which estimate CO₂, CH₄, and N₂O emissions from the combustion of coal. The Coal Mining subsector includes the following categories: Active Surface Mining, Active Underground Mining, Post-Mining, and Abandoned Coal Mines.

In 2023, Coal Mining emissions were 1.797 Tg CO₂eq, making up 1.9% of the Energy sector’s emissions. As shown in Figure 3.20, the Abandoned Coal Mines category contributed the highest emissions, accounting for 0.956 Tg CO₂eq (53.2%) of total emissions from Coal Mining. In contrast, Active Surface Mining had the smallest share, contributing 0.136 Tg CO₂eq (7.5%). Active Underground Mining and Post-Mining categories contributed 0.421 Tg CO₂eq (23.4%) and 0.285 Tg CO₂eq (15.8%), respectively, of the total Coal Mining emissions.

Figure 3.20. 2023 Coal Mining Emissions by Category.

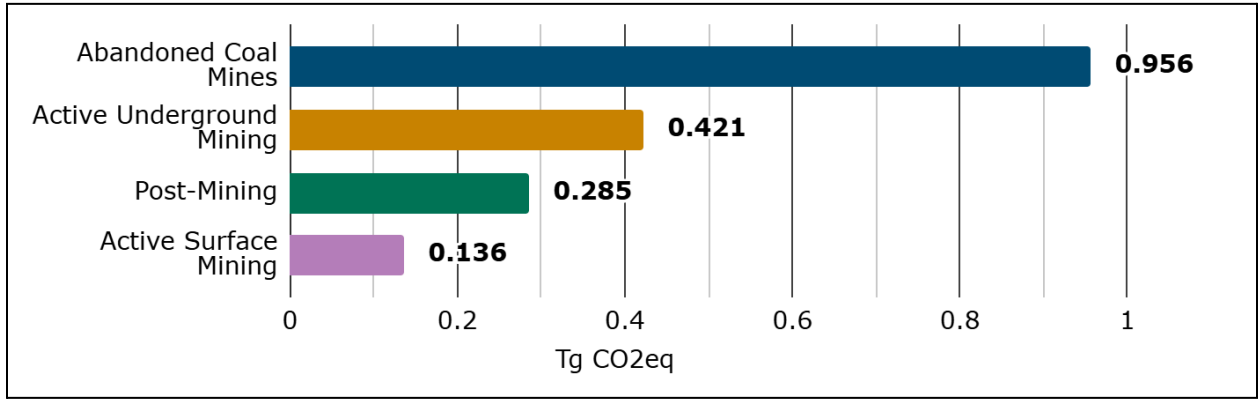


Figure 3.21 presents emissions from the Coal Mining subsector from 2005 through 2023. Compared to 2005 levels, emissions in the Coal Mining subsector decreased by 6.875 Tg CO₂eq (79.3%) in 2023. This decline was primarily driven by a reduction in emissions from the Active Underground Mining category, which decreased by 5.800 Tg CO₂eq (93.2%) in 2023 compared to 2005 levels. Emissions from Active Surface Mining fell the least, decreasing by about 0.141 Tg CO₂eq (51.1%) in 2023 relative to 2005 levels.

Emissions increased slightly early in the time series, peaking in 2010, and then declined through 2023. This trend was primarily driven by emissions from Active Underground Mining. Emissions from this category decreased substantially after 2010 and contributed the most to the subsector’s overall decline in total emissions. This change is most likely attributable to

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the closure of underground coal mines. Emissions from Abandoned Coal Mines have remained stable across the time series and were the largest source of emissions in the Coal Mining subsector from 2016 through 2023. Active Surface Mining and Post-Mining emissions contributed smaller shares to total emissions and generally decreased over time.

Figure 3.21. Historical Coal Mining Emissions by Category.

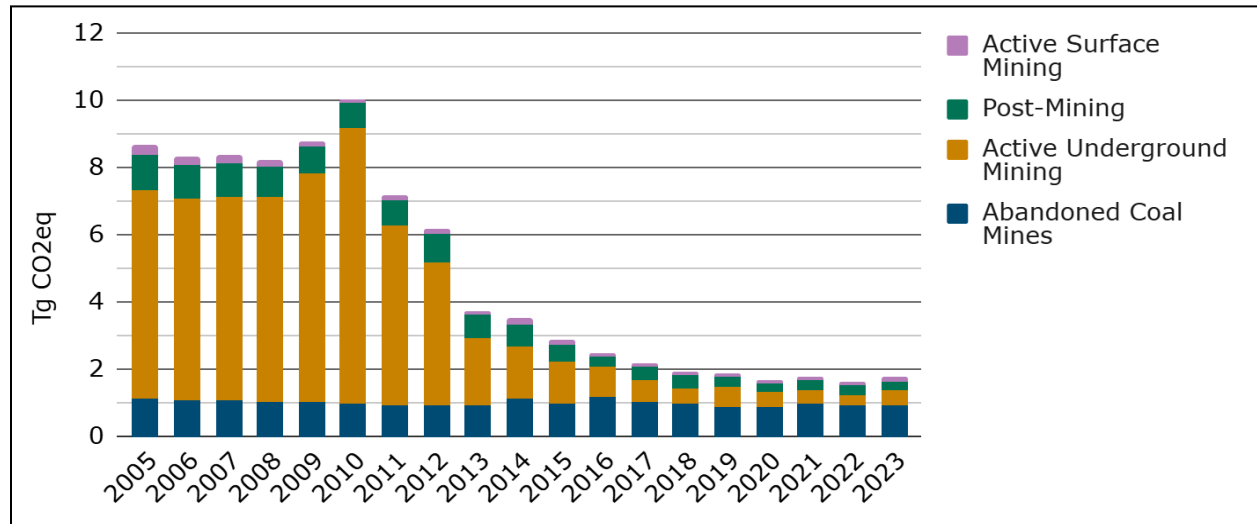


Table 3.19 details emissions by GHG and category from the Coal Mining subsector for select years. CH₄ was the dominant source of emissions throughout the time series, accounting for 95.7% of emissions from the Coal Mining subsector in 2023. CH₄ and CO₂ emissions fluctuated over the time series, ranging from 1.580 Tg CO₂eq in 2022 to 9.862 Tg CO₂eq in 2010 for CH₄ and from 0.064 Tg CO₂eq in 2020 to 0.287 Tg CO₂eq in 2005 for CO₂.

Table 3.19. Coal Mining Emissions by GHG and Category (Tg CO₂eq).

GHG	Category	2005	2010	2015	2020	2023
CO ₂	Active Surface Mining	0.007	0.004	0.004	0.003	0.004
CO ₂	Active Underground Mining	0.280	0.198	0.129	0.061	0.073
CO₂ Subtotal		0.287	0.201	0.133	0.064	0.077
CH ₄	Abandoned Coal Mines	1.136	0.980	1.011	0.886	0.956
CH ₄	Post-Mining	1.037	0.721	0.486	0.236	0.285
CH ₄	Active Surface Mining	0.270	0.136	0.154	0.103	0.132
CH ₄	Active Underground Mining	5.942	8.025	1.112	0.405	0.348
CH₄ Subtotal		8.385	9.862	2.763	1.630	1.720
Coal Mining Total		8.672	10.064	2.896	1.694	1.797

3.4.1. Active Surface Mining

Surface mining is the extraction of coal near the Earth's surface. It involves exposing buried coal seams by removing parts or all of mountaintops, and disposing of the excess overburden (rock above the coal seam) and interburden (rock between coal seams) in adjacent valleys. CH₄ and CO₂ are trapped in coal seams until mining operations take place. Surface coal mines liberate CH₄ and CO₂ as the overburden is removed and the coal is exposed to the atmosphere.³¹ CH₄ emissions are a function of depth and coal rank, a classification related to the percentage of carbon in the coal. Surface coal mines produce lower-rank coal and remove less than 250 feet of overburden, making their emissions lower than those of underground mines.³²

In 2023, emissions from Active Surface Mining were 0.136 Tg CO₂eq, accounting for 7.5% of total emissions from Coal Mining. Compared to 2005 levels, emissions decreased by 0.141 Tg CO₂eq (51.1%) in 2023. Emissions from Active Surface Mining decreased from 2005 to 2010, and then varied for the rest of the time series, as illustrated in Figure 3.22. Table 3.20 presents Active Surface Mining emissions by GHG for select years. CH₄ is the main source of emissions throughout the time series, with emissions ranging from 0.095 Tg CO₂eq in 2019 to 0.270 Tg CO₂eq in 2005. CO₂ emissions remained relatively stable over the period, ranging from 0.003 Tg CO₂eq in 2019 to 0.007 Tg CO₂eq in 2005.

Figure 3.22. Historical Active Surface Mining Emissions.

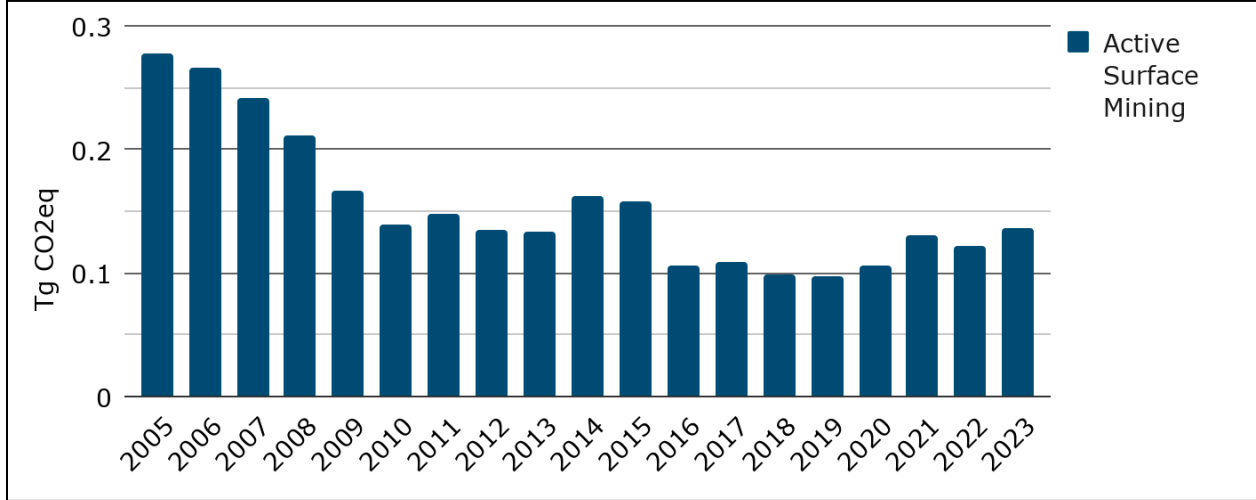


Table 3.20. Active Surface Mining Emissions by GHG (Tg CO₂eq).

GHG	2005	2010	2015	2020	2023
CH ₄	0.270	0.136	0.154	0.103	0.132
CO ₂	0.007	0.004	0.004	0.003	0.004
Active Surface Mining Total	0.277	0.140	0.158	0.106	0.136

³¹ [What is Surface Coal Mining?](#)

³² [2025 NIR](#), Chapter 3, Section 3.4, Introduction

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3.4.1.1. Methodology

This Inventory used the NI by State dataset to estimate emissions from the Active Surface Mining category. The EPA applied a Tier 2 methodology, in accordance with IPCC guidelines, to estimate CH₄ emissions. Basin-specific coal production data from the [EIA Annual Coal Report](#) were multiplied by basin-specific CH₄ content values and a 150% emission factor (to account for CH₄ from overburden and underburden).^{33,34,35} To determine state-level CH₄ emissions, national emissions were apportioned based on coal production in each state as reported in the EIA Annual Coal Report. The appropriate basin-specific CH₄ content for the coal produced in a state was assigned based on the coal basin within which the state is located.³⁶

CO₂ emissions were estimated using the IPCC Tier 1 approach in accordance with IPCC guidelines and using available data from the EIA Annual Coal Report. A default emission factor of 0.44 cubic meter per metric ton of coal produced and annual coal production from surface mines were used to estimate CO₂ emissions.³⁷ For state-level estimates, emissions were summed across individual mines within each state to produce state-level totals instead of aggregating to a national total as done in the NIR.

The Division reviewed the NI by State and the SIT for estimating emissions from Active Surface Mining. No emissions data are reported to the EPA or the Division, and no other FRD were considered. Emissions estimated with the SIT were generally lower than those in the NI by State while offering less disaggregation. The NI by State was determined to be more complete, rigorous, and detailed in its methods. Because of these reasons, and to present a more conservative estimate, the Division elected to use the NI by State's estimates.

3.4.1.2. Uncertainty

Though uncertainty was not quantified for state-level estimates, the NIR quantified uncertainty estimates for national emissions at the subsector level using the 2006 IPCC Guidelines Approach 2. The NIR did not assess uncertainty for the Active Surface Mining category and instead assessed uncertainty for the Coal Mining subsector as a whole. The uncertainty in the national estimates for Coal Mining in 2023 ranged from -10% to +21% for CH₄ and from -68% to +76% for CO₂.³⁸ State-level estimates may be more uncertain because national emissions estimates are apportioned across states.

³³ [2025 NIR](#), Chapter 3, Section 3.4, Methodology and Time-Series Consistency

³⁴ [EPA \(2005\) Surface Mines Emissions Assessment](#). Draft. U.S. Environmental Protection Agency.

³⁵ [EPA \(1996\) Evaluation and Analysis of Gas Content and Coal Properties of Major Coal Bearing Regions of the United States](#). EPA/600/R-96-065. U.S. Environmental Protection Agency.

³⁶ [2024 NI by State Methodology report](#), Chapter 2, Section 2.2.1.2.2, Surface Mining and Post-mining Activities

³⁷ [2025 NIR](#), Chapter 3, Section 3.4, Methodology and Time-Series Consistency

³⁸ [2025 NIR](#), Chapter 3, Section 3.4, Uncertainty

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3.4.1.3. QA/QC

For each emissions source category, the Division applied standard QA/QC procedures, including reviews of data connections and imports, unit conversions, GWP calculations, naming conventions, and data formatting.

The EPA applied QA/QC procedures consistent with the NIR QA/QC Management Plan, in accordance with IPCC Guidelines. The QA/QC Management Plan describes data and methodology checks, develops processes governing peer review and public comments, and provides guidance on conducting uncertainty analyses for the emissions estimates. The QA/QC Management Plan procedures also stress continual improvement, providing for corrective actions designed to improve the inventory estimates over time.³⁹ Trends in the time series were analyzed to determine if any corrective actions were needed.

3.4.1.4. Recalculations and Comparison to Prior Inventory

To estimate emissions from the Active Surface Mining category, this Inventory used the 2025 NI by State, whereas the prior Inventory used the 2022 NI by State. From 2005 through 2020, differences in historical emissions estimates between the 2023 and 2025 Inventories were de minimis.

3.4.1.5. Planned Improvements

For future Inventories, the Division will consider using FRD if available, either to represent the emissions from Active Surface Mining or in verification efforts. Active Surface Mining is not currently subject to FRD requirements at the state or federal level. Coordination and cooperation with other relevant external agencies and their data, including but not limited to the [Energy and Carbon Management Commission](#) (ECMC), could improve data quality and help develop Abandoned Coal Mines emissions factors and estimate emissions more accurately.

3.4.2. Active Underground Mining

An underground coal mine is a site where coal is extracted by tunneling into the earth to reach the coalbed. It is then mined using underground equipment like cutting machines, as well as longwall and shortwall mining machines, and transported to the surface. An underground coal mine is categorized as active if any of the following conditions are met: (1) Mine development is underway, (2) Coal has been produced within the last 90 days, (3) Mine personnel are present in the mine workings, (4) Mine ventilation fans are operative, or (5) The mine is designated as an “intermittent” mine by the [Mine Safety and Health Administration \(MSHA\)](#).⁴⁰

Active Underground Mining emissions include both CH₄ and CO₂, which occur naturally in coal seams and remain trapped until coal is mined. CH₄ is released or “liberated” from ventilation

³⁹ [NIR](#), Annex 8 QA/QC Procedures

⁴⁰ [Subpart FF, Underground Coal Mines](#)

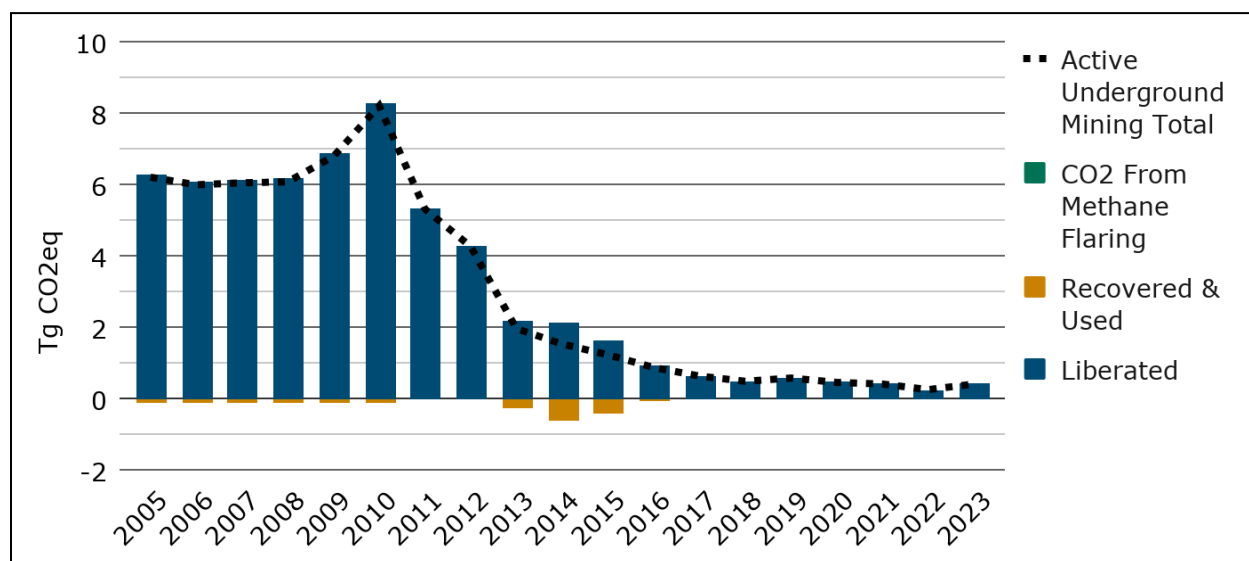
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and degasification systems. Some mines recover a portion of the CH₄ to utilize or flare, reducing the amount released to the atmosphere. This CH₄ is referred to as “recovered and used.” This CH₄ is not counted in the net emissions from Active Underground Mining, and is instead reported under the subsector in which it is used (e.g., stationary combustion of fossil fuel). Therefore, the total CH₄ emissions from Active Underground Mining are calculated as the CH₄ liberated from ventilation and degasification systems minus the CH₄ recovered and used. This method reflects an accounting practice to avoid double-counting emissions. The “negative” values represent mitigation of CH₄ that would have otherwise been liberated if not for the recovery and use, rather than the removal of CH₄ emissions.

Active Underground Mining emissions also include net fugitive CO₂ emissions. Coal seam gas that is recovered and used for on-site or off-site energy is reported under other sectors of the Inventory (e.g., stationary combustion of fossil fuel). This includes CO₂ entrained in the combusted coal seam gas. Therefore, the total CO₂ emissions from Active Underground Mining are calculated as the sum of CO₂ liberated during underground mining and CO₂ generated from CH₄ flaring, minus the CO₂ entrained in coal seam gas that is recovered and used for energy. The NI by State presents the CO₂ portion of coal seam gas that is recovered and used for energy separately from the CH₄ fraction (labeled ‘Recovered and Used for Energy’ and ‘Recovered and Used,’ respectively). For simplicity, this Inventory presents these recovered and used emissions as one combined category.

As shown in Figure 3.23, 2023 emissions from Active Underground Mining were 0.421 Tg CO₂eq, representing 23.4% of total emissions from Coal Mining. Compared to 2005 levels, emissions decreased by 5.800 Tg CO₂eq (93.2%) in 2023. Total emissions from Active Underground Mining peaked in 2010, then declined for the remainder of the time series. This decline was mainly due to decreases in liberated emissions, which accounted for the majority of total emissions. This change is most likely attributable to the closure of underground coal mines. CO₂ from Methane Flaring emissions were small and therefore not visible in the graph.

Figure 3.23. Historical Active Underground Mining Emissions by Category.



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Table 3.21 below displays liberated and recovered Active Underground Mining emissions by GHG and category for select years. CH₄ emissions accounted for the majority of emissions from 2005 through 2023, ranging from 0.182 Tg CO₂eq in 2022 to 8.025 Tg CO₂eq in 2010. CO₂ emissions remained relatively stable over the time series, ranging from 0.062 Tg CO₂eq in 2020 to 0.280 Tg CO₂eq in 2005. CO₂ emissions from CH₄ flaring were estimated only in 2020 and 2021, at 0.001 Tg CO₂eq in both years.

Table 3.21. Active Underground Mining Emissions by GHG and Category (Tg CO₂eq).

GHG	Category	2005	2010	2015	2020	2023
CH ₄	Liberated	6.033	8.117	1.530	0.410	0.348
CH ₄	Recovered & Used	-0.092	-0.092	-0.418	-0.005	0.000
CH₄ Subtotal		5.942	8.025	1.112	0.405	0.348
CO ₂	Liberated	0.280	0.198	0.129	0.061	0.073
CO ₂	CO ₂ From Methane Flaring	0.000	0.000	0.000	0.001	0.000
CO ₂	Recovered & Used	+	+	+	0.000	0.000
CO₂ Subtotal		0.280	0.198	0.129	0.062	0.073
Active Underground Mining Total		6.222	8.223	1.242	0.466	0.421

+ Absolute value of emissions does not exceed 0.0005 Tg CO₂eq.

3.4.2.1. Methodology

This Inventory used the NI by State dataset to estimate emissions from the Active Underground Mining category. State-level CH₄ and CO₂ emissions from Active Underground Mines were estimated using the same methodology applied at the national level. Instead of aggregating mine-level emissions to a national total, emissions were summed across individual mines within each state to produce state-level totals.⁴¹

To estimate CH₄ emissions, the EPA applied an IPCC Tier 3 methodology in accordance with IPCC guidelines and available data. CH₄ emissions from underground mines originate from both ventilation systems and degasification systems. Emissions from these systems were estimated using mine-specific data and summed to determine total CH₄ liberated.⁴² CH₄ emissions from ventilation systems were estimated using data from the GHGRP [Subpart FF, Underground Coal Mines](#), data provided by the MSHA, and site-specific data from additional sources. CH₄ emissions from degasification systems were estimated primarily using data reported under GHGRP Subpart FF.

CO₂ emissions from underground mining were estimated using the IPCC Tier 1 approach by applying a default emission factor (5.9 m³ per metric ton of coal produced) to annual coal production from underground mines. CO₂ emissions from coal seam gas recovered and utilized for energy purposes were estimated using the IPCC Tier 1 default emission factor (19.57 metric tons CO₂ per million cubic meters of coalbed CH₄ produced) and the quantity of CH₄ recovered and utilized. The GHGRP and state sale data provided data on the annual quantity

⁴¹ [2024 NI by State Methodology report](#), Chapter 2, Section 2.2.1.2.1, Active Underground Mines

⁴² [2025 NIR](#), Chapter 3, Section 3.4, Methodology and Time-Series Consistency

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of coal seam gas recovered and utilized. CO₂ emissions from CH₄ flaring were estimated using the IPCC Tier 1 method, which categorizes combustion-related CO₂ emissions from recovered gas used for non-energy purposes (e.g., flaring or catalytic oxidation) as fugitive emissions in underground coal mining.⁴³ Although these emissions result from stationary combustion processes, they are reported under fugitive CO₂ emission estimates in the Inventory.

Data reported under Subpart FF were not used directly in this Inventory due to incomplete coverage of the category and inconsistent reporting over time. Prior to 2020, underground coal mining facilities were subject to a reporting threshold of 25,000 metric tons CO₂eq under GHGRP Subpart FF, which limited coverage. Beginning in 2020, reporting requirements were expanded at the state level to include all underground coal mining operations regardless of emission magnitude. Due to these limitations, the Division did not rely solely on FRD for estimating emissions from active underground mines. However, Subpart FF data were used for comparison and were found to align closely with NI by State estimates in both magnitude and trends since 2014.

3.4.2.2. Uncertainty

Though uncertainty was not quantified for state-level estimates, the NIR quantified uncertainty estimates for national emissions at the subsector level. The EPA calculated uncertainties using the 2006 IPCC Guidelines Approach 2. The uncertainty associated with CH₄ emissions estimates from underground ventilation systems was relatively low because these emissions were based on actual measurement data from the EPA's GHGRP or MSHA. The NIR did not assess uncertainty for the Active Underground Mining category and instead assessed uncertainty for the Coal Mining subsector as a whole. The uncertainty in the national estimates for Coal Mining in 2023 ranged from -10% to +21% for CH₄ and from -68% to +76% for CO₂.⁴⁴ The uncertainty associated with underground mine emissions at the national level is assumed to be the same or similar for state-level underground mine emissions estimates.⁴⁵

3.4.2.3. QA/QC

For each emissions source category, the Division applied standard QA/QC procedures, including reviews of data connections and imports, unit conversions, GWP calculations, naming conventions, and data formatting.

The EPA applied QA/QC procedures consistent with the NIR QA/QC Management Plan, in accordance with IPCC Guidelines. The QA/QC Management Plan describes data and methodology checks, develops processes governing peer review and public comments, and provides guidance on conducting uncertainty analyses for the emissions estimates. The QA/QC Management Plan procedures also stress continual improvement, providing for corrective

⁴³ [2025 NIR](#), Chapter 3, Section 3.4, Methodology and Time-Series Consistency

⁴⁴ [2025 NIR](#), Chapter 3, Section 3.4, Uncertainty

⁴⁵ [2024 NI by State Methodology report](#), Chapter 2, Section 2.2.1.3, Uncertainty

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actions designed to improve the inventory estimates over time.⁴⁶ Trends in the time series were analyzed to determine if any corrective actions were needed.

3.4.2.4. Recalculations and Comparison to Prior Inventory

To estimate emissions from the Active Underground Mining category, this Inventory used the 2025 NI by State, whereas the prior Inventory used the 2022 NI by State. From 2005 through 2019, differences in historical emissions estimates between the 2023 and 2025 Inventories were de minimis. In 2020, emissions were less than 0.001 Tg CO₂eq or 1% lower in this Inventory than in the prior Inventory.

3.4.2.5. Planned Improvements

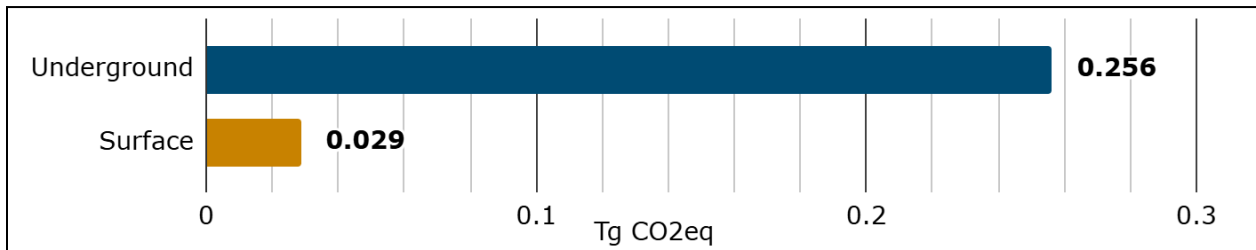
For future Inventories, the Division will consider using FRD, either to represent the emissions from Active Underground Mining activities or in verification efforts. Coordination and cooperation with other relevant external agencies and their data, including but not limited to the [ECMC](#), could improve data quality and help develop Active Underground Mining emissions factors and estimate emissions more accurately.

3.4.3. Post-Mining

Post-Mining activities are significant sources of CH₄ emissions that occur after coal has been extracted from the ground, including coal handling, storage, processing, and transportation.⁴⁷ The Post-Mining category encompasses emissions from coal obtained from both surface and underground mining. CO₂ emissions are not estimated for Post-Mining activities because the EPA performed a preliminary analysis and determined these emissions are negligible. Additionally, methods for estimating fugitive CO₂ emissions from abandoned coal mines are not provided in IPCC Guidelines.⁴⁸

In 2023, emissions from Post-Mining activities were 0.285 Tg CO₂eq, representing 15.8% of total emissions from Coal Mining. Post-Mining emissions from coal originating in underground mines were 0.256 Tg CO₂eq, and emissions from coal obtained in surface mines were 0.029 Tg CO₂eq, as shown in Figure 3.24.

Figure 3.24. 2023 Post-Mining Emissions.



⁴⁶ [NIR](#), Annex 8 QA/QC Procedures

⁴⁷ [2025 NIR](#), Chapter 3, Section 3.4, Introduction

⁴⁸ [NIR](#), Annex 5 Assessment of the Sources and Sinks of Greenhouse Gas Emissions Not Included

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As shown in Figure 3.25, emissions from Post-Mining activities generally declined between 2005 and 2023, with emissions decreasing by 0.753 Tg CO₂eq (72.5%) in 2023 compared to 2005 levels. CH₄ emissions were lowest in 2020, at 0.236 Tg CO₂eq, and highest in 2005, at 1.037 Tg CO₂eq. Post-Mining activities from coal sourced in underground mines were the main source of emissions throughout the time series, with levels ranging from 0.214 Tg CO₂eq in 2020 to 0.979 Tg CO₂eq in 2005. Post-Mining emissions from surface mined coal were more stable over the time series, ranging from 0.021 Tg CO₂eq in 2019 to 0.058 Tg CO₂eq in 2005.

Figure 3.25. Historical Post-Mining Emissions.

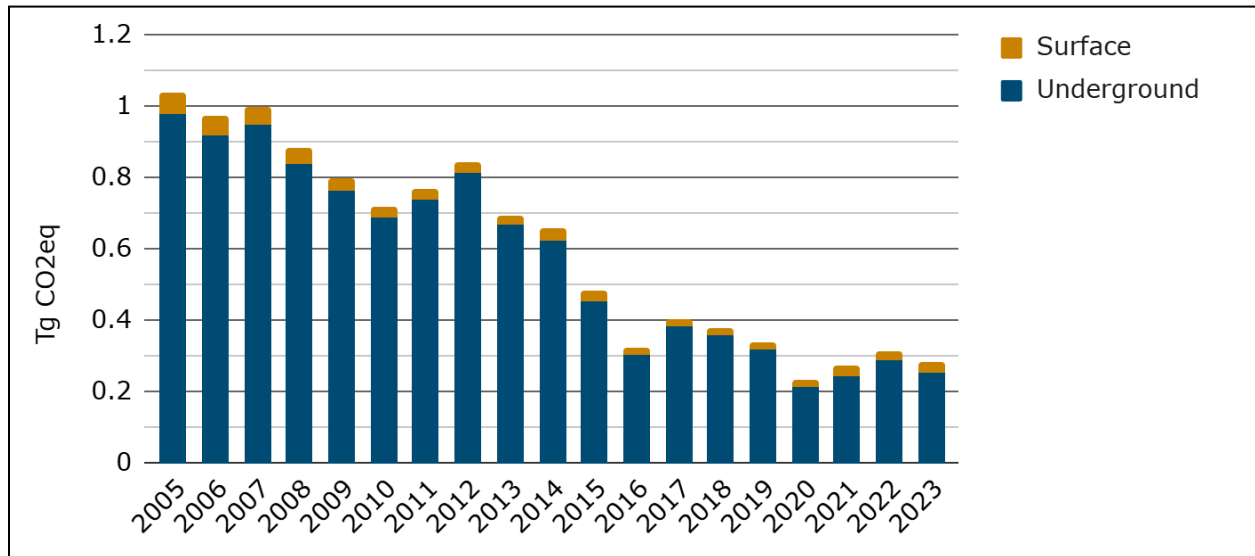


Table 3.22 presents CH₄ emissions for the Post-Mining by category for select years.

Table 3.22. Post-Mining Emissions by GHG and Category (Tg CO₂eq).

GHG	Category	2005	2010	2015	2020	2023
CH ₄	Surface	0.058	0.029	0.033	0.022	0.029
CH ₄	Underground	0.979	0.691	0.452	0.214	0.256
Post -Mining Total		1.037	0.721	0.486	0.236	0.285

3.4.3.1. Methodology

This Inventory used the NI by State dataset to estimate emissions from the Post-Mining category. The EPA applied a Tier 2 methodology, in accordance with IPCC guidelines, to estimate CH₄ emissions. CH₄ emissions were estimated by multiplying basin-specific coal production by basin-specific CH₄ contents and a 32.5% emission factor for CH₄ desorption during coal transportation and storage.⁴⁹ CO₂ emissions from Post-Mining activities are not estimated due to a lack of data and IPCC methodology. To determine state-level CH₄ emission estimates, national emissions were apportioned based on coal production in each state as reported in the EIA Annual Coal Report. The appropriate basin-specific CH₄ content for the

⁴⁹ [2025 NIR](#), Chapter 3, Section 3.4, Methodology and Time-Series Consistency

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coal produced in a state was assigned based on the coal basin within which the state is located. These emissions are assigned to the state where the coal was produced, even if a portion of such emissions may occur outside the state, such as during interstate transport and storage before use.⁵⁰

The Division reviewed the NI by State and the SIT for estimating Post-Mining emissions. No emissions data are reported to the EPA or the Division, and no other FRD were considered. The NI by State was determined to be more complete, rigorous, and detailed in its methods. Because of these reasons, and to present a more conservative estimate, the Division elected to use the NI by State's estimates.

3.4.3.2. Uncertainty

Though uncertainty was not quantified for state-level estimates, the NIR quantified uncertainty estimates for national emissions. The EPA calculated uncertainties using the 2006 IPCC Guidelines Approach 2. The NIR did not assess uncertainty for the Post-Mining category and instead assessed uncertainty for the Coal Mining subsector as a whole. The levels of uncertainty in the national estimates for Coal Mining in 2023 ranged from -10% to +21% for CH₄ and from -68% to +76% for CO₂.⁵¹ State-level estimates may be more uncertain because national emissions estimates are apportioned across states.

3.4.3.3. QA/QC

For each emissions source category, the Division applied standard QA/QC procedures, including reviews of data connections and imports, unit conversions, GWP calculations, naming conventions, and data formatting.

The EPA applied QA/QC procedures consistent with the NIR QA/QC Management Plan, in accordance with IPCC Guidelines. The QA/QC Management Plan describes data and methodology checks, develops processes governing peer review and public comments, and provides guidance on conducting uncertainty analyses for the emissions estimates. The QA/QC Management Plan procedures also stress continual improvement, providing for corrective actions designed to improve the inventory estimates over time.⁵² Trends in the time series were analyzed to determine if any corrective actions were needed.

3.4.3.4. Recalculations and Comparison to Prior Inventory

To estimate emissions from the Post-Mining category, this Inventory used the 2025 NI by State, whereas the prior Inventory used the 2022 NI by State. From 2005 through 2020, differences in historical emissions estimates between the 2023 and 2025 Inventories were de minimis.

⁵⁰ [2024 NI by State Methodology report](#), Chapter 2, Section 2.2.1.2.2 Surface Mining and Post-mining Activities

⁵¹ [2025 NIR](#), Chapter 3, Section 3.4, Uncertainty

⁵² [NIR](#), Annex 8 QA/QC Procedures

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3.4.3.5. Planned Improvements

For future Inventories, the Division will consider using FRD if available, either to represent the emissions from Post-Mining activities or in verification efforts. Post-Mining activities are not currently subject to FRD requirements at the state or federal level. Coordination and cooperation with other relevant external agencies and their data, including but not limited to the [ECMC](#), could improve data quality and help develop Abandoned Coal Mines emissions estimates.

3.4.4. Abandoned Coal Mines

Abandoned Coal Mines continue to release CH₄ after closure, even though active mining operations have ceased. Mines are typically closed and abandoned once coal seams have been fully extracted. Some abandoned mines are vented to prevent CH₄ buildup, which can migrate to surface structures through overburden fractures.⁵³ Though CO₂ is likely entrained in the coal seam gas released from abandoned coal mines, CO₂ emissions from Abandoned Coal Mines were not estimated in the NI by State and are not presented in this Inventory. The EPA performed a preliminary analysis of CO₂ emissions from Abandoned Coal Mines based on gas composition data from two underground mines in two states and determined that this emission source is likely negligible. Additionally, methods for estimating fugitive CO₂ emissions from abandoned coal mines are not provided in IPCC Guidelines.⁵⁴

Although work within the mine stops, CH₄ liberation does not cease. While emissions decline after abandonment, they may continue at a near-steady rate for many years. In mines that are flooded by the intrusion of groundwater or surface water, emissions may last only a few years until the mine is fully flooded. Gas can reach the surface through inadequately sealed conduits, and diffuse emissions may occur through cracks and fissures in overlying strata. The magnitude of CH₄ emissions from abandoned mines depends on several factors, including the time since abandonment, coal gas content and adsorption characteristics, mine flooding, the presence of vent holes, and the effectiveness of mine seals. Abandoned mines may have a status of vented, in which all CH₄ is vented freely to the atmosphere, sealed (50%, 80%, or 95% sealed), or flooded (no emissions). A list of abandoned underground coal mines in Colorado, including the county, the operator, and their status as either abandoned (A) or abandoned and sealed (A & S), is provided in Appendix D.⁵⁵

Abandoned Coal Mine emissions comprise total liberated CH₄ (from both ventilation and degasification systems), minus the quantity of CH₄ that was recovered and used. As discussed in Active Underground Mining (Section 3.4.2), recovered and used CH₄ is accounted for in the category in which it is used (e.g., stationary combustion of fossil fuel). Therefore, these emissions are treated as a negative value in the Abandoned Coal Mine category as an accounting practice to avoid double-counting. In 2023, emissions from Abandoned Coal Mines were 0.956 Tg CO₂eq, representing 53.2% of Coal Mining emissions. This accounts for

⁵³ [2025 NIR](#), Chapter 3, Section 3.5, Introduction

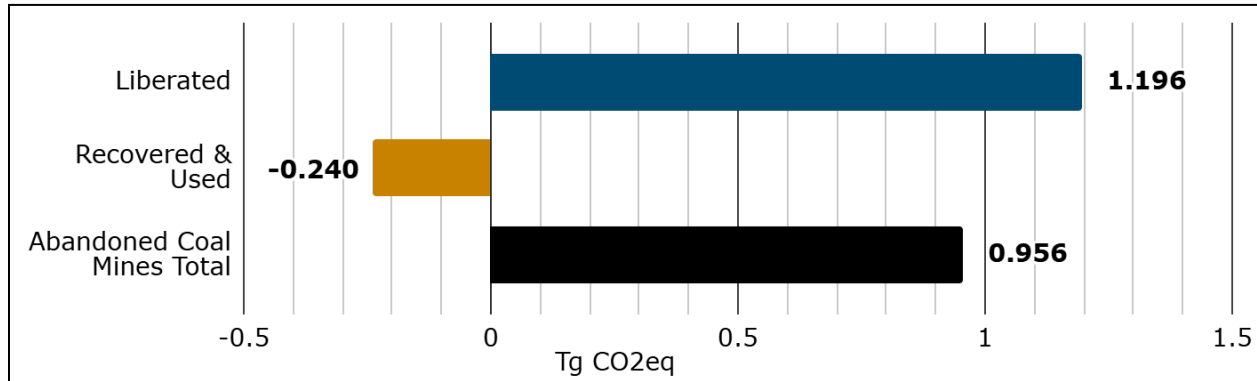
⁵⁴ [NIR](#), Annex 5 Assessment of the Sources and Sinks of Greenhouse Gas Emissions Not Included

⁵⁵ [MSHA Mine Data Retrieval System](#)

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approximately 1.196 Tg CO₂eq liberated, and -0.240 Tg CO₂eq recovered and used, as shown in Figure 3.26. Compared to 2005 levels, Abandoned Coal Mines emissions decreased by 0.181 Tg CO₂eq (15.9%) in 2023.

Figure 3.26. 2023 Abandoned Coal Mines Emissions.



Historical emissions from Abandoned Coal Mines are shown in Figure 3.27. From 2005 through 2023, total emissions from Abandoned Coal Mines fluctuated, peaking in 2016, and then declining afterward. Table 3.23 presents CH₄ emissions for the Abandoned Coal Mines by category for select years. CH₄ emissions ranged from 0.886 Tg CO₂eq in 2020 to 1.192 Tg CO₂eq in 2016.

Figure 3.27. Historical Abandoned Coal Mines Emissions by Category.

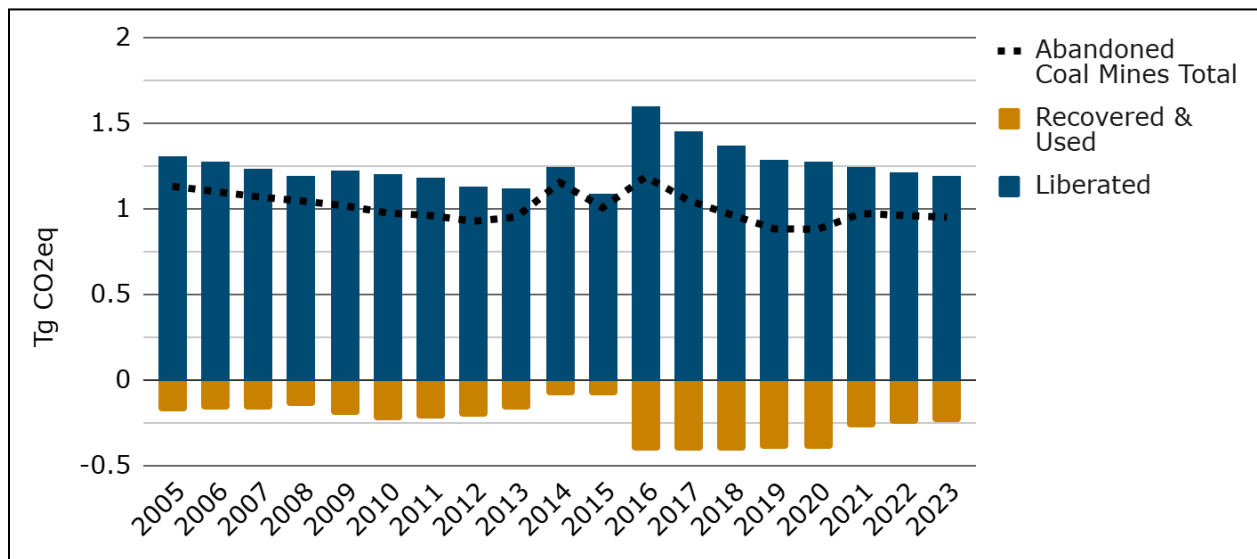


Table 3.23. Abandoned Coal Mines Emissions by GHG and Category (Tg CO₂eq).

GHG	Category	2005	2010	2015	2020	2023
CH ₄	Liberated	1.316	1.209	1.096	1.283	1.196
CH ₄	Recovered & Used	-0.180	-0.229	-0.085	-0.397	-0.240
Abandoned Coal Mines Total		1.136	0.980	1.011	0.886	0.956

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3.4.4.1. Methodology

This Inventory used the NI by State dataset to estimate emissions from the Abandoned Coal Mines category. The EPA used multiple data sources to estimate CH₄ emissions from Abandoned Coal Mines, including MSHA reports, field measurements, state and basin-level coal production data, EPA abandoned mine databases, and recovery project data from the [Coalbed Methane Outreach Program \(CMOP\)](#), the [Global Methane Initiative \(GMI\)](#), and the [American Carbon Registry \(ACR\)](#).⁵⁶

For 2020 through 2023, state-level emissions were estimated alongside the national totals using an updated version of the Abandoned Mine Methane model, which includes state identifiers as a standard output. Previously, the model generated estimates based only on coal basin identifiers. The Abandoned Mine Methane model has been enhanced to produce both national- and state-level estimates for each inventory year. State-level emissions estimates for 2005 through 2019 were estimated by disaggregating national emissions using allocation factors based on CH₄ emissions at the time of abandonment, mine status (i.e., flooded, sealed, vented, and unknown), date of abandonment, and mine location. State allocation factors were first based on basin and mine status using average daily emissions at the state and basin levels. Annual emissions by coal basin were developed and normalized using the Abandoned Mine Methane model. Then the state allocation factors were multiplied by the basin totals to estimate state emissions.⁵⁷

Estimating CH₄ emissions from Abandoned Coal Mines requires projecting emissions from the time of closure through the inventory year. Emissions depend on the mine's characteristics at abandonment, including prior CH₄ production rates, the extent of flooding, and sealing. Decline curves, similar to hyperbolic curves used in natural gas well production, are applied to model the decrease in emissions over time. Mines may be vented, partially or fully sealed, or flooded, with each condition affecting the rate and duration of CH₄ liberation.

The EPA used a hybrid Tier 2/Tier 3 approach to model Abandoned Coal Mines through Monte Carlo simulations. Tier 3 calculations were applied for mines with known post-abandonment status, where decline curves could directly estimate emissions. Tier 2 calculations were used for mines with unknown status to estimate basin-level emissions. Decline functions account for vented, flooded, and sealed mines, with sealed mines modeled using reduced initial emission rates corresponding to the degree of closure. Historical mine data, production rates, and basin-specific characteristics were applied to all abandoned coal mines, and totals were adjusted to account for CH₄ recovery projects.

The Division reviewed the NI by State and the SIT for estimating emissions from Abandoned Coal Mines. No emissions data are reported to the EPA or the Division, and no other FRD were considered. Emissions estimated with the SIT were generally lower than those in the NI by State while offering less disaggregation. The NI by State was determined to be more

⁵⁶ [2025 NIR](#), Chapter 3, Section 3.5, Methodology and Time-Series Consistency

⁵⁷ [2024 NI by State Methodology report](#), Chapter 2, Section 2.2.2.2, Methods/Approach

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complete, rigorous, and detailed in its methods. Because of these reasons, and to present a more conservative estimate, the Division elected to use the NI by State's estimates.

3.4.4.2. Uncertainty

Though uncertainty was not quantified for state-level estimates, the NIR quantified uncertainty estimates for national emissions. The EPA calculated uncertainties using the 2006 IPCC Guidelines Approach 2. For Abandoned Coal Mines, levels of uncertainty in the national estimates in 2023 ranged from -20 to +24% for CH₄.⁵⁸ State-level estimates may be more uncertain because national emissions estimates are apportioned across states.

3.4.4.3. QA/QC

For each emissions source category, the Division applied standard QA/QC procedures, including reviews of data connections and imports, unit conversions, GWP calculations, naming conventions, and data formatting.

The EPA applied QA/QC procedures consistent with the NIR QA/QC Management Plan, in accordance with IPCC Guidelines. The QA/QC Management Plan describes data and methodology checks, develops processes governing peer review and public comments, and provides guidance on conducting uncertainty analyses for the emissions estimates. The QA/QC Management Plan procedures also stress continual improvement, providing for corrective actions designed to improve the inventory estimates over time.⁵⁹ Trends in the time series were analyzed to determine if any corrective actions were necessary.⁶⁰

3.4.4.4. Recalculations and Comparison to Prior Inventory

To estimate emissions from the Abandoned Coal Mines category, this Inventory used the 2025 NI by State, whereas the prior Inventory used the 2022 NI by State. From 2005 through 2019, differences in historical emissions estimates between the 2023 and 2025 Inventories were de minimis. In 2020, emissions were less than 0.001 Tg CO₂eq or 1% lower in this Inventory than in the prior Inventory.

3.4.4.5. Planned Improvements

For future Inventories, the Division will consider using FRD if available, either to represent the emissions from Abandoned Coal Mines or in verification efforts. Abandoned Coal Mines are not currently subject to FRD requirements at the state or federal level. Coordination and cooperation with other relevant external agencies and their data, including but not limited to the [Energy & Carbon Management Commission](#) (ECMC), could improve data quality and help develop Abandoned Coal Mines emissions factors and estimate emissions more accurately.

⁵⁸ [2025 NIR](#), Chapter 3, Section 3.5, Uncertainty

⁵⁹ [NIR](#), Annex 8 QA/QC Procedures

⁶⁰ [2025 NIR](#), Chapter 3, Section 3.5, QA/QC and Verification

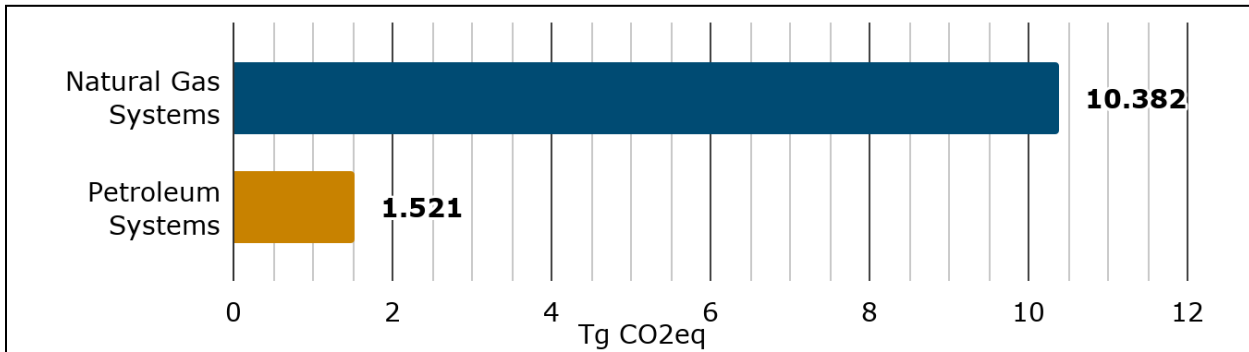
3.5. Natural Gas and Petroleum Systems

The Natural Gas and Petroleum Systems (NGPS) subsector includes predominantly CH₄ and CO₂ emissions from various equipment, activities, and segments of the oil and natural gas supply chain. These emissions include fugitives or leaks, as well as emissions from venting and flaring. The segments of Natural Gas Systems include exploration, production and gathering, processing, distribution, transmission and storage, gas post-meter, and abandoned wells.⁶¹ The segments of Petroleum Systems include exploration, production and upgrading, crude oil transportation, and abandoned wells.⁶²

Combustion emissions, such as those from fuel used in engines or heaters, that occur within the oil and natural gas supply chain (with the exception of the destruction of waste gas streams via flaring) are captured elsewhere, either in Industrial Fuel Use (RCI Fuel Use, Section 3.2.3) or Transportation emissions (Section 3.3). Emissions from refining are captured in the IPPU sector (Chapter 4).

In 2023, NGPS emissions were 11.093 Tg CO₂eq, accounting for 12.3% of Energy emissions. About 87% of NGPS emissions were from Natural Gas Systems (10.382 Tg CO₂eq) and 12% from Petroleum Systems (1.521 Tg CO₂eq), as detailed in Figure 3.28 below.

Figure 3.28. 2023 Natural Gas and Petroleum Systems Emissions.



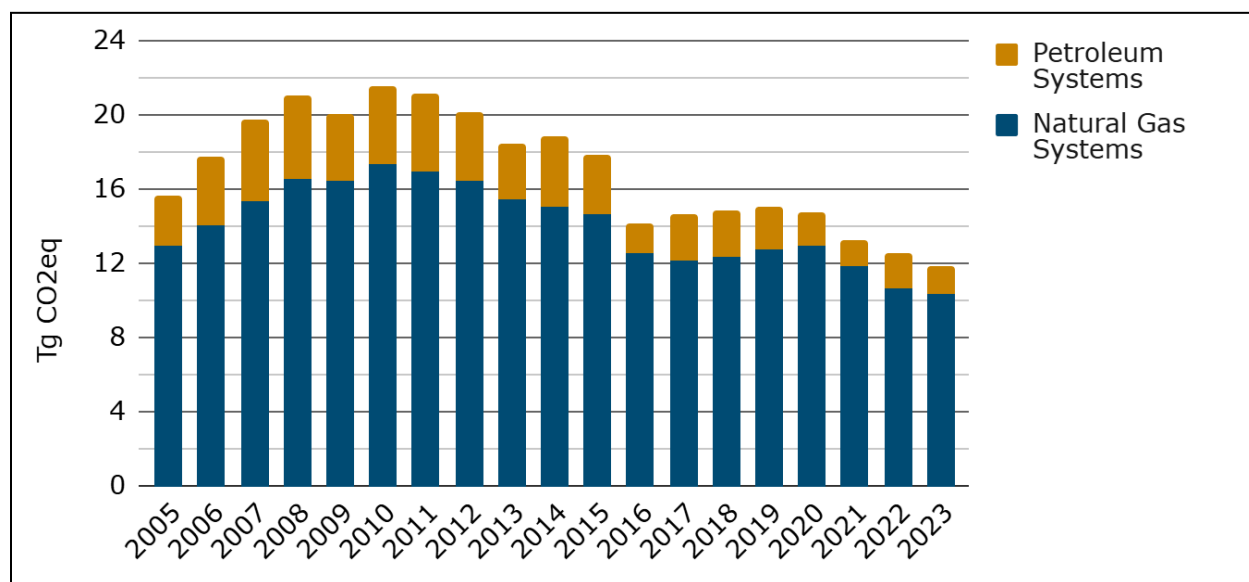
From 2005 through 2010, NGPS emissions generally increased; since 2020, though, NGPS emissions generally decreased through 2023. Historical emissions and reductions were mostly driven by Natural Gas Systems, as shown in Figure 3.29 below. NGPS emissions in 2023 were 3.830 Tg CO₂eq or 24.3% below 2005 emissions. This corresponds to reductions of 2.616 Tg CO₂eq or 20.1% for Natural Gas Systems and 1.214 Tg CO₂eq or 44.4% for Petroleum Systems.

⁶¹ [2025 NIR](#), Chapter 3, Section 3.7, Natural Gas Systems Introduction

⁶² [2025 NIR](#), Chapter 3, Section 3.6, Petroleum Systems Introduction

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Figure 3.29. Historical Natural Gas and Petroleum Systems Emissions.



Across the time series, CH₄ emissions made up the majority of NGPS emissions (83% to 91%), as detailed in Table 3.24 below. From 2005 through 2023, CH₄ emissions decreased by 4.013 Tg CO₂eq or 28.7%. However, CH₄ emissions peaked in 2010 at 19.469 Tg CO₂eq; since that peak, CH₄ emissions decreased 9.496 Tg CO₂eq or 48.8%. CO₂ emissions increased since 2005 but remained below the 2015 peak of 2.733 Tg CO₂eq. N₂O emissions increased slightly from 2005, but the emissions remain de minimis (less than 0.001 Tg CO₂eq).

Table 3.24. NGPS Emissions by GHG and System (Tg CO₂eq).

GHG	System	2005	2010	2015	2020	2023
CO ₂	Natural Gas Systems	1.664	1.969	2.228	2.029	1.644
CO ₂	Petroleum Systems	0.082	0.192	0.505	0.310	0.285
CO₂ Subtotal		1.746	2.161	2.733	2.339	1.929
CH ₄	Natural Gas Systems	11.333	15.397	12.423	10.939	8.738
CH ₄	Petroleum Systems	2.653	4.072	2.735	1.535	1.236
CH₄ Subtotal		13.986	19.469	15.157	12.475	9.973
N ₂ O	Natural Gas Systems	+	0.001	0.001	+	+
N ₂ O	Petroleum Systems	+	+	+	+	0.001
N₂O Subtotal		+	0.001	0.001	0.001	0.001
NGPS Total		15.733	21.631	17.891	14.814	11.903

+ Emissions do not exceed 0.0005 Tg CO₂eq.

This Inventory incorporated new data sources and methodologies for estimating emissions from Natural Gas and Petroleum Systems. These updates resulted in changes to the emissions estimates and also the categorical structure as compared with the prior two inventories. The new methodology utilized the overall categorical structure of the NI by State and much of its data, supplemented with FRD and modifications for specific segments or categories.

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Additional detail about the methodological change in comparison to the previous inventory is available in [Section 3.5.4](#) for NGPS as a whole.

3.5.1. Natural Gas Systems

Emissions from Natural Gas Systems include those from leaks, venting, and flaring, per IPCC guidelines. These emissions may occur across several segments or activities, including Exploration, Production and Gathering, Processing, Transmission and Storage, Distribution, Post-Meter, and Abandoned Wells.⁶³

Exploration segment emissions include those from well drilling, testing, and completion.

Production segment emissions include leaks, vents, and flaring from wells and from well-site equipment such as pneumatic controllers and separators, and activities such as liquids unloading of wells and equipment blowdowns for maintenance operations. Production emissions also include those from gathering and boosting sources, including compressor stations (with multiple emissions sources) and pipeline operations.

Processing segment emissions result from equipment used to remove natural gas liquids and various other constituents from the raw gas, resulting in “pipeline quality” gas, which is sent into the transmission system.

Transmission and Storage segment emissions result from high-pressure, large-diameter pipelines that transport gas long distances from field production and processing areas to distribution systems or large-volume customers such as power plants or chemical plants. Leaks and venting emissions also result when natural gas is injected and stored in underground formations, or liquefied and stored in above-ground tanks, during periods of low demand (e.g., summer), and withdrawn, processed, and distributed during periods of high demand (e.g., winter).

Distribution segment emissions include leaks from pipelines that take high-pressure gas from the transmission system at “city gate” stations, reduce the pressure, and distribute the gas to individual end users.

Post-Meter emissions include leaks from residential and commercial appliances, industrial facilities, power plants, and natural gas-fueled vehicles. These emissions represent fugitives or leaks from natural gas consumption. Combustion emissions are captured under RCI Fuel Use, Electric Power, and Transportation.

Abandoned Wells represent orphaned and other non-producing wells that are still capable of fugitive emissions. These wells include those with no recent production that are not plugged (i.e., inactive or dormant), those with no recent production and no responsible operator (i.e., orphaned or deserted), and those that have been plugged.⁶⁴

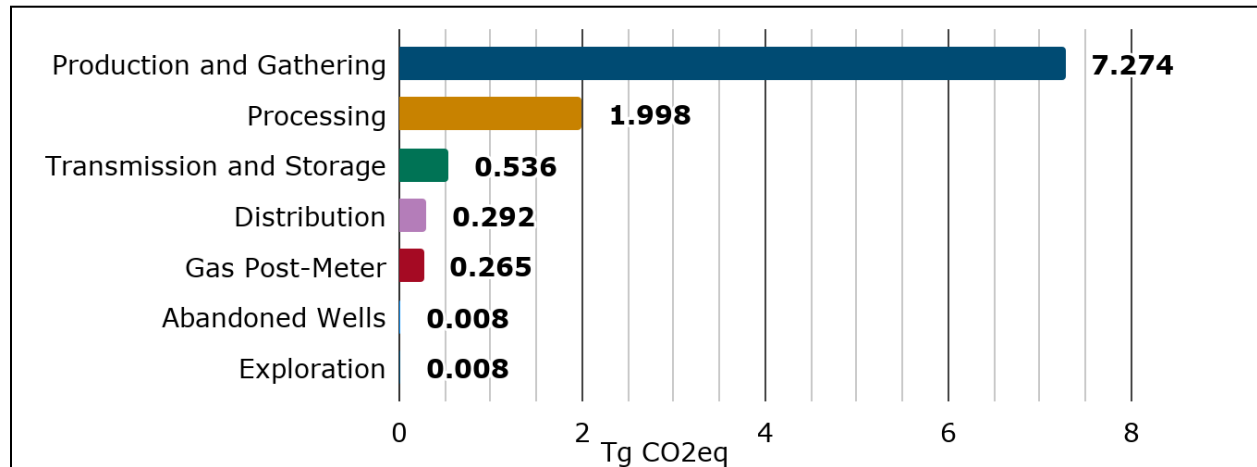
⁶³ [2025 NIR](#), Chapter 3, Section 3.7, Natural Gas Systems Introduction

⁶⁴ [2025 NIR](#), Chapter 3, Section 3.8, Abandoned Oil and Gas Wells, Methodology

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In 2023, Natural Gas Systems emissions were 10.832 Tg CO₂eq. Production and Gathering emissions (7.274 Tg CO₂eq) accounted for 70%, and Processing emissions (1.998 Tg CO₂eq) accounted for about 19% of total Natural Gas Systems Emissions, as illustrated in Figure 3.30 below. Emissions from Transmission and Storage (0.536 Tg CO₂eq), Distribution (0.292 Tg CO₂eq), and Gas Post-Meter (0.265 Tg CO₂eq) together accounted for about 10%, while emissions from Exploration (0.008 Tg CO₂eq) and Abandoned Wells (0.008 Tg CO₂eq) together accounted for less than 1% of the 2023 emissions.

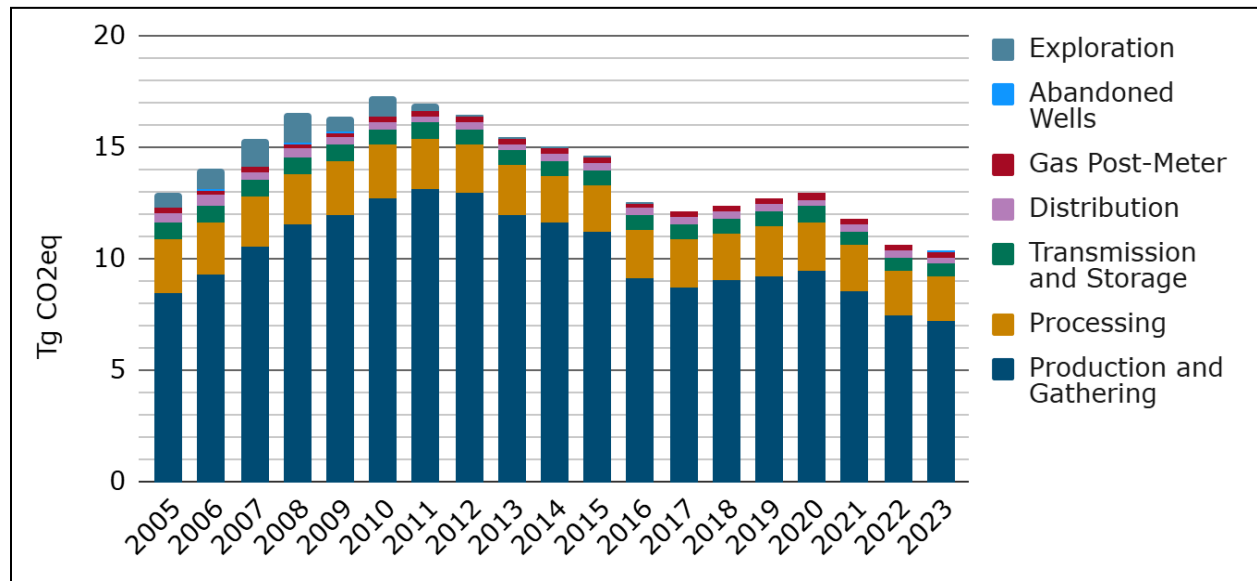
Figure 3.30. 2023 Natural Gas Systems Emissions.



From 2005 through 2023, Natural Gas Systems emissions decreased by 2.616 Tg CO₂eq or 20.1%. As shown in Figure 3.31 below, emissions from Exploration fell significantly through 2011 and remained less than 0.100 Tg CO₂eq from 2012 through 2023. Exploration emissions in 2023 were 0.653 Tg CO₂eq or 99% below 2005 levels. Across the time series, Production and Gathering emissions were highest among the segments. Production and Gathering emissions increased from 2005 to about 2011 but generally decreased since then, falling to 1.254 Tg CO₂eq or 14.7% below 2005 levels in 2023. Emissions from Processing, Transmission and Storage, and Distribution all decreased from 2005 through 2023, by 0.348 Tg CO₂eq (14.8%), 0.216 Tg CO₂eq (28.7%), and 0.184 Tg CO₂eq (38.6%), respectively.

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Figure 3.31. Historical Natural Gas Systems Emissions by Segment.



As detailed in Table 3.25 below, most of the Natural Gas Systems emissions were CH₄, which changed more significantly than other GHGs over the time series. CH₄ emissions increased from 2005 to a peak of 15.397 Tg CO₂eq in 2010, and decreased since then. 2023 emissions were 2.596 Tg CO₂eq (22.9%) below 2005 emissions. CH₄ emissions were mostly from Production and Gathering across the time series; from 2005 through 2023, this segment accounted for 73% to 86% of Natural Gas Systems CH₄ emissions. Since 2005, CH₄ emissions from Production and Gathering decreased by 1.387 Tg CO₂eq (16.7%).

CO₂ emissions changed somewhat over the time series, peaking at 2.228 Tg CO₂eq in 2015, but 2023 emissions were just 0.020 Tg CO₂eq (1.2%) below 2005 emissions. Most of the Natural Gas Systems CO₂ emissions (61% to 83%) were from the Processing segment, which includes acid gas removal units (also called amine units). Acid gas removal units are used to remove CO₂ from or “sweeten” natural gas. CO₂ emissions from Processing decreased by 0.117 Tg CO₂eq (8.4) from 2005 through 2023.

N₂O emissions were very low from 2005 through 2023, peaking at just 0.001 Tg CO₂eq in 2014. N₂O emissions in 2023 were less than 0.0005 Tg CO₂eq below 2005 emissions, though this corresponds to a 44.0% change. N₂O emissions were mostly from Production and Gathering.

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Table 3.25. Natural Gas Systems Emissions by GHG and Segment (Tg CO₂eq).

GHG	Segment	2005	2010	2015	2020	2023
CO ₂	Abandoned Wells	+	+	+	+	+
CO ₂	Distribution	0.001	+	+	+	+
CO ₂	Exploration	0.042	0.061	0.042	0.001	0.002
CO ₂	Gas Post-Meter	+	+	+	+	+
CO ₂	Processing	1.390	1.477	1.298	1.429	1.273
CO ₂	Production and Gathering	0.231	0.429	0.886	0.586	0.364
CO ₂	Transmission and Storage	0.001	0.001	0.001	0.012	0.005
CO₂ Subtotal		1.664	1.969	2.228	2.029	1.644
CH ₄	Abandoned Wells	0.020	0.021	0.019	0.009	0.008
CH ₄	Distribution	0.476	0.303	0.316	0.313	0.292
CH ₄	Exploration	0.618	0.886	0.029	0.005	0.006
CH ₄	Gas Post-Meter	0.215	0.225	0.236	0.262	0.265
CH ₄	Processing	0.957	0.895	0.763	0.786	0.725
CH ₄	Production and Gathering	8.297	12.334	10.385	8.878	6.910
CH ₄	Transmission and Storage	0.751	0.733	0.673	0.686	0.531
CH₄ Subtotal		11.333	15.397	12.423	10.939	8.738
N ₂ O	Distribution	0.000	0.000	0.000	0.000	0.000
N ₂ O	Exploration	+	+	+	+	+
N ₂ O	Gas Post-Meter	0.000	0.000	0.000	0.000	0.000
N ₂ O	Processing	+	+	+	+	+
N ₂ O	Production and Gathering	+	+	0.001	+	+
N ₂ O	Transmission and Storage	+	+	+	+	0.000
N₂O Subtotal		+	0.001	0.001	+	+
Natural Gas Systems Total		12.998	17.367	14.651	12.969	10.382

+ Emissions do not exceed 0.0005 Tg CO₂eq.

3.5.1.1. Methodology

This Inventory estimated emissions for all categories using the NI by State to varying degrees. For Exploration, Production and Gathering, and Abandoned Wells, the NI by State was modified for 2005 through 2015 and used “as is” for 2016 through 2023. For Distribution and Transmission and Storage, FRD was used in recent years, and the NI by State was used to inform historical trends. For Processing and Post-Meter, the NI by State was used “as is” from 2005 through 2023.

The EPA generally estimated national Natural Gas Systems emissions using emission factors and activity data. For most sources, this approach involved equipment- or activity-specific emission factors which may vary over time as a result of changes to technologies and operating practices. These emission factors are used to calculate net emissions directly. For

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other sources, the approach used what are considered “potential factors” and emission reduction data to calculate net emissions. Emission factors for these sources represent “potential” emissions without consideration of technologies or practices that reduce emissions, informed by data from relevant voluntary and federally-required regulatory reductions, such as those from the [Natural Gas STAR Program](#), [New Source Performance Standards \(NSPS\)](#) and [National Emission Standards for Hazardous Air Pollutants \(NESHAP\)](#) regulations. The estimated emission reductions were subtracted from potential emissions to calculate net emissions. The adjustments do not account for reductions from state actions or programs.

The estimates were developed with an IPCC Tier 2 approach; Tier 1 approaches were not used. When [Subpart W](#) data from the GHGRP were available (generally starting in 2010, 2011, or 2015, depending on the segment), they were used to develop emission factors. Emission factors were calculated by dividing net emissions by activity data. For years prior to when Subpart W data were available, emission factors were interpolated, extrapolated, calculated using interpolated or extrapolated emissions and activity data, or pulled from published literature, including from a [1996 study published by the Gas Research Institute \(GRI\) and the EPA](#). Activity data were sourced from several datasets, including those published by Enverus, the Federal Energy Regulatory Commission (FERC), the EIA, and the Pipeline and Hazardous Materials Safety Administration (PHMSA).⁶⁵ Annex Tables 3.6-6, 3.6-13, and 3.6-16, respectively, detail the CH₄, CO₂, and N₂O emission factor data sources and methodologies by segment and source for national Natural Gas Systems emissions estimates. Annex Table 3.6-8 details the activity data sources and methodologies by segment and source for national Natural Gas Systems emissions estimates.

The national estimates were then disaggregated by the EPA to the state level for the NI by State. For 2011 through 2023, emissions from most sources were disaggregated based on data collected through [Subpart W](#) of the federal GHGRP. The GHGRP data include emissions for individual sources or equipment, such as pneumatic controllers or storage tanks, in several segments. Emissions and activity data are reported at the facility level or the basin level, depending on the segment. When facility-level data were available, they were used to develop state-level emission estimates. When only basin-level data were available, the national-level emission estimates were disaggregated to states using proxy data, such as well counts or gas production volumes from [Enverus](#).⁶⁶ For 2005 through 2010 when GHGRP data does not exist, national emissions were disaggregated to the state level using proxy data.. Although national emissions were published by equipment, state-level estimates are available by segment only.

For 2005 through 2015, this Inventory estimated emissions by adjusting the NI by State emissions with annual scaling factors for the Exploration, Production and Gathering, and Abandoned Wells segments. This approach was developed to reflect findings from research published in the last decade which consistently found “bottom up” methods like those used in

⁶⁵ [2025 NIR](#), Annex 3.6, Methodology for Estimating CH₄, CO₂, and N₂O Emissions from Natural Gas Systems

⁶⁶ [2024 NI by State Methodology Report](#), Chapter 2, Section 2.2.4.2, Methods/Approach

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the NIR underestimate CH₄ emissions in comparison to atmospheric observations, particularly in the “production” segment.⁶⁷ For 2016 through 2023, these segments’ emissions were estimated using the NI by State without modifications based on the methodology and data changes incorporated into the NI by State around 2015 and to reflect Colorado’s oil and gas regulatory timeline.

Rutherford et al. 2021 recalculated national production-segment CH₄ emission estimates for calendar year 2015 using component level emissions measurement data.⁶⁸ The recalculated emissions were about 85% higher than the emissions estimates published in the [2020 NIR](#). This result aligned (within uncertainty) to a study by Alvarez et al. 2018 which evaluated the 2015 emissions published in the [2017 NIR](#) and found observed CH₄ emissions to be 60% higher.⁶⁹ The Alvarez et al. 2018 study synthesized a decade of measurement efforts, consistently finding that inventory-based methods, including those in the NIR, consistently underestimate oil and gas CH₄ emissions, especially in the production and gathering segment.

The Division reviewed the 2017 and 2020 NIRs referenced by these academic studies, and compared their emissions and trends with the 2025 NIR emissions and trends (which informed the 2025 NI by State). Based on the Rutherford and Alvarez findings that measured or modeled emissions are larger than those based on bottom-up inventories, the Division chose to increase the NI by State emissions estimates with annual emissions scaling factors for 2005 through 2015. These scaling factors were increased linearly from 2015 through 2005 by 2% each year (i.e., 60% in 2015, 62% in 2014, 64% in 2013, and so on, culminating at 80% in 2005). The NI by State emissions in each year were increased by that year’s scaling factor. The assumed 2005 through 2015 trend in annual scaling factors created an emissions time series more representative of expectations based on production and operations in the state and attempted to address some of the uncertainty and lower confidence in the methodologies and data available for estimating 2005 emissions. The scale factors were applied to the following NI by State categories: Natural Gas Exploration, Natural Gas Production and Gathering, Natural Gas Abandoned Wells, Petroleum Exploration, Petroleum Production and Upgrading, and Petroleum Abandoned Wells. Although the literature informing the annual scale factors only studied CH₄ emissions, the same annual scale factors were also applied to CO₂ and N₂O emissions in this Inventory. As NGPS emissions are from leaks, venting, and flaring (rather combustion emissions, which are accounted for under Industrial Fuel Use discussed in Section 3.2.3), the underestimation of CH₄ emissions likely translates to an underestimation of CO₂ and N₂O as well, particularly before 2015, after which flaring controls and other regulatory measures became more widely adopted by [Regulation Number 7](#).

⁶⁷ The “production” segment may be defined differently across literature, but it generally aligns with the “upstream” segment and includes all activities and operations associated with drilling new wells (exploration) and extraction of hydrocarbon fluids from wells (production), as well as abandoned wells which are no longer producing but may still emit small amounts of CH₄.

⁶⁸ Rutherford et al., 2021, Closing the methane gap in US oil and natural gas production emissions inventories. Nat Commun 12, 4715 (2021). <https://doi.org/10.1038/s41467-021-25017-4>

⁶⁹ Ramón A. Alvarez et al., Assessment of methane emissions from the U.S. oil and gas supply chain. Science 361, 186-1 (2018). <https://www.science.org/doi/10.1126/science.aar7204>

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For 2016 through 2023, this Inventory estimated Natural Gas Exploration, Production and Gathering, Processing, Gas Post-Meter, and Abandoned Wells emissions using the NI by State without modification. Emissions in this period were more generally based on or estimated using reported data compared to earlier years before Subpart W data were available. Starting in 2014, Colorado began a series of significant updates to AQCC Regulation Number 7 to control the emissions of hydrocarbons (including CH₄) from oil and gas operations. This included prohibitions on venting, compulsory leak detection and repair surveys, additional requirements for the design of new facilities as well as expanded protocols for operation and maintenance of air pollution control equipment. Several additional updates to Regulation 7 occurred in the late 2010s, increasing in frequency to at least annual updates in the subsequent decade.⁷⁰ Because the NI by State does not exhaustively account for state actions, emissions are likely overestimated for states like Colorado with robust regulatory programs that exceed federal requirements. This Inventory took a conservative approach by assuming a delayed implementation schedule for state regulations of a year or two and not downscaling emissions for any years 2016 through 2023.

Exploration

For 2005 through 2015, this Inventory estimated Natural Gas Exploration emissions using the NI by State and annualized scale factors, as described above. For 2016 through 2023, this Inventory estimated Natural Gas Exploration emissions using the NI by State.

National emissions for the Exploration segment of Natural Gas Systems were estimated for: non-completion well testing, hydraulically fractured gas well completions (with venting or flaring, and with or without reduced emissions completions (RECs)), non-hydraulically fractured gas well completions (with venting or flaring), well blowouts, and well drilling. Emissions estimates for well completions were mostly developed for each basin that reported to Subpart W. Emissions estimates for well testing, blowouts, and drilling utilized multiple datasets.⁷¹

National Exploration emissions were disaggregated to the state level using proxy data, including total well counts and the number of gas wells drilled. The disaggregation occurred from the basin level for well completions and from the national level for the remaining sources.⁷²

Production and Gathering

For 2005 through 2015, this Inventory estimated Natural Gas Production and Gathering emissions using the NI by State and annualized scale factors, as described above for the Exploration segment. For 2016 through 2023, this Inventory estimated Natural Gas Production and Gathering emissions using the NI by State.

⁷⁰ [Oil and gas regulation in Colorado](#)

⁷¹ [2025 NIR](#), Annex 3.6, Methodology for Estimating CH₄, CO₂, and N₂O Emissions from Natural Gas Systems

⁷² [2024 NI by State Methodology Report](#), Chapter 2, Section 2.2.4.2.1, Exploration and Production

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National emissions for the Production (and Gathering) segment of Natural Gas Systems were estimated for a variety of equipment and activities, including: gas wells, well pad equipment, well workovers, normal operations (e.g., pneumatic devices), condensate tank vents, well clean-ups, blowdowns, upsets, and gathering and boosting.⁷³ For many equipment types, emissions and activity data from GHGRP Subpart W were used when available (from 2015 or 2016 on). For years before GHGRP data were available, emission factors and activity data were either held at a set value (such as setting equal to 2015 values) or interpolated between the latest GHGRP data point and a published value in literature. CH₄ emission factors for other equipment, including separators, dehydrators, and compressors, were from the [1996 GRI/EPA Study](#). For these equipment types, CO₂ emission factors were estimated based on the CH₄ emission factors and CO₂ content in natural gas.⁷⁴

National Production emissions were disaggregated to the state level using basin-level GHGRP data for pneumatic controllers, storage tanks, equipment leaks (i.e., from separators, dehydrators, heaters, compressors, and meters/piping), liquids unloading, chemical injection pumps, and workovers (Approach 1). For other sources, national emissions were disaggregated using proxy data (Approach 2), such as natural gas production data and well counts from Enverus.⁷⁵

Processing

This Inventory estimated Natural Gas Processing emissions using the NI by State for all years.

National emissions for the Processing segment of Natural Gas Systems were estimated for a variety of equipment and activities at gas processing plants, including: leaks, compressors, flares, dehydrators, acid gas removal units or vents, pneumatic devices, and blowdowns.

When available, facility-specific GHGRP Subpart W data were used to develop emission factors for leaks, compressors, flares, dehydrators, and blowdowns/venting. For leaks, compressors, flares, and dehydrators, CH₄ emission factors from Subpart W in 2015 were also used for 2011 through 2014. From 2005 through 2010, CH₄ emission factors were linearly interpolated between the GHGRP data and data from the [GRI/EPA 1996 study](#). Activity data, such as the count or number of equipment types, were also based on either GHGRP data or the GRI/EPA 1996 study.⁷⁶

National Processing emissions were disaggregated to the state level based on the fraction of national onshore marketed natural gas production occurring in each state from EIA.^{77,78}

⁷³ The NIR refers to this segment as the “production” segment, but the NI by State refers to it as the “production and gathering” segment.

⁷⁴ [2025 NIR](#), Annex 3.6, Methodology for Estimating CH₄, CO₂, and N₂O Emissions from Natural Gas Systems

⁷⁵ [2024 NI by State Methodology Report](#), Chapter 2, Section 2.2.4.2.1, Exploration and Production

⁷⁶ [2025 NIR](#), Annex 3.6, Methodology for Estimating CH₄, CO₂, and N₂O Emissions from Natural Gas Systems

⁷⁷ [EIA, Natural Gas Gross Withdrawals and Production, Marketed Production](#)

⁷⁸ [2024 NI by State Methodology Report](#), Chapter 2, Section 2.2.4.2.1, Processing

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Transmission and Storage

This Inventory estimated Natural Gas Transmission and Storage emissions using FRD and the NI by State. Emissions data are reported to the Division through annual [Natural Gas Transmission and Storage Segment-Wide Reports](#). There is no reporting threshold for these reports, so they are considered to fully represent the segment's emissions. This Inventory incorporated those reported emissions from 2021 through 2023. For 2005 through 2020, the reported GHG-specific emissions from 2021 through 2024 were overlapped with the NI by State emissions using Equation 5.1 from the IPCC 2006 Guidelines.⁷⁹ Equation 5.1 creates a consistent time series by aligning older estimates with newer estimates during years where data overlap.⁸⁰ The overlap method, compared to a linear or other simple extrapolation to 2005, allowed for annual variability and incorporation of expected trends in emissions while accounting for the state-specific emissions and trends in recent years.

National emissions for the Transmission and Storage segment of Natural Gas Systems were estimated for pipeline leaks, transmission compressor stations, storage compressor stations, dehydrators, flaring, pneumatic devices, station venting, pipeline venting, and liquefied natural gas (LNG) storage. When available, GHGRP Subpart W data were used to develop emission factors for all years of the time series for pneumatic controllers, LNG storage and transmission pipeline blowdowns. For other activities and equipment, CH₄ emission factors were mostly from the [GRI/EPA 1996 study](#) and a Zimmerle et al. 2015 study.⁸¹ As needed, emission factors were linearly interpolated between sources. Activity data were sourced from several datasets, including transmission stations calculated from [FERC data](#), storage stations calculated from Zimmerle et al., and transmission pipeline miles from [PHMSA](#).⁸²

National Transmission and Storage segment emissions were disaggregated to the state level based on the fraction of national transmission pipeline mileage occurring in each state.⁸³

Distribution

This Inventory estimated Natural Gas Distribution emissions using FRD and the NI by State.

While the EPA's GHGRP requires local distribution companies (LDCs) to report under [Subpart W](#) only if their emissions exceed 25,000 metric tons (0.025 Tg) CO₂eq in any given year, Colorado's [AQCC Regulation Number 22, Part A](#) does not have this threshold for LDCs. For years when Colorado collected this Subpart W data (2021 through 2023), this Inventory used those emissions as reported. For 2011 through 2023, Subpart W emissions were used when available. For years when data were unavailable, such as years in which an LDC was below the reporting threshold, emissions were linearly interpolated or extrapolated by GHG. The

⁷⁹ This Inventory presents emissions through 2023, but reported emissions through 2024 were used for the overlap method.

⁸⁰ [IPCC, 2006 Guidelines, Volume 1, Chapter 5, Time Series Consistency](#)

⁸¹ Zimmerle et al. (2015) Methane Emissions from the Natural Gas Transmission and Storage System in the United States. Environmental Science & Technology, 49: 9374-9383. DOI: [10.1021/acs.est.5b01669](#)

⁸² [2025 NIR](#), Annex 3.6, Methodology for Estimating CH₄, CO₂, and N₂O Emissions from Natural Gas Systems

⁸³ [2024 NI by State Methodology Report](#), Chapter 2, Section 2.2.4.2.1, Processing

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gap-filling efforts included reported emissions for 2024, though this Inventory only includes emissions through 2023.

For 2005 through 2010, the emissions from Subpart W were overlapped with the NI by State emissions by GHG using Equation 5.1 from the IPCC 2006 Guidelines.⁸⁴ Equation 5.1 creates a consistent time series by aligning older estimates with newer estimates during years where data overlap.⁸⁵ The overlap method, compared to a linear or other simple extrapolation to 2005, allowed for annual variability and incorporation of expected trends in emissions while accounting for the state-specific emissions and trends in recent years. The trends of the Subpart W and NI by State data were similar in common years.

National emissions for the Distribution segment of Natural Gas Systems were estimated for leaks from pipeline mains, services, metering and regulating stations, customer meters, and routine maintenance.

For most sources, CH₄ emission factors were sourced from Lamb et al. 2015 for 2011 through 2023 and interpolated back to the 1992 estimates from the [GRI/EPA 1996 study](#).⁸⁶ A few sources were not included in the Lamb et al. study, such as pipeline blowdowns or upsets; for those sources, the CH₄ emission factors from the GRI/EPA 1996 study were used across the time series.⁸⁷ When a CO₂-specific emission factor was not available for a source, the CO₂ emission factors were derived from the corresponding source CH₄ emission factors using default gas composition data. CO₂ emission factors are estimated by multiplying the CH₄ emission factors by the ratio of the CO₂-to-CH₄ gas content. N₂O emission factors were calculated using GHGRP data. Activity data were sourced from several datasets, including natural gas consumers from EIA, miles of distribution pipeline from PHMSA, and station or service counts from PHMSA.⁸⁸

National Distribution segment emissions were disaggregated to the state level based on the relative pipeline mileage by material (cast iron, unprotected or protected steel, plastic) in each state, the relative number of natural gas consumers (residential, commercial, and industrial) from the EIA, and the number of above- and below-grade stations in each state as reported to the GHGRP and scaled up by the EPA using the ratio of PHMSA to GHGRP pipeline mileage data to account for non-reporters.⁸⁹

⁸⁴ This Inventory presents emissions through 2023, but reported emissions through 2024 were used for the overlap method.

⁸⁵ [IPCC, 2006 Guidelines, Volume 1, Chapter 5, Time Series Consistency](#)

⁸⁶ Lamb et al. (2015) Direct Measurements Show Decreasing Methane Emissions from Natural Gas Local Distribution Systems in the United States. Environmental Science and Technology, 49: 5161-5169. DOI: [10.1021/es505116p](#)

⁸⁷ Zimmerle et al. (2015) Methane Emissions from the Natural Gas Transmission and Storage System in the United States. Environmental Science & Technology, 49: 9374-9383. DOI: [10.1021/acs.est.5b01669](#)

⁸⁸ [2025 NIR](#), Annex 3.6, Methodology for Estimating CH₄, CO₂, and N₂O Emissions from Natural Gas Systems

⁸⁹ [2024 NI by State Methodology Report](#), Chapter 2, Section 2.2.4.2.1, Processing

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Gas Post-Meter

This Inventory estimated Natural Gas Post-Meter emissions using the NI by State for all years.

National emissions for the Post-Meter segment of Natural Gas Systems were estimated for leaks from residential, commercial, industrial, and electric utility customers and from natural gas vehicles. These emissions only represent leaks; combustion emissions are captured under RCI Fuel Use, Electric Power, and Transportation.

Default IPCC 2019 CH₄ and CO₂ emission factors were used for all Post-Meter estimates except residential. The residential CH₄ emission factor was from the Fischer et al. 2018 study. N₂O emissions were not estimated. Activity data were sourced from several datasets, including residential natural gas consumers from the [U.S. Census](#), commercial natural gas consumers and appliance counts from the [EIA](#), natural gas consumption in the industrial and utility sectors from the EIA, and national counts of natural gas-burning vehicles from [MOVES3](#). Annex Tables 3.6-6, 3.6-13, and 3.6-16, respectively, detail the CH₄, CO₂, and N₂O emission factor data sources and methodologies by segment and source for national Natural Gas Systems emissions estimates. Annex Table 3.6-8 details the activity data sources and methodologies by segment and source for national Natural Gas Systems emissions estimates.⁹⁰

National Post-Meter segment emissions were disaggregated to the state level using proxy data that were generally consistent with the activity data used in emissions estimates, including the relative number of residential and commercial natural gas customers in each state from EIA, relative industrial and electric utility consumption volumes of natural gas from the EIA SEDS, and the relative number of compressed natural gas vehicles in each state, derived from the MOVES model.⁹¹

Abandoned Wells

For 2005 through 2015, this Inventory estimated Natural Gas Abandoned Wells emissions using the NI by State and annualized scale factors, as described above for the Exploration segment. For 2016 through 2023, this Inventory estimated Natural Gas Abandoned Wells emissions using the NI by State.

To quantify emissions from abandoned wells, the EPA used an IPCC Tier 2 method based on the number of plugged and unplugged wells and associated emission factors. Plugged and unplugged abandoned well CH₄ emission factors were developed at the national level using emission data from Townsend-Small et al. These emission factors were applied to counts of abandoned wells. The EPA developed abandoned well CO₂ emission factors using the CH₄ emission factors and an assumed ratio of CO₂-to-CH₄ gas content. For abandoned gas wells, EPA used the Natural Gas Systems default production segment CH₄ and CO₂ gas content values

⁹⁰ [2025 NIR](#), Annex 3.6, Methodology for Estimating CH₄, CO₂, and N₂O Emissions from Natural Gas Systems

⁹¹ [2024 NI by State Methodology Report](#), Chapter 2, Section 2.2.4.2.1, Processing

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to develop a ratio of 0.044 metric tons CO₂/metric ton CH₄.⁹² The same emission factors were applied for each year of the time series.⁹³

National estimates were disaggregated to the state level using Enverus data. State-level counts of abandoned wells were based on the counts of abandoned wells in Enverus compared to estimates of wells not in the Enverus dataset; the wells not included in the Enverus dataset were informed by a variety of agencies, including the U.S. Geological Survey. The state-level abandoned well population was then split into plugged and unplugged wells based on an assumption that all abandoned wells were unplugged in 1950, and using Enverus data to calculate the fraction of plugged abandoned wells in 2023. The fraction of plugged wells was linearly interpolated between 1950 and 2023.⁹⁴ The full methodology is provided in the 2018 Abandoned Wells Memo.⁹⁵

3.5.1.2. Uncertainty

This Inventory did not quantify the uncertainty associated with NGPS estimates. The NI by State also did not quantify uncertainty for state-level estimates, but the NIR quantified uncertainty estimates for national emissions. The EPA presented the results of the uncertainty analyses for Natural Gas Systems and Petroleum Systems, but not for any deeper segments or activities.

The EPA conducted a quantitative uncertainty analysis using the IPCC Approach 2 methodology (Monte Carlo Simulation technique) to characterize the uncertainty for Natural Gas Systems. For the CH₄ uncertainty analysis, EPA focused on the six highest-emitting sources for the year 2023, which together emitted 51% of CH₄ from natural gas systems in 2023, and extrapolated the estimated uncertainty for the remaining sources. For the CO₂ uncertainty analysis, EPA focused on the highest-emitting source for the year 2023, which emitted 50% of CO₂ from natural gas systems in 2023, and extrapolated the estimated uncertainty for the remaining sources. To estimate uncertainty for N₂O, EPA applied the uncertainty bounds calculated for CO₂. Uncertainty levels in national 2023 Natural Gas Systems emissions ranged from -10% to 11%, -14% to 17%, and -14% to 17% for CH₄, CO₂, and N₂O, respectively.⁹⁶ The use of FRD, where available, likely constrained the uncertainty for the Distribution segment and Transmission and Storage segment of Natural Gas Systems.

The EPA also conducted a quantitative uncertainty analysis using the IPCC Approach 2 methodology (Monte Carlo Simulation technique) to characterize the uncertainty for Abandoned Wells. Uncertainty in national 2023 Abandoned Gas Wells emissions ranged from -83% to 255% for both CH₄ and CO₂.⁹⁷ Uncertainty in this Inventory's estimates is likely to be similar, as the national estimates were based on state-level data.

⁹² Gas Technology Institute (GTI) 2001, Gas Resource Database: Unconventional Natural Gas and Gas Composition Databases. Second Edition. GRI-01/0136.

⁹³ [2025 NIR](#), Chapter 3, Section 3.7, Natural Gas Systems, Introduction

⁹⁴ [2025 NIR](#), Chapter 3, Section 3.7, Natural Gas Systems, Introduction

⁹⁵ [Inventory of US Greenhouse Gas Emissions and Sinks 1990-2016: Abandoned Oil and Gas Wells | EPA](#)

⁹⁶ [2025 NIR](#), Chapter 3, Section 3.7, Natural Gas Systems, Uncertainty

⁹⁷ [2025 NIR](#), Chapter 3, Section 3.8, Abandoned Oil and Gas Wells, Uncertainty

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Some uncertainty in the estimates stems from possible double-counting with Industrial Fuel Use (within RCI Fuel Use). Though the NGPS subsector is meant to include leaks, venting, flaring, and other fugitive emissions, some of the NI by State estimates may include a portion of combustion emissions that could not be disentangled. For example, the gathering and boosting emissions may include small amounts of emissions from CH₄ slip in engines. Without an equipment-level emissions dataset (which is not provided in the NI by State), the existence or extent of double counting cannot be validated.

The scaling factor methodology applied to the 2005 through 2015 emissions estimates carries an additional level of uncertainty that is difficult to constrain or quantify. The Rutherford et al. estimates carried a relative uncertainty of $\pm 7.5\%$ at a 95% confidence interval, although that uncertainty only considers the uncertainty due to resampling of the emissions distributions in their model. Alvarez et al. expressed a relative uncertainty in their production emissions estimate ranging from -21% to +25%. These values were expressed with respect to the national emissions, and extrapolation to the state level would carry an additional unknown level of uncertainty.

3.5.1.3. QA/QC

For each emissions source category, the Division applied standard QA/QC procedures, including reviews of data connections and imports, unit conversions, GWP calculations, naming conventions, and data formatting.

The EPA applied QA/QC procedures consistent with the NIR QA/QC Management Plan, in accordance with IPCC Guidelines. The QA/QC Management Plan describes data and methodology checks, develops processes governing peer review and public comments, and provides guidance on conducting uncertainty analyses for the emissions estimates. The QA/QC Management Plan procedures also stress continual improvement, providing for corrective actions designed to improve the inventory estimates over time.⁹⁸

The EPA's GHGRP reporting tool includes built-in verification processes for facility reports to identify potential errors and ensure that submitted data are accurate, complete, and consistent. After submission, the EPA employs additional statistical, algorithm, and reporting trends checks, and may follow up with facilities to resolve mistakes.⁹⁹ The Division also reviewed FRD for accuracy and reasonability.

3.5.1.4. Recalculations and Comparisons to Prior Inventory

Due to the categorical structure changes in this Inventory, recalculations and comparisons to the prior Inventory were not possible for Natural Gas Systems. [Section 3.5.3](#) provides a discussion for recalculations and comparisons for the NGPS subsector as a whole.

⁹⁸ [NIR](#), Annex 8 QA/QC Procedures

⁹⁹ [GHGRP Methodology and Verification | US EPA](#)

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3.5.1.5. Planned Improvements

The Division will continue to prioritize FRD and evaluate opportunities to incorporate more facility or operation-specific data into the estimates for Distribution and Transmission and Storage. The Division will also evaluate opportunities for leveraging FRD in other segments, such as Production and Gathering, Processing, or Post-Meter, for directly estimating emissions or for validation of other estimates. Additional planned improvements for the NGPS subsector as a whole are outlined in [Section 3.5.4.](#)

3.5.2. Petroleum Systems

Emissions from Petroleum Systems include emissions from leaks, venting, and flaring, per IPCC guidelines. These emissions may occur across several segments or activities, including Exploration, Production and Upgrading, Crude Oil Transportation, and Abandoned Wells.¹⁰⁰

Exploration emissions include those from well drilling, testing, and completion.

Production emissions include leaks, vents, and flaring from wells and from well-site equipment, such as pneumatic controllers, storage tanks, and chemical injection pumps.

Emissions from Crude Oil Transportation (or “Transport”) primarily include leaks and vents from tanks and truck loading operations. Emissions from marine loading are not relevant for Colorado.¹⁰¹

Abandoned Wells are orphaned wells and other non-producing wells. Non-producing wells include those with no recent production that are not plugged (i.e., inactive or dormant), with no recent production and no responsible operator (i.e., orphaned or deserted), and those that have been plugged.¹⁰²

Combustion emissions (except for flaring emissions) are captured elsewhere, either in Industrial Fuel Use (RCI Fuel Use in Section 3.2.3) or Transportation emissions. Emissions from refining are captured in the IPPU sector (Chapter 4).

In 2023, emissions from Petroleum Systems were 1.521 Tg CO₂eq, accounting for 12.8% of the NGPS emissions and 1.6% of Energy emissions. Of these emissions, 95.6% were from Production and Upgrading (1.454 Tg CO₂eq). Emissions from Abandoned Wells (0.042 Tg CO₂eq), Exploration (0.022 Tg CO₂eq), and Crude Oil Transportation (0.265 Tg CO₂eq) together accounted for less than 5% of the 2023 emissions.

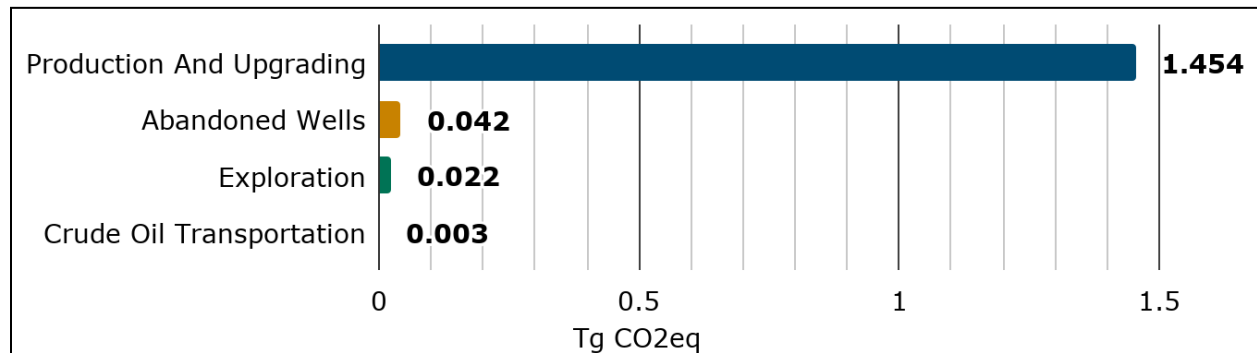
¹⁰⁰ [2025 NIR](#), Chapter 3, Section 3.6, Petroleum Systems Introduction

¹⁰¹ [2025 NIR](#), Chapter 3, Section 3.7, Petroleum Systems, Introduction

¹⁰² [2025 NIR](#), Chapter 3, Section 3.6, Petroleum Systems, Introduction

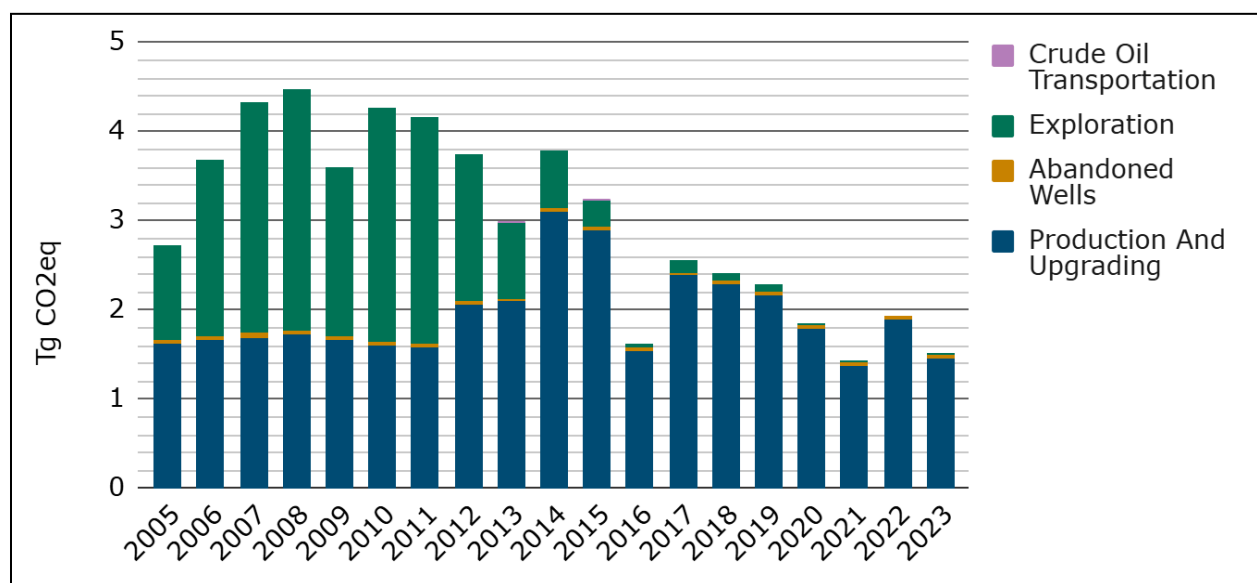
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Figure 3.32. 2023 Petroleum Systems Emissions.



From 2005 through 2023, Petroleum Systems emissions decreased by 1.214 Tg CO₂eq or 44.4%. As shown in Figure 3.33 below, emissions from Exploration were significant (up to 61% of all Petroleum Systems emissions in 2011), but decreased and have remained low since about 2014. Reduced emissions completions (RECs) became an operating requirement in 2016, though some operations began the practice earlier. RECs employ specially designed equipment to capture gas during well completions and workovers that reduce emissions and the need for flaring.¹⁰³ Exploration emissions in 2023 were 1.044 Tg CO₂eq or 98% below 2005 levels. Production and Upgrading emissions were relatively stable from 2005 through 2010, then fluctuated through 2023. Since about 2017, emissions have been decreasing, falling to 0.161 Tg CO₂eq or 9.9% below 2005 emissions in 2023. Emissions from Abandoned Wells did not fluctuate much over time, and 2023 emissions were just 0.011 Tg CO₂eq (21.4%) below 2005 emissions. Crude Oil Transportation was the only segment with higher emissions in 2023 compared to 2005, but the difference was just 0.002 Tg CO₂eq.

Figure 3.33. Historical Petroleum Systems Emissions by Segment.



¹⁰³ [EPA, Reduced Emission Well Completions and Workovers](#)

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As detailed in Table 3.26 below, most of the Petroleum Systems emissions (80% to 98%) were CH₄, and CH₄ emissions changed the most over the time series. CH₄ emissions increased from 2005 to a peak of 4.336 Tg CO₂eq in 2008, but decreased since then. 2023 emissions were 1.417 Tg CO₂eq (53.4%) below 2005 emissions. In most years from 2005 through 2011, CH₄ emissions were mostly from Exploration, with similar but slightly lower emissions from Production and Upgrading. Since about 2012, CH₄ emissions have been mostly from Production and Upgrading, as Exploration emissions have decreased significantly. From 2005 through 2023, Exploration CH₄ emissions decreased 1.040 Tg CO₂eq (99.6%), and Production and Upgrading CH₄ emissions decreased 0.368 Tg CO₂eq (23.7%).

CO₂ emissions increased from 2005 to a peak of 0.528 Tg CO₂eq in 2019, but generally decreased since then. By 2023, emissions were 0.203 Tg CO₂eq (248%) above 2005 emissions. Most of the Petroleum Systems CO₂ emissions (50% to 98%) were from Production and Upgrading, where most flaring occurs. CO₂ emissions from Production and Upgrading increased by 0.207 Tg CO₂eq (347%) from 2005 through 2023.

N₂O emissions were very low from 2005 through 2023, peaking at just 0.006 Tg CO₂eq in 2018. N₂O emissions in 2023 were less than 0.001 Tg CO₂eq below 2005 emissions. N₂O emissions were mostly from Production and Upgrading.

Table 3.26. Petroleum Systems Emissions by GHG and Segment (Tg CO₂eq).

GHG	Segment	2005	2010	2015	2020	2023
CO ₂	Abandoned Wells	+	+	+	+	+
CO ₂	Crude Oil Transportation	+	+	+	+	+
CO ₂	Exploration	0.022	0.095	0.074	0.005	0.018
CO ₂	Production And Upgrading	0.060	0.097	0.431	0.305	0.267
CO₂ Subtotal		0.082	0.192	0.505	0.310	0.285
CH ₄	Abandoned Wells	0.053	0.044	0.043	0.041	0.042
CH ₄	Crude Oil Transportation	0.001	0.001	0.004	0.003	0.003
CH ₄	Exploration	1.044	2.514	0.223	0.002	0.004
CH ₄	Production And Upgrading	1.555	1.512	2.464	1.489	1.187
CH₄ Subtotal		2.653	4.072	2.735	1.535	1.236
N ₂ O	Crude Oil Transportation	+	+	+	+	+
N ₂ O	Exploration	+	+	+	+	+
N ₂ O	Production And Upgrading	+	+	+	+	0.001
N₂O Subtotal		+	+	+	+	0.001
Petroleum Systems Total		2.735	4.264	3.240	1.846	1.521

+ Emissions do not exceed 0.0005 Tg CO₂eq.

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3.5.2.1. Methodology

This Inventory estimated emissions for all categories using the NI by State to varying degrees. For Exploration, Production and Upgrading, and Abandoned Wells, the NI by State was used with modifications for 2005 through 2015. For 2016 through 2023, the NI by State was used “as is,” or without modifications. The modifications and development of scale factors are described in detail in [Section 3.5.1.1, Natural Gas Systems Methodology](#). For Crude Oil Transportation, the NI by State was used “as is” from 2005 through 2023.

The EPA estimated national Petroleum Systems emissions generally using emission factors and activity data. For most sources, this approach used technology-specific emission factors or emission factors that vary over time and take into account changes to technologies and practices, which are used to calculate net emissions directly. The estimates were developed with an IPCC Tier 2 approach; Tier 1 approaches were not used. When [Subpart W](#) data from the GHGRP were available, they were used to develop emission factors. Before the Subpart W data were available, emission factors were based on emissions trends and published literature, including from the American Petroleum Institute (API) and from a 1996 study published by the GRI and the EPA. Activity data were sourced from several datasets, including Enverus and historical EPA publications.¹⁰⁴ Annex Tables 3.5-6, 3.6-9, and 3.6-12 respectively detail the CH₄, CO₂, and N₂O emission factor data sources and methodologies by segment and source for national Petroleum Systems emissions estimates. Annex Table 3.5-8 details the activity data sources and methodologies by segment and source for national Petroleum Systems emissions estimates.

The national emissions were then disaggregated to the state level. For 2011 through 2023, emissions from most sources were based on data collected through Subpart W of the federal GHGRP. The GHGRP data include emissions for individual sources or equipment, such as pneumatic controls, in several segments. Emissions and activity data may be reported at the facility level or the basin level. When facility-level data were available, they were used to develop state-level emissions estimates. When only basin-level data were available, the national-level emissions estimates were disaggregated to states using proxy data, such as well counts from Enverus or crude oil production from EIA.¹⁰⁵ For 2005 through 2010, national emissions were disaggregated to the state level using proxy data. Though national emissions estimates were available at the equipment level, state emissions were only published at the segment level.

Exploration

For 2016 through 2023, emissions were estimated using the NI by State. For 2005 through 2015, emissions were estimated using the NI by State and applied scale factors to account for expected underestimation from the NI by State. The details of the scale factors and modifications are described in [Section 3.5.1.1, Natural Gas Systems Methodology](#). The same

¹⁰⁴ [2025 NIR](#), Annex 3.5, Methodology for Estimating CH₄, CO₂, and N₂O Emissions from Petroleum Systems

¹⁰⁵ [2024 NI by State Methodology Report](#), Chapter 2, Section 2.2.3.2, Methods/Approach

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approach was applied to Natural Gas Exploration, Natural Gas Production and Gathering, Abandoned Gas Wells, Petroleum Exploration, Petroleum Production and Upgrading, and Abandoned Oil Wells.

The EPA generally estimated national emissions as the product of emission factors and activity data. National emissions for the Exploration segment of Petroleum Systems were estimated for: non-completion well testing (with venting or flaring), well completion venting, hydraulically fractured well completions (with venting or flaring, and with or without RECs), and well drilling.

CH₄ emission factors and activity data for well completions and non-completion well testing were developed using data reported to Subpart W. CH₄ emission factors for other sources were cited from literature, including the GRI/EPA 1996 study. Counts of well completions were sourced from EIA, and total oil wells drilled were calculated using Enverus data.¹⁰⁶

National Exploration emissions were disaggregated to the state level using proxy data, including total well counts and the number of oil wells drilled. The disaggregation occurred from the basin level for well completions and from the national level for the remaining sources.¹⁰⁷

Production and Upgrading

For 2016 through 2023, emissions were estimated using the NI by State. For 2005 through 2015, emissions were estimated using the NI by State and applied scale factors to account for expected underestimation from the NI by State. The details of the scale factors and modifications are described in [Section 3.5.1.1, Natural Gas Systems Methodology](#).

The EPA generally estimated national emissions as the product of emission factors and activity data. Petroleum Production and Upgrading emissions include those from a variety of activities and equipment types, such as oil wells, tanks, pneumatic devices, chemical injection pumps, well workovers, separators, and flaring, among others. When GHGRP Subpart W data were available, such as for pneumatic controllers or associated gas venting and flaring, the EPA developed basin-level CH₄ emission factors for 2011 through 2023 or 2015 through 2023. For many sources, the 2011 or 2015 emission factors were used for earlier years before the GHGRP data were available. In the absence of any GHGRP data, published reports or research, such as the GRI/EPA 1996 study, were used to develop emission factors. For non-reporting basins, Subpart W averages were used. Emission factors for CO₂ were estimated using the same methods as CH₄ emission factors (e.g., from GHGRP data) or by multiplying the CH₄ emission factors by a conversion factor, which is the ratio of CO₂ content and CH₄ content in whole crude post-separator.¹⁰⁸ Activity data for Petroleum Production and Upgrading included a variety of data points, such as counts of compressors, tank volumes, and workover events in

¹⁰⁶ [2025 NIR](#), Chapter 3, Section 3.7, Petroleum Systems, Methodology

¹⁰⁷ [2024 NI by State Methodology Report](#), Chapter 2, Section 2.2.3.2.1, Exploration and Production

¹⁰⁸ API 4697: Production Tank Emissions Model - A Program For Estimating Emissions From Hydrocarbon Production Tanks - E&P Tank Version 2.0. American Petroleum Institute. Washington, DC.

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a given year. Much of this data was from Subpart W data, Enverus well counts, and the EIA's [Petroleum Supply Annual data](#).¹⁰⁹

National Petroleum Production and Upgrading emissions were disaggregated to the state level using activity data reported under Subpart W or proxy data. For pneumatic controllers, storage tanks, equipment leaks, chemical injection pumps, and workovers with hydraulic fracturing, basin-specific emission factors and/or activity data were used to estimate basin and state emissions. For other sources, national emissions were allocated to states primarily based on the fraction of oil wells in each state relative to national totals and the fraction of oil production in each state relative to national totals. Both of these were largely based on data from Enverus.¹¹⁰

Crude Oil Transportation

This Inventory estimated Crude Oil Transportation emissions using the NI by State for all years.

The EPA estimated national emissions as the product of emission factors and activity data. Crude Oil Transportation emission factors for CH₄ were largely developed using data from historical studies.^{111, 112, 113} Emission factors for CO₂ were estimated by multiplying the CH₄ emission factors by a conversion factor, which is the ratio of CO₂ content and CH₄ content in whole crude post-separator.¹¹⁴ Activity data for Crude Oil Transportation included a variety of data points, such as crude oil volumes delivered to refineries, volumes loaded onto trucks and rail, pump stations, and pipeline miles. Much of this data was from the EIA's [Petroleum Supply Annual data](#), the EIA's [Refinery Capacity Reports](#), and Pipeline Economics reports from the [Oil and Gas Journal](#).¹¹⁵

National Crude Oil Transportation emissions were disaggregated to the state level using proxy data. Vented emissions from truck and rail loading were allocated based on onshore oil production in each state. Vented emissions from marine loading were allocated to states based on offshore production from Enverus, though that was not applicable to Colorado. Other transport emissions were allocated based on the relative state counts of oil refineries from GHGRP Subpart Y (2010 and on) and EIA atmospheric crude oil distillation capacity (before 2010).¹¹⁶

¹⁰⁹ [2025 NIR](#), Annex 3.5, Methodology for Estimating CH₄, CO₂, and N₂O Emissions from Petroleum Systems

¹¹⁰ [2024 NI by State Methodology Report](#), Chapter 2, Section 2.2.3.2.2, Crude Oil Transport

¹¹¹ Compilation of Air Pollutant Emission Factors, AP-42. Office of Air Quality Planning and Standards, U.S. Environmental Protection Agency. Research Triangle Park, NC. October 1997.

¹¹² Global Emissions of Methane from Petroleum Sources. American Petroleum Institute, Health and Environmental Affairs Department, Report No. DR140, February 1992.

¹¹³ Methane Emissions from the U.S. Petroleum Industry. Prepared by Radian International. U.S. Environmental Protection Agency. February 1999.

¹¹⁴ API 4697: Production Tank Emissions Model - A Program For Estimating Emissions From Hydrocarbon Production Tanks - E&P Tank Version 2.0. American Petroleum Institute. Washington, DC.

¹¹⁵ [2025 NIR](#), Annex 3.5, Methodology for Estimating CH₄, CO₂, and N₂O Emissions from Petroleum Systems

¹¹⁶ [2024 NI by State Methodology Report](#), Chapter 2, Section 2.2.3.2.2, Crude Oil Transport

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Abandoned Wells

For 2016 through 2023, emissions were estimated using the NI by State. For 2005 through 2015, emissions were estimated using the NI by State and applied scale factors to account for expected underestimation from the NI by State. The details of the scale factors and modifications are described in [Section 3.5.1.1, Natural Gas Systems Methodology](#).

To quantify emissions from abandoned wells, the EPA used an IPCC Tier 2 method based on the number of plugged and unplugged wells and associated emission factors. Plugged and unplugged abandoned well CH₄ emission factors were developed at the national level using emission data from Townsend-Small et al. These emission factors were applied to counts of abandoned wells. The EPA developed abandoned well CO₂ emission factors using the CH₄ emission factors and an assumed ratio of CO₂-to-CH₄ gas content. For abandoned oil wells, EPA used the Petroleum Systems default production segment associated gas ratio of 0.020 metric tons CO₂/metric ton CH₄, derived through API TankCalc modeling runs. The same emission factors were applied for each year of the time series.¹¹⁷

National estimates were disaggregated to the state level using Enverus data. State-level counts of abandoned wells were based on the counts of abandoned wells in Enverus compared to estimates of wells not in the Enverus dataset; the wells not included in the Enverus dataset were informed by a variety of agencies, including the U.S. Geological Survey. The state-level abandoned well population was then split into plugged and unplugged wells based on an assumption that all abandoned wells were unplugged in 1950, and using Enverus data to calculate the fraction of plugged abandoned wells in 2023. The fraction of plugged wells was linearly interpolated between 1950 and 2023.¹¹⁸ The full methodology is provided in the 2018 Abandoned Wells Memo.¹¹⁹

3.5.2.2. Uncertainty

This Inventory did not quantify the uncertainty associated with NGPS estimates. The NI by State also did not quantify uncertainty for state-level estimates, but the NIR quantified uncertainty estimates for national emissions. The EPA presented the results of the uncertainty analyses for Natural Gas Systems and Petroleum Systems but not for any deeper segments or activities.

The EPA conducted a quantitative uncertainty analysis using the IPCC Approach 2 methodology (Monte Carlo Simulation technique) to characterize the uncertainty for Petroleum Systems. For the CH₄ uncertainty analysis, EPA focused on the three highest-emitting sources for the year 2023, which together emitted 51% of CH₄ from Petroleum Systems in 2023, and extrapolated the estimated uncertainty for the remaining sources. For the CO₂ uncertainty analysis, EPA focused on the four highest-emitting sources for the year 2023, which emitted 52% of CO₂ from Petroleum Systems in 2023, and extrapolated the estimated uncertainty for the remaining sources. To estimate uncertainty for N₂O, EPA applied the uncertainty bounds

¹¹⁷ [2025 NIR](#), Chapter 3, Section 3.8, Abandoned Oil and Gas Wells, Methodology

¹¹⁸ [2025 NIR](#), Chapter 3, Section 3.8, Abandoned Oil and Gas Wells, Methodology

¹¹⁹ [Inventory of US Greenhouse Gas Emissions and Sinks 1990-2016: Abandoned Oil and Gas Wells | EPA](#)

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calculated for CO₂. Uncertainty in national 2023 Petroleum Systems emissions ranged from -13% to 24%, -18% to 22%, and -18% to 22% for CH₄, CO₂, and N₂O, respectively.¹²⁰

The EPA also conducted a quantitative uncertainty analysis using an IPCC Approach 2 methodology (Monte Carlo Simulation technique) to characterize the uncertainty for Abandoned Wells. Uncertainty in national 2023 Abandoned Oil Wells emissions ranged from -83% to 223% for both CH₄ and CO₂.¹²¹ Uncertainty in this Inventory's estimates is likely to be similar as the national estimates were based on state-level data.

The Division introduced some additional uncertainty with its scaling factors, although they were informed by published literature. Discussion related to the scaling factors is in Section 3.5.1.2, Uncertainty for Natural Gas Systems.

3.5.2.3. QA/QC

For each emissions source category, the Division applied standard QA/QC procedures, including reviews of data connections and imports, unit conversions, GWP calculations, naming conventions, and data formatting.

The EPA applied QA/QC procedures consistent with the NIR QA/QC Management Plan, in accordance with IPCC Guidelines. The QA/QC Management Plan describes data and methodology checks, develops processes governing peer review and public comments, and provides guidance on conducting uncertainty analyses for the emissions estimates. The QA/QC Management Plan procedures also stress continual improvement, providing for corrective actions designed to improve the inventory estimates over time.¹²²

3.5.2.4. Recalculations and Comparisons to Prior Inventory

Due to the categorical structure changes in this Inventory, recalculations and comparisons to the prior Inventory were not possible for Petroleum Systems. [Section 3.5.3](#) provides a discussion for recalculations and comparisons for the NGPS subsector as a whole.

3.5.2.5. Planned Improvements

The Division will continue to evaluate best available methods and data for estimating Petroleum systems emissions. This may include leveraging FRD, incorporating more state-specific inputs to emissions estimates, or validating estimates against measured or modeled estimates from published research. Additional planned improvements for the NGPS subsector as a whole are outlined in [Section 3.5.4](#).

3.5.3. Recalculations and Comparisons to Prior Inventory

This Inventory (2025) used the 2025 NI by State (with some modifications) and FRD to estimate NGPS emissions. The prior Inventory (2023) used the Division's internal analysis tool,

¹²⁰ [2025 NIR](#), Chapter 3, Section 3.6, Petroleum Systems, Uncertainty

¹²¹ [2025 NIR](#), Chapter 3, Section 3.8, Abandoned Oil and Gas Wells, Uncertainty

¹²² [NIR](#), Annex 8 QA/QC Procedures

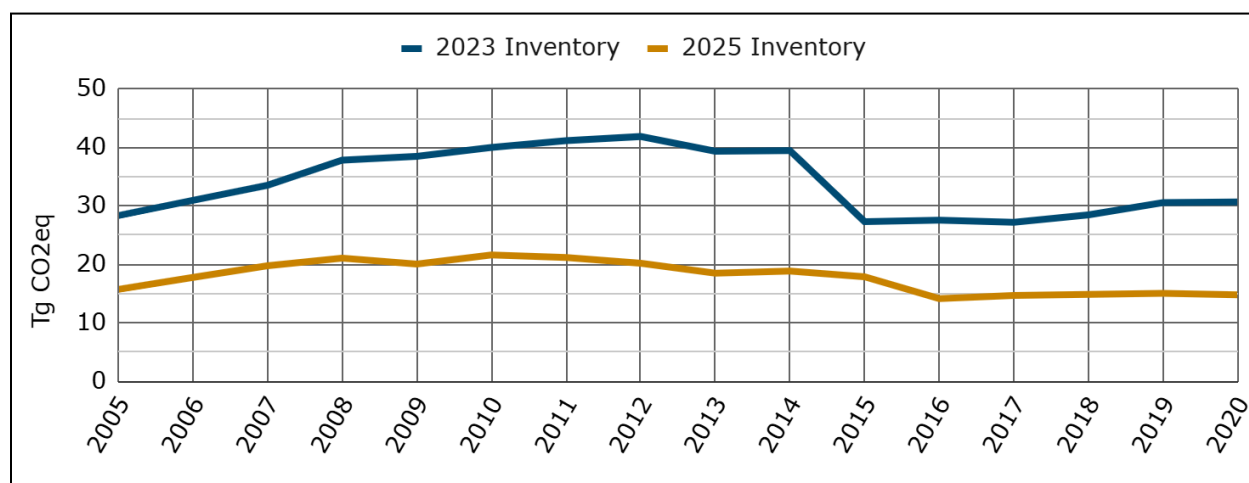
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the NI by State, and FRD. As a result of these changes, the categorization differs between the two Inventories. This Inventory splits NGPS emissions into Natural Gas Systems and Petroleum Systems, while the prior Inventory presented Natural Gas and Oil (NGO) Systems emissions together. The NGPS emissions are further disaggregated into segments, such as Exploration, Exploration, Production (and Gathering or Upgrading), Processing, Distribution and Abandoned Wells, that are specific to Natural Gas Systems or Petroleum Systems. The prior Inventory disaggregated NGO emissions into Upstream and Midstream, Downstream, Fugitive, Venting & Flaring, Acid Gas Removal Units, and Abandoned Wells.

Though the emissions under the two categorization schemes cannot be directly compared, there are some similarities and alignment. The prior Inventory's "Upstream and Midstream" emissions correspond to the current Exploration, Production, and Processing emissions. Each of these segments also includes, to varying degrees, the Fugitive and Venting & Flaring emissions, as defined in the prior Inventory. The prior Inventory's emissions from Acid Gas Removal Units are now captured under the Processing segment of Natural Gas Systems. The Downstream segment in the prior Inventory corresponds to the current Inventory's Natural Gas Transmission and Storage, Natural Gas Distribution, and Crude Oil Transport. Both the 2023 and 2025 Inventories used NI by State data to estimate emissions from Abandoned Wells, although the approach differed slightly. This Inventory applied a scaling factor to 2005 through 2015 emissions, while the prior Inventory used NI by State without modifications. The greater disaggregation and confidence in categorical definitions contributed to the change in methodologies, utilizing the NI by State and FRD rather than the previously utilized Division analysis. The NI by State and FRD also represent years of research and continual improvement, including from Colorado researchers and facilities.

Due to the difference in categorical structure, recalculations and direct comparisons at categorical or sub-sector levels are not possible. Across the common time series (2005 through 2020), total annual NGPS CO₂eq emissions in this Inventory were 9.431 Tg CO₂eq to 21.655 Tg CO₂eq (34.5% to 53.0%) lower than those in the prior Inventory, though the trends are similar, as shown in Figure 3.34 below.

Figure 3.34. Comparison of NGPS Emissions in the 2023 and 2025 Inventory.



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The decision to change data and methodologies for the current Inventory was informed by the increased levels of disaggregation offered by the NI by State, the methodological rigor and review of the NI by State by experts in oil and gas research and modeling communities, the opportunity to incorporate more FRD, and in particular as a result of comparing prior Inventory Upstream and Midstream estimates against recently published literature and research. These adjustments primarily impacted Upstream and Midstream CH₄ emissions, which accounted for 77% to 83% of all oil and gas CO₂eq emissions in Colorado in the prior Inventory.

The previous Division analysis estimated Upstream and Midstream emissions using assumed Statewide loss rates (called “leak rates” in the analysis) and natural gas and oil production data reported to the ECMC.¹²³ Emissions were calculated as the product of the loss rates and natural gas production volumes, with conversion factors accounting for the molecular composition and density of natural gas. Loss rates ranged from 4% for 2005 through 2014, 2.5% for 2015 through 2017, and 2.4% for 2018 through 2020. The loss rate trend was based on a pair of academic studies of the Denver-Julesburg Basin using data collected in 2008 and 2012, as well as assumptions for how the loss rate would change over time.^{124,125} These studies used measured data (via ambient tower monitoring, mobile surveys, and aerial mass balance surveys) to estimate loss rates for all oil and gas sources in the Denver-Julesburg Basin. A similar evaluation of recent (2019 through 2024) CH₄ emission monitoring efforts was published by the Division in the [2026 State Default Intensity Verification Factor Report](#) as part of the [Upstream Greenhouse Gas Intensity Program](#). One important conclusion from that evaluation was that loss rate estimates varied significantly (+/- 100%) across studies depending on the types of measurements collected, methodologies for data evaluation, and temporal variability, even for measurements conducted days apart. With so few measurement studies available prior to 2019, the degree of accuracy of defining a basin-wide or regional loss rate of any one individual study is limited. The evaluation also found the assumed loss rates in the Division’s analysis for recent years (2.5% and 2.4%) were much higher than would be expected based on measurement data (0.9 +/- 0.3%).

The loss rates applied to represent statewide Upstream and Midstream emissions were based on measurements of oil and gas emissions in the Denver-Julesburg Basin. While segments and activities within the oil and natural gas supply chain can be distinguished with inventory accounting and “bottom-up” calculations (i.e., at the equipment level), satellite, tower, and aircraft measurements and mass balance methods are typically limited to total supply chain emissions within the study region. Applying the “total” loss rate to just the Upstream and Midstream segment may have introduced double-counting and overestimation of emissions, as the other segments’ and activities’ emissions (e.g., Downstream, Venting and Flaring) were estimated separately and added to the Upstream and Midstream emissions. This

¹²³ <https://ecmc.colorado.gov/data-maps/production-by-county>

¹²⁴ Pé on, G., et al. (2012), Hydrocarbon emissions characterization in the Colorado Front Range: A pilot study, J. Geophys. Res., 117, D04304, doi:[10.1029/2011JD016360](https://doi.org/10.1029/2011JD016360)

¹²⁵ Peischl, J., et al. (2018). Quantifying methane and ethane emissions to the atmosphere from central and western U.S. oil and natural gas production regions. Journal of Geophysical Research: Atmospheres, 123, 7725-7740, doi:[10.1029/2018JD028622](https://doi.org/10.1029/2018JD028622)

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double-counting may have also included combustion emissions, which should be captured under Industrial Fuel Use ([Section 3.2.3](#), within RCI Fuel Use).

In addition to the uncertainty introduced by using a leak rate, the production data used in the Division analysis included production from wells within the boundaries of Tribal reservations. Between 2010 and 2020, that production accounted for about 11% to 23% of total statewide production. This means the previous analysis likely overestimated emissions based on production because the Inventory is meant to exclude emissions occurring within Tribal reservations. Although uncertainty remains in this Inventory's estimates, the data and methodology changes in this Inventory likely reduced the uncertainty and improved the accuracy of the NGPS emissions estimates compared to the prior Inventory.

3.5.4. Planned Improvements

The Division will continue to evaluate best available data and methodologies, including current research and measurement-based methods, to estimate NGPS emissions. Current oil and gas CH₄ emissions research is focused on the development of “measurement informed inventories”, whereby measurement data can augment or validate certain sectors or components of an inventory initially developed using bottom-up approaches. This approach is more sophisticated than the simple loss rate assumption used in the prior 2023 inventory since it retains information regarding the source of the emissions and improves the granularity of inventory estimates.

In the next iteration of the inventory, the Division will aim to incorporate more information from the [Oil and Natural Gas Annual Emission Inventory Reporting \(ONGAEIR\)](#) program, which mandates emissions reporting for all NGPS activities upstream of or at a natural gas processing plant. While the Division evaluated the recent ONGAEIR data for inclusion in this inventory, internal analysis found significant amounts of missing, incomplete, or erroneous reporting across both operators and source categories. These issues with the 2021-2024 ONGAEIR reports rendered it not useful for the developing sub-sector level estimates in this inventory.

The Division will also engage subject matter experts, researchers, operators, and other stakeholders to ensure the estimates represent Colorado's operations as closely as possible while fulfilling the needs and requirements of the Inventory.

3.5.5. Consumption of Colorado Natural Gas and Petroleum Products

The GHG emissions associated with the complete combustion of natural gas and petroleum (oil) products produced in Colorado were estimated for 2023. In reality, these emissions may not necessarily occur, or occur in Colorado, as natural gas and oil may be refined into products and/or exported for use elsewhere. Some of these products store a portion of their carbon for a period of time before being oxidized, while others may essentially store the carbon indefinitely (see Non-Energy Use of Fossil Fuels, Section 3.6). The analysis below

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summarizes estimated potential emissions for 2023, though the actual emissions will vary by year depending on the production volumes, refinery yields, and annual storage factors for each product. This analysis does not attempt to quantify the amount of Colorado production that directly supplies Colorado consumers. There are multiple interstate oil, natural gas, and refined product pipelines that traverse Colorado, and petroleum products are both imported to and exported from the state through these pipelines.

To estimate the GHG emissions associated with petroleum produced in Colorado, national production of refinery products was scaled to Colorado production using 2025 national refinery output data in gallons of product per barrel of crude oil published by EIA and emission factors published by the U.S. EPA.^{126,127} Table 3.27 outlines the fuel products, scaled gallons of each based on Colorado oil production rates, emission factors for CO₂, CH₄, and N₂O, and total CO₂eq emissions. Emissions were calculated as the product of fuel- or product-specific emission factors and production volumes, with unit conversions as needed.

Emissions from supplied natural gas were calculated as the product of GHG emission factors and natural gas produced in Colorado, as published by the EIA.¹²⁸ Table 3.28 outlines the products, the MMscf natural gas produced in Colorado, and emission factors for CO₂, CH₄, and N₂O, and total CO₂eq emissions.

In 2023, Colorado produced about 7,010 million gallons (167 million barrels) of oil, scaled to 7,442 million gallons of petroleum products to account for processing gain. The estimated GHG emissions from the consumption or combustion of these products were 70.732 Tg CO₂eq. In 2023, Colorado produced 1,826,767 million standard cubic feet (MMscf) of natural gas, corresponding to estimated emissions of 99.550 Tg CO₂eq. The estimated potential emissions from the complete combustion of both natural gas and petroleum products produced in Colorado in 2023 were 170.282 Tg CO₂eq.

¹²⁶ [EIA Refining Crude Oil Inputs and Outputs](#)

¹²⁷ [GHG Emission Factors Hub](#)

¹²⁸ [Colorado Natural Gas Gross Withdrawals](#)

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Table 3.27. Potential Emissions from Complete Combustion of Supplied Petroleum Products (Tg CO₂eq).

Product	Million gallons	g CO ₂ /gallon	Mg CO ₂	g CH ₄ /gallon	Mg CH ₄	g N ₂ O/gallon	Mg N ₂ O	Tg CO ₂ eq
Asphalt and road oil*	140	11,910	1,669,784	0.470	66	0.090	13	1.675
Distillate fuel oil	2,081	10,450	21,749,641	0.423	881	0.083	173	21.820
Finished aviation gasoline	7	8,310	55,479	0.360	2	0.070	0	0.056
Finished motor gasoline	3,266	8,780	28,678,385	0.380	1,241	0.080	261	28.782
Hydrocarbon gas liquids	259	6,060	1,567,739	0.276	71	0.053	14	1.573
Kerosene	7	10,150	67,763	0.410	3	0.080	1	0.068
Kerosene-type jet fuel	736	9,750	7,176,498	0.410	302	0.080	59	7.201
Lubricants*	63	10,690	678,001	0.430	27	0.090	6	0.680
Miscellaneous products*	35	10,590	371,180	0.420	15	0.080	3	0.372
Naptha for feedstocks*	57	8,500	482,355	0.380	22	0.080	5	0.484
Other oils for feedstocks*	35	8,880	311,244	0.380	13	0.080	3	0.312
Petroleum coke	344	14,640	5,033,588	0.430	148	0.090	31	5.046
Residual fuel oil	119	10,740	1,272,717	0.435	52	0.085	10	1.277
Special naphthas*	13	9,040	120,706	0.380	5	0.080	1	0.121
Still gas	280	4,500	1,261,802	0.090	25	0.010	3	1.263
Totals	7,442	-	70,496,883	-	2,873	-	582	70.732

* These products have non-energy uses (NEUs) and may not be fully combusted.

Table 3.28. Potential Emissions from Complete Combustion of Supplied Natural Gas (Tg CO₂eq).

Product	MMscf	g CO ₂ /scf	Mg CO ₂	g CH ₄ /scf	Mg CH ₄	g N ₂ O/scf	Mg N ₂ O	Tg CO ₂ eq
Natural Gas	1,826,767	54	99,449,195	0.001	1,882	0.000	183	99.550

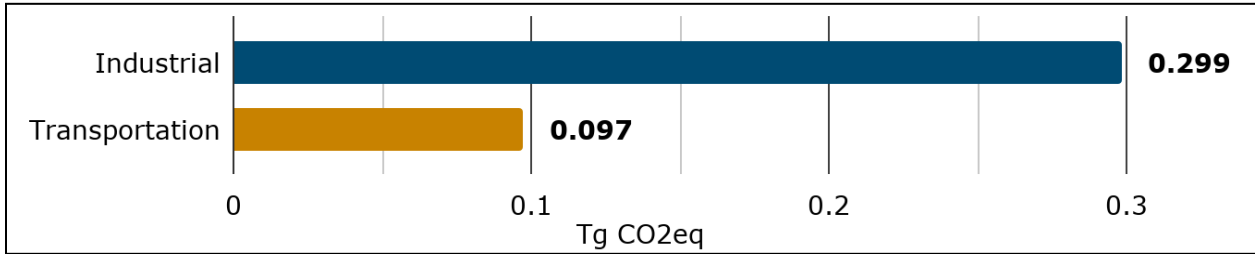
3.6. Non-Energy Use of Fossil Fuels

In addition to being combusted for energy, fossil fuels may be consumed for non-energy uses (NEUs). The non-energy applications of these fuels are diverse, including feedstocks for manufacturing, reducing agents for producing various metals, and products such as lubricants. CO₂ emissions may occur during production and/or throughout a product’s lifetime. The amount of fuel used for non-energy purposes is subtracted from the total fuel consumption data that would otherwise be combusted.

In Colorado, emissions result from Industrial and Transportation NEU of fossil fuels (NEUFFs). Fuels with Industrial NEUs include coal, natural gas, and petroleum. Industrial petroleum includes a variety of fuels: aviation gasoline blending components, asphalt and road oil, crude oil, distillate fuel, feedstocks, hydrocarbon gas liquids, kerosene, lubricants, motor gasoline, motor gasoline blending components, pentanes plus, petroleum coke, residual fuel, special naphthas, still gas, unfinished oils, and waxes. Fuels with Transportation NEUs include petroleum, specifically lubricants.

Emissions from NEUFFs were 0.396 Tg CO₂eq in 2023, accounting for 0.4% of the Energy Sector and 0.3% of statewide emissions (including or excluding LULUCF). As shown in Figure 3.35 below, most of the 2023 NEUFFs emissions were Industrial. About 75% of 2023 NEUFFs emissions were Industrial (0.299 Tg CO₂eq), and about 25% were Transportation (0.097 Tg CO₂eq). Lubricants (across both categories) accounted for over half of all NEUFFs emissions, totaling 0.227 Tg CO₂eq in 2023.

Figure 3.35. 2023 Non-Energy Uses of Fossil Fuels Emissions by Category.



NEUFFs emissions fluctuated from 2005 through 2023, as shown in Figure 3.36. Emissions peaked in 2006 at 0.597 Tg CO₂eq, then reached their lowest point in 2009 at 0.350 Tg CO₂eq. Emissions increased from about 2009 to about 2018 before falling slightly. In 2023, total NEUFFs emissions were 0.097 Tg CO₂eq (19.6%) below 2005 levels. Though Industrial NEUFFs emissions fluctuated more than Transportation NEUFFs emissions, both categories’ emissions decreased from 2005 through 2023. Industrial NEUFFs emissions decreased by 0.049 Tg CO₂eq (14.2%), and Transportation NEUFFs emissions decreased by 0.047 Tg CO₂eq (32.6%).

While all of the Transportation NEUFFs emissions are from lubricants, the Industrial NEUFFs emissions result from a variety of fuels. From 2005 through 2023, most of the emissions were associated with lubricants, hydrocarbon gas liquids, and natural gas. Emissions from lubricants

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and natural gas both increased, by 0.044 Tg CO₂eq (51.0%) and 0.012 Tg CO₂eq (13.0%), respectively. Emissions from hydrocarbon gas liquids decreased by 0.077 Tg CO₂eq (70.9%).

Figure 3.36. Historical Non-Energy Uses of Fossil Fuels Emissions by Category.

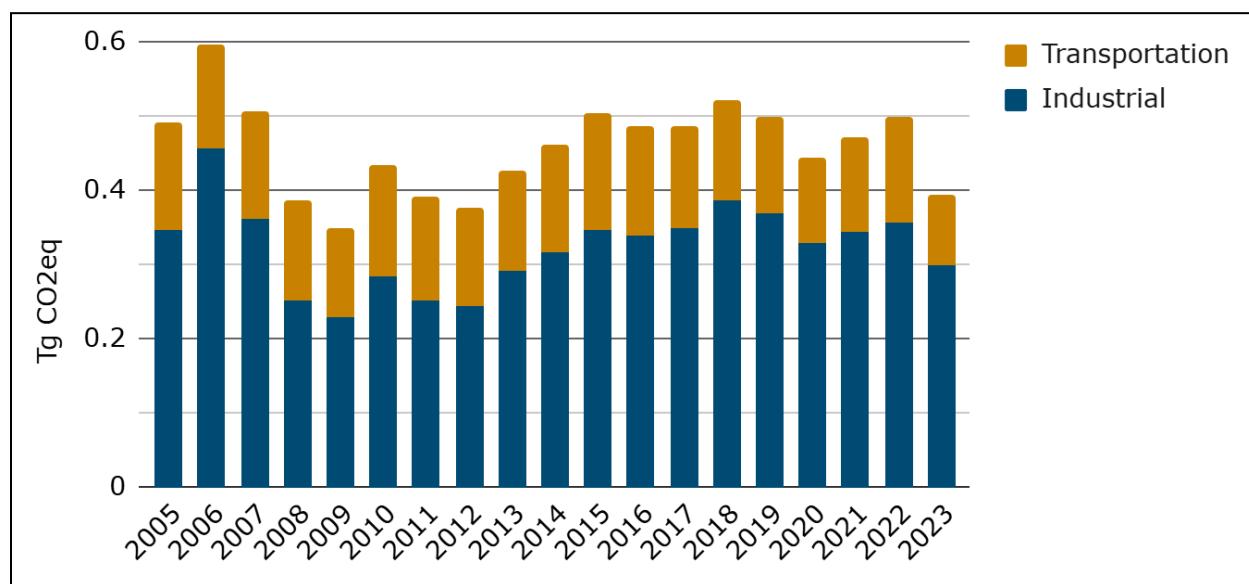


Table 3.29. Non-Energy Uses of Fossil Fuels Emissions by GHG and Category (Tg CO₂eq).

GHG	Category	2005	2010	2015	2020	2023
CO ₂	Industrial	0.348	0.284	0.348	0.330	0.299
CO ₂	Transportation	0.145	0.150	0.156	0.114	0.097
NEUFFs Total		0.492	0.434	0.504	0.444	0.396

3.6.1. Methodology

This Inventory estimated NEUFFs emissions using the SIT CO2FFC Module with some modifications. The CO2FFC Module estimates emissions from NEUFFs using a combination of national data on fossil fuel consumption for NEUs and carbon storage factors for each fuel (as applicable). The carbon storage factor represents the amount of carbon in NEUs that typically remains stored for longer than 20 years. This non-energy consumption is then subtracted from the total consumption to yield the net combustible consumption. Emissions from the net combustible consumption are presented under [Section 3.3 for Transportation](#) and [Section 3.2.3 for Industrial Fuel Use](#). The emissions were calculated as the product of the non-energy consumption, the storage factor, and the carbon content, along with unit conversions.

The SIT Module available at the time of this analysis included data only through 2022. To estimate emissions for 2023, the Division incorporated updated consumption data from the EIA's SEDS through 2023 and extrapolated inputs as needed. Default inputs were used where available for storage factors and carbon content coefficients. Where the inputs were variable across the time series, 2023 values were assumed equal to 2022 values.

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For Transportation-related NEUs, default data was used to estimate emissions from NEUFFs through the use of lubricants. The CO2FFC Module categorized all consumption of lubricants in the Transportation subsector as non-energy consumption and applied a storage factor of 9%, meaning 91% of the carbon is not retained in the lubricants.

For Industrial-related NEUs, default data was used to estimate emissions from NEUFFs for all fuels except natural gas. The fuels identified as having non-energy consumption included: asphalt and road oil, distillate fuel, hydrocarbon gas liquids, lubricants, petroleum coke, still gas, special naphthas, and natural gas. Each of these fuels had its own storage factor, ranging from 9.8% for lubricants to 99.56% for asphalt and road oil. For many fuels, the storage factors also varied over time. The same methodology was applied for natural gas, but it was only applied to the “other industrial” consumption and not to the upstream or midstream segment consumption. In Colorado, natural gas consumed in upstream and midstream operations is most likely to be burned as a fuel rather than used for other processes. Thus, the non-energy consumption deductions were not applicable to the plant and lease fuel sales data that inform upstream and midstream natural gas consumption. See Section 3.2.3 for more details on the disaggregation of natural gas consumption for Industrial Fuel Use.

3.6.2. Uncertainty

The SIT Modules do not quantify uncertainty. The NI by State did not quantify uncertainty for state-level estimates, but the NIR provides uncertainty ranges for national emissions estimates from NEUFFs. The uncertainty in the estimates presented in this Inventory is likely similar to the national uncertainty, as the carbon contents, emission factors, and fuel consumption were generally the same.

The EPA performed Monte Carlo uncertainty analyses by source for CO₂. Uncertainty in total national 2023 NEUFFs CO₂ emissions ranged from -36% to +62%. Uncertainty was assessed for fuel sources, but the fuels were not attributed to transportation or industrial uses. By fuel sources, uncertainty ranges were -42% to +77% for feedstocks, -58% to +116% for asphalt, -17% to +16% for lubricants, -26% to +103% for waxes, and -81% to +17% for other fuels. The primary sources of uncertainty identified were fuel consumption by U.S. territories, and the carbon content of fuels, and the products.¹²⁹

3.6.3. QA/QC

For each emissions source category, the Division applied standard QA/QC procedures, including reviews of data connections and imports, unit conversions, GWP calculations, naming conventions, and data formatting.

The Division also compared the emissions estimates from this Inventory using the SIT to the emissions estimates provided in the NI by State. Between the two, emissions estimates follow similar trends, though the estimates in this Inventory ranged from 0.051 Tg CO₂eq (15.1%) to 0.131 Tg CO₂eq (38.5%) higher than those in the NI by State. These differences were likely due

¹²⁹ [2025 NIR](#), Chapter 3, Section 3.1, CO₂ from Fossil Fuel Combustion, Uncertainty

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to the different data sources and methodologies used to disaggregate fuel consumption data. The NIR sourced fuel consumption data from several sources, including the [Monthly Energy Review](#), then the NI by State attributed consumption to sectors and states using SEDS, compared to the SIT, which mainly used SEDS directly for state-level consumption.

3.6.4. Recalculations and Comparisons to Prior Inventory

This Inventory and the prior Inventory used mostly the same methods, relying on the SIT CO2FFC, though this Inventory implemented updated consumption data. Unlike the prior Inventory, this Inventory did not back out the emissions from natural gas feedstock consumption for the production of hydrogen. The total natural gas feedstock consumption in FRD exceeded the NEU consumption of natural gas in the SIT, indicating emissions were likely not double-counted.

Across the common time series (2005 through 2020), emissions in this and the prior Inventories followed similar trends, but the emissions estimates in this Inventory were higher. Emissions in this Inventory ranged from 0.017 Tg CO₂eq to 0.075 Tg CO₂eq, higher than the estimates in the prior Inventory. Overall, the percent differences ranged from 3.7% to 24.8%. The majority of those differences come from Industrial emission estimates, specifically natural gas and petroleum. Estimates of Transportation NEUFFs emissions differences were minimal across the time series, with a maximum of 0.001 Tg CO₂eq, or 0.8% higher than the estimates in the prior Inventory.

3.6.5. Planned Improvements

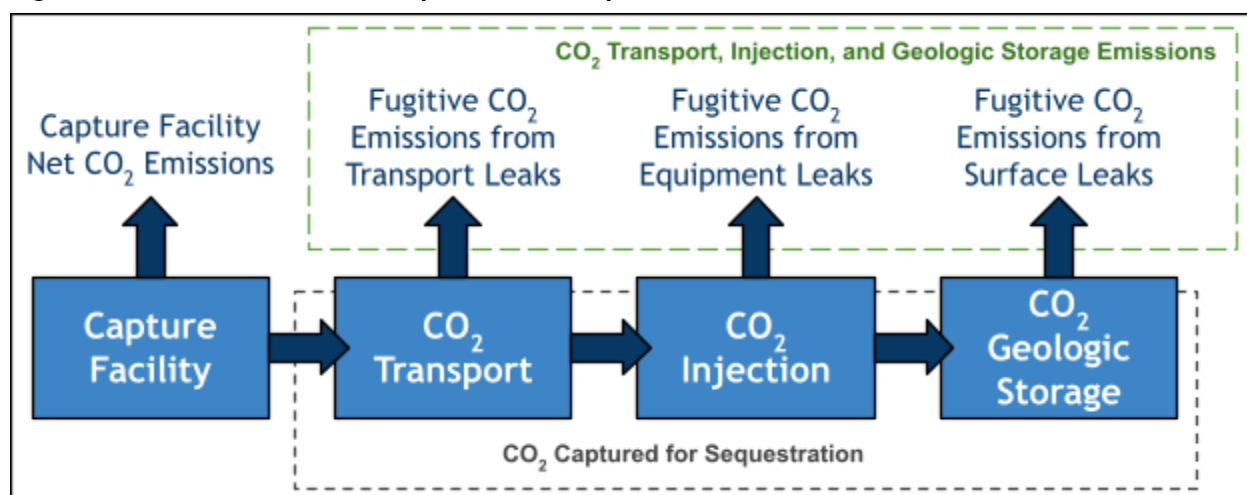
The Division will review the possibility of using FRD, the NI by State, or other new data sources to estimate the emissions from NEUFFs as applicable. This effort will include continued review of possible sources of double-counted emissions. Coordination and cooperation with other relevant external agencies and their data could improve data quality and assist in developing emissions factors specific to fuel use in Colorado. The methodologies and data sources that inform Transportation and Industrial Fuel Use estimates will be considered as well.

3.7. CO₂ Transport, Injection, and Geologic Storage

Carbon Capture and Storage (CCS) is one carbon management technology that Colorado is pursuing to achieve net-zero GHG emissions by 2050.¹³⁰ A CO₂ stream, either captured from an emission stream before being released into the atmosphere or generated from direct air capture, can be compressed and transported to an injection site where it is sequestered in geologic formations.¹³¹ The CO₂ Transport, Injection, and Geologic Storage (TIGS) subsector accounts for fugitive emissions occurring downstream of the capture facility during CCS. This includes fugitive emissions from CO₂ transport from the source to the injection site, activities at the injection site, and any surface leakage from the geologic reservoir. Surface leakage occurs when the injected CO₂ moves from the injection zone to the surface, and into the atmosphere, indoor air, oceans, or surface water. CO₂ may also leak or be vented from surface equipment.

The accounting of CO₂ emissions across the transport, injection, and storage chain is depicted in Figure 3.37. In this approach, a capture facility's net CO₂ emissions would be calculated by subtracting the amount of CO₂ captured for sequestration from its gross emissions. The net facility emissions would be accounted for under other Inventory categories applicable to the type of emission. For example, if a cement manufacturing facility captured some portion of its gross process emissions and transported them off-site, its net process emissions (equaling its gross process emissions minus the amount of captured CO₂) would be accounted for in the Cement Production category of the IPPU sector (Chapter 4, Section 4.1.1). The same approach is used for allocating biogenic CO₂ captured from ethanol plants, with the sequestered CO₂ accounted for in the IPPU sector (Section 4.6.3). If the source of the sequestered CO₂ is unknown or if multiple sources are listed, it is allocated across the source categories reported under [Subpart PP, Suppliers of CO₂](#).

Figure 3.37. Flow of CO₂ in Capture and Sequestration Processes.



¹³⁰ [Colorado Carbon Management Roadmap](#)

¹³¹ [DOE Explains: Direct Air Capture](#)

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CO₂ can be transported by pipeline or containers. While trucks and rail are used to move small quantities of CO₂, pipelines are often the safest, most efficient, and most cost-effective transport method.¹³² The NIR assumes all transport is via pipeline, and therefore, fugitive emissions from CO₂ transport are associated solely with pipeline leakage. Injection emissions include equipment leaks and venting of CO₂ from equipment between the flow meter used to measure injection quantity and the injection wellhead. Emissions from geologic storage arise if there are surface leaks, where the sequestered CO₂ is able to migrate back to the surface. Including these fugitive emissions from injection and storage has the potential to overestimate emissions from CCS, as they may already be accounted for in the calculation of the total amount of CO₂ sequestered, but due to the uncertainty in these estimates and the potential for other sources of CO₂ that aren't considered (e.g., from natural domes), it is considered appropriate to include them in this subsector.¹³³

In addition to CCS projects, where CO₂ is injected specifically for geologic storage, the process of enhanced oil recovery (EOR) can incidentally result in CO₂ sequestration. During EOR, CO₂ is injected to produce additional oil that would otherwise be unrecoverable. A portion of the injected CO₂ remains trapped in the reservoir, similar to CO₂ injected for sequestration, and the rest is produced along with hydrocarbons and brine through the production well. The methods used in the U.S. to estimate fugitive emissions from oil and gas production are not able to distinguish between CO₂ from EOR and CO₂ occurring in the produced natural gas. The emission factors used include CO₂ that was originally injected and is emitted through the production well, in addition to CO₂ from leaks, venting, and flaring. As a result, EOR CO₂ emitted through these pathways is included in the NGPS subsector.

Emissions from injection and geologic storage are based on data reported under [Subpart RR, Geologic Sequestration of Carbon Dioxide](#). Colorado did not have any subpart RR reporters across the entire time series, and therefore, no emissions were included for either category in the NI by State. Transport emissions are included for Colorado, as the state has CO₂ pipelines.¹³⁴ Transport emissions are below 0.0005 Tg CO₂eq for the entire time series, as shown in Figure 3.38. Table 3.30 details the emissions for select years. Because the emissions are less than 0.005 Tg from 2005 through 2023, the figure and table below show emissions in megagrams (Mg).¹³⁵

¹³² [Colorado Carbon Management Roadmap](#)

¹³³ [2025 NIR](#), Chapter 3, Section 3.9, CO₂ Transport, Injection, and Geological Storage, Methodology

¹³⁴ [National Pipeline Mapping System \(NPMS\) Public Map Viewer](#)

¹³⁵ 1 Tg = 10⁶ Mg

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Figure 3.38. Historical CO₂ TIGS Emissions.

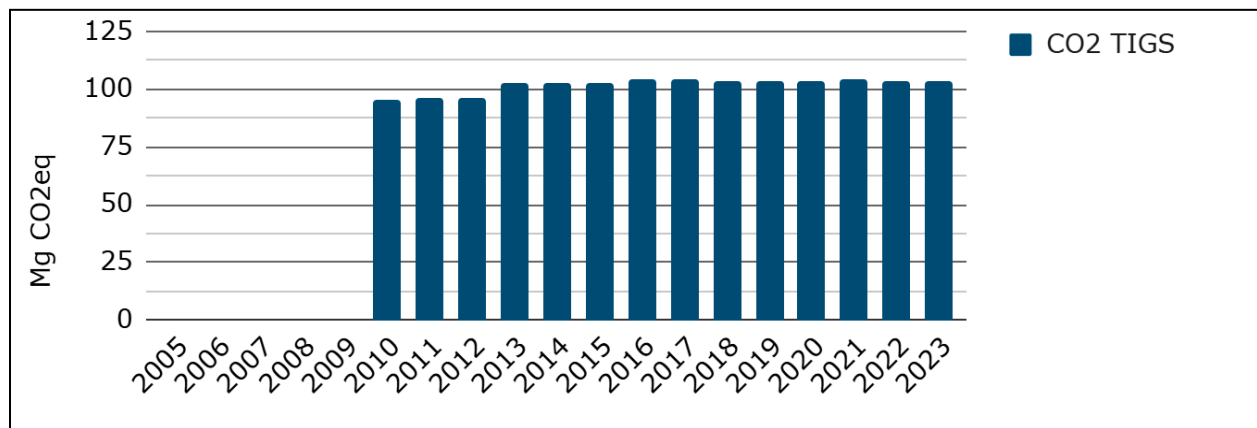


Table 3.30. CO₂ TIGS Emissions by GHG and category (Mg CO₂eq).

GHG	Category	2005	2010	2015	2020	2023
CO ₂	Transport	0	96	103	104	104
CO ₂ Transport, Injection and Geologic Storage Total		0	96	103	104	104

3.7.1. Methodology

This Inventory used the NI by State to estimate emissions from CO₂ TIGS. The NI by State relied on data provided by the EPA GHGRP, and did not include estimates for CO₂ sequestered prior to 2010.

The EPA estimated CO₂ emissions from transport based on leakage from pipelines using the IPCC Tier 1 default factor.¹³⁶ Pipeline leakage emissions were assumed to be independent of the throughput of CO₂ and were based only on the length of the pipeline. Though leakage rates depend on the type and size of equipment in the pipeline system, the national estimates were calculated using the average default fugitive emissions rate of 0.265 metric tons CO₂/kilometer (km) pipeline, based on empirical data and analysis.¹³⁷

The NIR calculated transport emissions based on the entire length of operational CO₂ pipelines in the country and the average of the default fugitive emissions rate range (0.265 metric tons CO₂/km pipeline). Pipeline length was obtained using data published by [PHMSA](#), the federal agency responsible for regulating the construction, operation, and maintenance of CO₂ pipelines.¹³⁸ The data are collected at the state level and summed to obtain national values, so disaggregation to estimate state-level emissions was not required.

¹³⁶ [IPCC, 2006 Guidelines, Volume 2, Chapter 5, Carbon Dioxide Transport, Injection and Geological Storage](#)

¹³⁷ [2025 NIR](#), Chapter 3, Section 3.9, CO₂ Transport, Injection, and Geological Storage, Methodology

¹³⁸ [2010+ Pipeline Miles and Facilities](#) Data Explorer

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3.7.2. Uncertainty

Though uncertainty was not quantified for state-level estimates, the NIR quantified uncertainty estimates for national emissions. The EPA calculated uncertainties for combined CO₂ TIGS emissions using the 2006 IPCC Guidelines Approach 2, which uses Monte Carlo simulations.¹³⁹ Disaggregated uncertainties for each category (e.g. transport) were not calculated. Uncertainty in national 2023 CO₂ TIGS emissions ranged from -51% to +51%.

3.7.3. QA/QC

For each emissions source category, the Division applied standard QA/QC procedures, including reviews of data connections and imports, unit conversions, GWP calculations, naming conventions, and data formatting.

The EPA applied QA/QC procedures consistent with the NIR QA/QC Management Plan, in accordance with IPCC Guidelines. The QA/QC Management Plan describes data and methodology checks, develops processes governing peer review and public comments, and provides guidance on conducting uncertainty analyses for the emissions estimates. The QA/QC Management Plan procedures also stress continual improvement, providing for corrective actions designed to improve the inventory estimates over time.¹⁴⁰

The Division also validated the NI by State data for CO₂ transport emissions using PHMSA data. PHMSA reported 243 miles (391 km) of CO₂ pipelines in Colorado in 2023, which results in estimated fugitive emissions of 104 Mg CO₂eq using an emissions rate of 0.265 metric tons CO₂/km pipeline. This is consistent with Colorado's transport emissions estimated in the NI by State for 2023.

3.7.4. Recalculations and Comparisons to Prior Inventory

Emissions from CO₂ TIGS were not accounted for in the prior Inventory. These emissions are new and additional to the entire time series.

3.7.5. Planned Improvements

Updates to the EPA's GHGRP that went into effect on January 1, 2025 may impact the CO₂ TIGS subsector in future Inventories. The EPA added [Subpart VV, Geologic Sequestration of Carbon Dioxide With Enhanced Oil Recovery Using ISO 27916](#). This created a reporting pathway for quantifying CO₂ sequestered during EOR operations when using the [ISO 27916:2019 standard](#). This may provide more data on CO₂ sequestered during EOR operations, analogous to the Subpart RR data used in this Inventory.¹⁴¹

Subpart PP was also updated to add reporting requirements that will provide a more direct link between carbon capture and CO₂ sequestration, including for CO₂ captured through direct

¹³⁹ [2025 NIR](#), Chapter 3, Section 3.9, CO₂ Transport, Injection, and Geological Storage, Uncertainty

¹⁴⁰ [NIR](#), Annex 8 QA/QC Procedures

¹⁴¹ [2025 NIR](#), Chapter 3, Section 3.9, CO₂ Transport, Injection, and Geological Storage, Planned Improvements

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air capture. Facilities that supply CO₂ to Subpart RR (geologic sequestration) or Subpart VV (EOR) facilities must now report the identification numbers for each facility to which CO₂ is transferred and the annual quantity of CO₂ transferred. This data would improve allocation of sequestered CO₂ across categories. To prevent double counting, the updates also clarify that wells reported under Subpart RR and VV are not also counted under [Subpart UU, Injection of Carbon Dioxide](#).

This Inventory does not include data for the CO₂ TIGS subsector prior to 2010. Future Inventories may explore alternate data sources to estimate emissions from 2005 to 2010, such as from the Regional Carbon Sequestration Partnerships ([RCSPs](#)) identified by the EPA.

3.8. Sector Comparison to Prior Inventory

For the Energy sector, this Inventory (2025) implemented several methodological and data updates since the prior Inventory (2023). The most significant changes were the addition of a new category (CO₂ TIGS) and in the NGPS subsector. The changes to NGPS emissions estimates accounted for most of the differences discussed below. Energy emissions in the prior Inventory were presented in million metric tons (MMT) CO₂eq, whereas this Inventory presents emissions in Tg CO₂eq (1 MMT = 1 Tg).

As shown in Figure 3.39 below, the emissions trends of the Energy sector were similar in both Inventories, though the estimates in this Inventory are generally lower. Emissions and trends of the Energy sector in this Inventory are similar to those in the prior Inventory, as shown in Figure 3.39. From 2005 through 2020, Energy emission estimates in this Inventory ranged from 20.844 Tg CO₂eq (15.0%) to 8.907 Tg CO₂eq (7.3%) lower than those in the prior Inventory. Although the emissions were different, Energy emissions peaked in 2010 in both Inventories, at 146.137 Tg CO₂eq in the prior Inventory and 128.784 Tg CO₂eq in this Inventory.

From 2005 through 2020, CH₄ emissions in this Inventory ranged from 22.225 Tg CO₂eq (52.9%) to 10.745 Tg CO₂eq (35.4%) below estimates in the prior Inventory. CO₂ emissions in this Inventory ranged from 0.961 Tg CO₂eq (1.2%) to 1.838 Tg CO₂eq (2.0%) higher than estimates in the prior Inventory. N₂O emissions in this Inventory ranged from 0.028 Tg CO₂eq (3.9%) lower to 0.011 Tg CO₂eq (2.6%) higher than estimates in the prior Inventory.

Figure 3.39. Comparison of 2023 and 2025 Inventory Energy Sector Emissions.

